

System Startup Guide CD-ROM

SW11-600

**Implementation
Startup & Reconfiguration - 1**

System Startup Guide
CD-ROM

**SW11-600
Release 681
10/2011**

Notices and Trademarks

Copyright 2008 by Honeywell International Srl.
October, 2011

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customers.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell, PlantScape, Experion PKS, and **TotalPlant** are registered trademarks of Honeywell International Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA
1-800 822-7673

About This Publication

This publication provides an example of the startup of a **TotalPlant** Solution (TPS) system for Release 630 and later, using Honeywell-supplied CD-ROMs. TPS is the evolution of TDC 3000^X.

The tasks in this publication are intended to show you how a system is initially configured. You should have filled in configuration forms based on the system you are working with, so you know exactly what information needs to be entered.

If you are upgrading to a newer release of software, consult the *Customer Release Guide* for procedures. If you are reconfiguring an existing LCN network, refer to the *Network Data Entry* manual (see References).

You should know how to use the Command Processor, the Data Entity Builder, the Picture Editor, the Text Editor, etc., before attempting this procedure. In addition, you should know Global User Station (GUS) and Windows NT navigation. This information is explained in the manuals listed in the References section of this manual.

Support and Other Contacts

United States and Canada

Contact: Honeywell Solution Support Center
Phone: 1-800-822-7673
Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Facsimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC-EMEA
Phone: +32-2-728-2345
Facsimile: +32-2-728-2696
Mail: TAC-BE02
Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Facsimile: +61-8-9362-9564
Mail: Honeywell Limited Australia
5 Kitchener Way
Burswood 6100, Western Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-9400
Facsimile: +91-20- 6603-9800
Mail: Honeywell Automation India Ltd
56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Facsimile: +82-2-792-9015
Mail: Honeywell Co., Ltd
4F, Sangam IT Tower, , B4-4Block
1590, DMC Sangam-dong, Mapo-gu
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6580-3500
Facsimile: +65-6580-3501
+65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Taiwan

Contact: Honeywell Global TAC – Taiwan
Phone: +886-7-536-2567
Facsimile: +886-7-536-2039
Mail: Honeywell Taiwan Ltd.
17F-1, No. 260, Jhongshan 2nd Road.
Cianjhen District
Kaohsiung, Taiwan, ROC
Email: Global-TAC-Taiwan@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Phone: +81-3-6730-7160
Facsimile: +81-3-6730-7228
Mail: Honeywell Japan Inc.
New Pier Takeshiba, South Tower Building,
20th Floor, 1-16-1 Kaigan, Minato-ku,
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

Elsewhere

Call your nearest Honeywell office.

World Wide Web

Honeywell Solution Support Online:

<http://www.honeywell.com/ps>

Training Classes

Honeywell Automation College:

<http://www.automationcollege.com>

Contents

SECTION 1 - INTRODUCTION.....	15
1.1 Scope of Coverage and Suggestions.....	15
1.2 How This Startup Guide is Organized	16
1.3 How to Use the Procedures in this Document	17
1.4 What to Do before Starting-up the System.....	18
Format History Modules	18
Have Reference Material Available	18
Understand Various Responses after the Enter Key Is Pressed.....	18
Understand the Function of the HELP key	18
Understand the Concept of Work Files	18
Understand Time Delays	18
Printer Use.....	18
Understand How to Respond to Messages.....	18
Know What Files You Are Copying or Deleting.....	18
1.5 References	19
1.6 Sample System this Document is based upon.....	20
SECTION 2 - OVERVIEW	21
SECTION 3 - INSTRUCTIONS	23
3.1 General Instructions	23
3.2 System Startup Task Record	25
3.3 Node Identification	27
3.4 CD-ROM Contents	28
SECTION 4 – PROCEDURE 1/PREPARE FOR NETWORK CONFIGURATION	31
4.1 Task 1 – Reset Nodes and Create Emulated Disks.....	31
4.2 Task 2 – Load the GUS with the UNP Personality	33
4.3 Task 3 – Set Time and Date.....	34
4.4 Task 4 – Create Volumes on Emulated Disks.....	35
4.5 Task 5 – Copy Files	37
SECTION 5 – PROCEDURE 2/CONFIGURE THE NETWORK.....	39
5.1 Task 6 – Establish Customer NCF	40
5.2 Task 7 – Configure Unit, Area, Console Names	41
5.3 Task 8A – Configure System Wide Values	43
5.4 Task 8B – Copy GUS Custom Backplane Software	46
5.5 Task 9A – Configure LCN Nodes	47
5.6 Task 9B – Configure System Wide Values For NGs.....	51
5.7 Task 10 – Configure HM Volumes	53
5.8 Task 11 – Install the NCF	57

SECTION 6 – PROCEDURE 3/PREPARE THE HISTORY MODULE(S)	59
6.1 Task 12 – Reload the GUS Personality and Set Time and Date	60
6.2 Task 13 – Load the History Module with the HM Initialization Personality	61
6.3 Task 14 – Initialize the History Module.....	63
6.4 Task 15 – Copy Software from Emulated Disks to the History Module	64
6.5 Task 16 – Copy GUS External Load Modules to the History Module	69
6.6 Task 17 – Copy Optional Software to the History Module	70
 SECTION 7 – PROCEDURE 4/CONFIGURE PROCESS NETWORKS AND HIWAYS.....	 71
7.1 Task 18 – Load On-Process Personalities to Gateways and Modules	72
7.2 Task 19 – Configure Hiways	74
7.3 Task 20 – Configure Boxes and Slots	75
7.4 Task 21 – Configure Process Networks and Hiways	77
7.5 Task 22 – Configure Logic Blocks.....	82
7.6 Task 23 – Configure UCN Nodes and Subsystems	85
 SECTION 8 – PROCEDURE 5/CONFIGURE CUSTOM FUNCTIONS.....	 95
8.1 Task 24 – Build and Compile Pictures	96
8.2 Task 25 – Build and Compile Buttons	101
8.3 Task 26 – Build and Load HG Libraries	102
8.4 Task 27 – Build and Compile Free Format Logs.....	103
8.5 Task 28 – Build and Load HM History Groups.....	106
8.6 Task 29 – Build, Compile and Load CL Programs	107
 SECTION 9 – PROCEDURE 6/CONFIGURE AREA DATABASE	 109
9.1 Task 30 – Build, Select, and Install Area Database	110
 SECTION 10 – PROCEDURE 7/LOAD AREA DATABASE.....	 123
10.1 Task 31 – Change Area Database on GUS Stations	123
 SECTION 11 – PROCEDURE 8/MODIFY THE DATABASE	 125
11.1 Task 32 – Change Network Configuration.....	125
 SECTION 12 – ERROR RECOVERY PROCEDURES.....	 127
12.1 Task 33 – Error Conditions, Reasons, and Recovery	127

APPENDIX A – LOAD GUS PERSONALITY ON ANOTHER GUS.....	129
A.1 Task A – Load the GUS Personality into Another Global User Station.....	129
APPENDIX B – SAVE HG OR PLCG CHECKPOINTS	131
B.1 Task B – Save Checkpoint Data for Hiway Gateway and Its Boxes.....	131
APPENDIX C – SAVE AM CHECKPOINT	133
C.1 Task C – Save Checkpoint Data for AM or CM Modules.....	133
APPENDIX D – EXCEPTION BUILDING	135
D.1 Task D – Build Points by Exception for HG, AM, A ^X M, CM, PLCG, and PM/APM/HPM Points.....	135
APPENDIX E – CREATE EXCEPTION BUILD FILES FROM IDFS.....	137
E.1 Task E – Create Exception Build Files from IDFs	137

List of Tables

Figure 1 The Sample System This Document is Based Upon	20
Figure 2 Overview of the System Startup Process	21
Figure 3 The Standard Format of System Startup Instructions	24
Figure 4 Node Identification.....	27
Figure 5 Overview of Configuring the Network	39
Figure 6 Prepare the History Module(s)	59
Figure 7 Configure Process Networks and Hiways.....	71
Figure 8 Configure Custom Functions	95
Figure 9 Picture Building.....	97
Figure 10 Picture Building Continued.....	98
Figure 11 Free Format Log Example.....	104
Figure 12 CL Source File Example	108
Figure 13 Build and Install the Area Database.....	109

List of Figures

Figure 1 The Sample System This Document is Based Upon	20
Figure 2 Overview of the System Startup Process.....	21
Figure 3 The Standard Format of System Startup Instructions.....	24
Figure 4 Node Identification	27
Figure 5 Overview of Configuring the Network.....	39
Figure 6 Prepare the History Module(s).....	59
Figure 7 Configure Process Networks and Hiways	71
Figure 8 Configure Custom Functions.....	95
Figure 9 Picture Building.....	97
Figure 10 Picture Building Continued	98
Figure 11 Free Format Log Example	104
Figure 12 CL Source File Example.....	108
Figure 13 Build and Install the Area Database	109

Section 1 - Introduction

1.1 Scope of Coverage and Suggestions

This "how-to" document covers the engineering tasks necessary to initially configure a History Module-based **TotalPlant** Solution (TPS) system. Its step-by-step instructions cover all file and emulated disk manipulations necessary for initial system startup. It does not cover all permutations of data that can be entered. For instance, in developing the guide, we configured one Unit, one Area, a few points, and one Group; otherwise, the step-by-step instructions would have been far too bulky.

During Tasks 6-11 configure as much of the network as possible. At the very least, you will have to use the Online Network Reconfiguration procedures if you later wish to change the Network Configuration. Pay particular attention to accurately and completely configuring the History Modules; otherwise, you will have to perform time-consuming procedures if you have to change the History Module's volume configuration.

Refer to the Customer Release Guide to determine exactly what functions are included in the current software.

The procedures described in this publication require at least one Global User Station (GUS) with emulated disks from which configuration operations are performed.

1.2 How This Startup Guide is Organized

This guide is organized as follows:

Section (Number and Title)	You Should Use the Information to...
1. INTRODUCTION	Learn what this document is all about
2. OVERVIEW	Understand the overall System Startup process
3. INSTRUCTIONS General Instructions System Startup Task Record Node Identification CD-ROM Contents	Take action necessary to start up your system Understand the format of standard instructions Check off tasks completed and record information Determine what types of nodes you have Find necessary software
4. Prepare for Network Configuration	Tasks 1-5
5. Configure the Network	Tasks 6-11
6. Prepare the History Modules(s)	Tasks 12-17
7. Configure Process Networks and Hiways	Tasks 18-23
8. Configure Custom Functions	Task 24-29
9. Configure Area Database	Task 30
10. Load Area Database	Task 31
11. Modify the database	Task 32
12. Error Recovery Procedures	Solve a problem that you have encountered
A. APPENDIX	Load the GUS Personality into another Global Universal Station (GUS)
B. APPENDIX	Save Hiway Gateway checkpoint data in the History Module and on emulated disks
C. APPENDIX	Save Application Module checkpoint data in the History Module and on emulated disks
D. APPENDIX	Build points by exception
E. APPENDIX	Create exception build files from IDFs

1.3 How to Use the Procedures in this Document

The procedures in this manual were developed on a sample system. Use them as a guide for starting up a system.

One person can easily perform these procedures, but if two people are available, the most efficient method is to have one person read the step-by-step instructions while another person keys in the data.

1.4 What to Do before Starting-up the System

Format History Modules

The system's hardware must be installed and checked out before performing system startup. Each History Module (HM) must be formatted before system startup. This is normally done at the factory. See your maintenance technician or Honeywell support engineer if there is any question.

Have Reference Material Available

You should have access to the publications listed under References during the entire system startup process.

Understand Various Responses after the Enter Key Is Pressed

In the Command Processor function, when the <ENTER> key is pressed, the command entered is redisplayed with all characters in the command in uppercase. In the Network Configuration functions, when the <ENTER> key is pressed, the data is entered and reappears in upper case and in cyan. Also in the Network Configuration functions, when a target (pick) is selected, it usually fills with full intensity white. After the <ENTER> key is pressed, the target reappears filled with half-intensity white. In Point Build, HG Library, and Area Database Parameter Entry Displays (PEDs), when the <ENTER> key is pressed the data entered reappears in cyan and in upper case.

Understand the Function of the HELP key

When you are entering configuration data, help information is available. If you get a red error message or if a data entry region reappears in red after you press the <ENTER> key, press the <HELP> key and any available help information is displayed.

Understand the Concept of Work Files

Work files accumulate configuration data that you enter until you perform an "install" function, which renames the work file to become your permanent site-specific configuration file.

Understand Time Delays

As you perform some of the procedures in this guide, time delays can occur. You may have to wait up to several minutes for a few of the responses listed in the instructions.

Printer Use

It is an advantage to have a printer attached to the Global User Station (GUS) where the configuration tasks are being performed. This allows you to print the display and perform file-print operations on that printer.

Understand How to Respond to Messages

The *TPS Messages Directory* gives you a definition of the message and error recovery procedures, if available.

Know What Files You Are Copying or Deleting

If you add "-D" at the end of command lines such as copy, delete, and so on, the software lets you see exactly which files are being copied or deleted. Using this feature allows you to know that all file operations are completed successfully.

1.5 References

Publication Title	Binder
<i>Advanced Process Manager Control Function and Algorithms</i>	Implementation/Advanced Process Manager – 1
<i>Advanced Process Manager Parameter Reference Dictionary</i>	Implementation/Advanced Process Manager – 2
<i>Application ModuleX User Guide</i>	Application ModuleX
<i>Button Configuration Data Entry</i>	Implementation/Engineering Operations – 1
<i>Command Processor Operation</i>	Implementation/Startup & Reconfiguration – 1
<i>Control Language/Application Module Data Entry</i>	Implementation/Application Module – 3
<i>Control Language/MC Reference Manual</i>	Implementation/Hiway Gateway – 2
<i>Control Language/MC Data Entry</i>	Implementation/Hiway Gateway – 2
<i>Configuration Data Collection Guide</i>	Implementation/Startup & Reconfiguration – 2
<i>Data Entity Builder Manual</i>	Implementation/Engineering Operations – 1
<i>Engineer's Reference Manual</i>	Implementation/Startup & Reconfiguration – 2
<i>Free Format Log Data Entry</i>	Implementation/Engineering Operations – 1
<i>High-Performance Process Manager Control Function and Algorithms</i>	Implementation/High-Performance Process Manager – 1
<i>High-Performance Process Manager Parameter Reference Dictionary</i>	Implementation/High-Performance Process Manager – 2
<i>History Module Service</i>	LCN Service – 2
<i>HM History Group Form Instructions</i>	Implementation/Engineering Operations – 1
<i>Logic Manager Control Functions</i>	Implementation/Logic Manager
<i>Logic Manager Parameter Reference Dictionary</i>	Implementation/Logic Manager
<i>Messages Directory</i>	Implementation/Startup & Reconfiguration – 3
<i>Network Form Instructions</i>	Implementation/Startup & Reconfiguration – 1
<i>Network Gateway Implementation and Operation</i>	Implementation/Network Gateway
<i>Picture Editor Data Entry</i>	Implementation/Engineering Operations – 3
<i>PLC Gateway Control Functions</i>	Implementation/PLC Gateway
<i>Process Manager Control Functions and Algorithms</i>	Implementation/Process Manager – 1
<i>Process Manager Parameter Reference Dictionary</i>	Implementation/Process Manager – 2
<i>Control Language/Process Manager Data Entry</i>	Implementation/Process Manager – 2
<i>Safety Manager Module Control Functions</i>	Implementation/Safety Manager
<i>Safety Manager Module Parameter Reference Dictionary</i>	Implementation/Safety Manager
<i>Text Editor Operation</i>	Implementation/Engineering Operations – 3
<i>GUS Customer Release Guide</i>	
<i>Customer Release Guide</i>	TPS Network

1.6 Sample System this Document is based upon

Figure 1 shows the system that this guide is based upon. Please memorize the definitions and be aware that your node numbers may be different. A Programmable Logic Controller Gateway (PLCG) and its associated controllers was also temporarily connected.

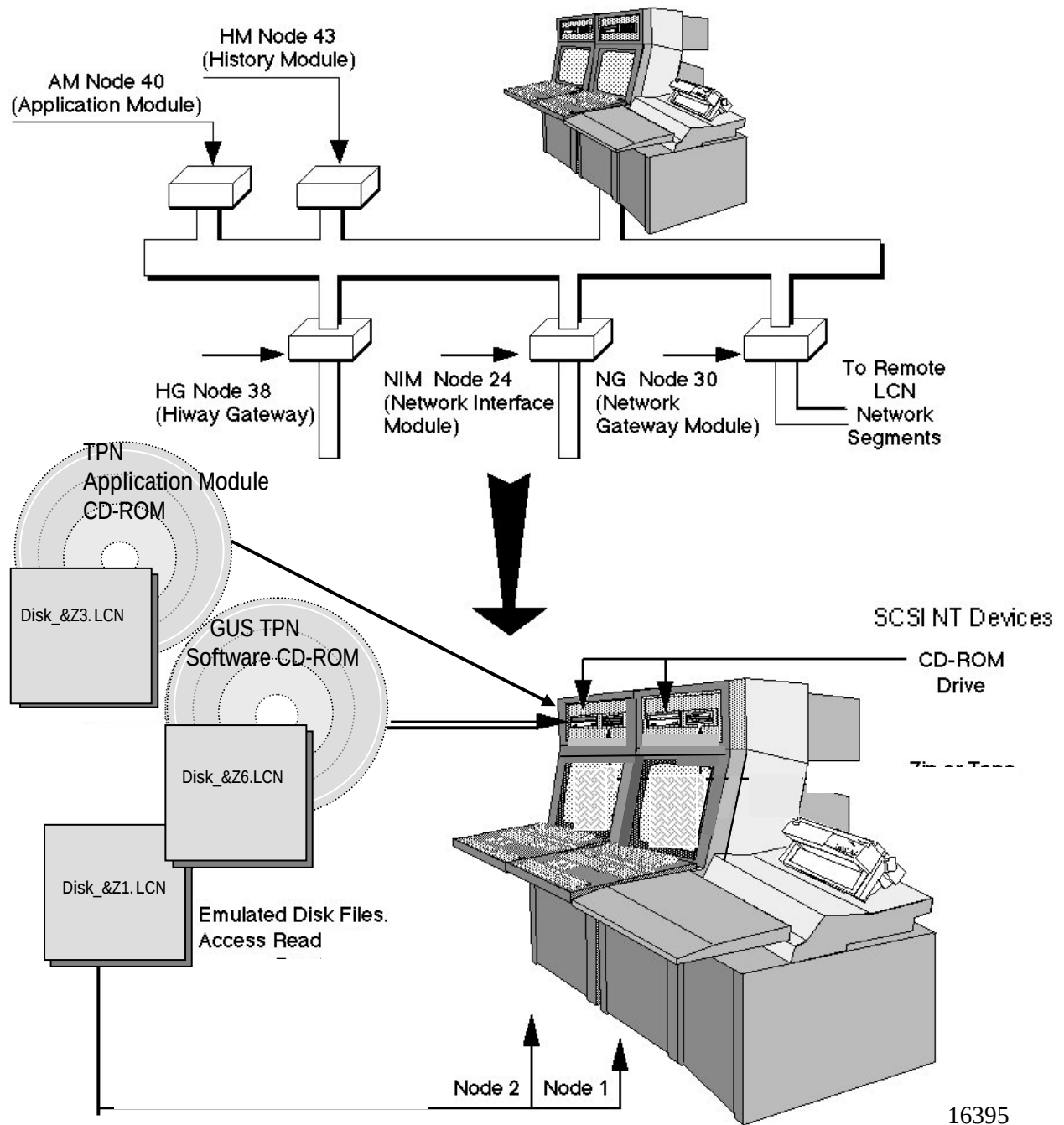


Figure 1 The Sample System This Document is Based Upon

Section 2 - Overview

Figure 2 shows an overview of the build and modification sequences involved in the TotalPlant Solution system startup process.

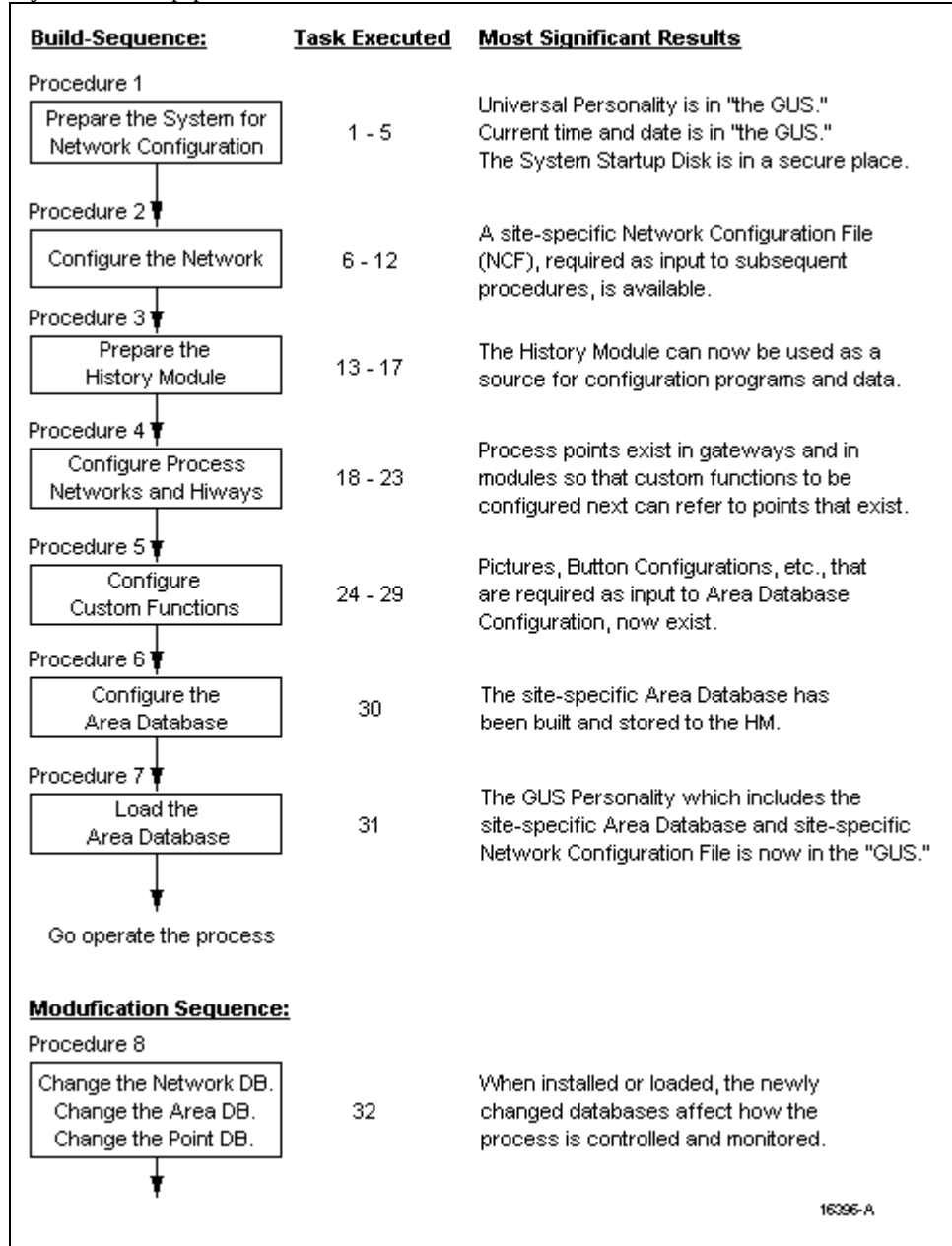


Figure 2 Overview of the System Startup Process

Section 3 - Instructions

3.1 General Instructions

The process of starting up a system is really the process of following step-by-step instructions that tell you how you use configuration programs and configuration data to specify the specifics of your unique site.

Program and Data Sources—During Tasks 2 through 17, the program and data sources are emulated disks. After the History Module (HM) has been loaded in Task 17, (if it is necessary to restart the system) the GUS Personality must be loaded from W (for workstation); however, most other data and program sources can be read from the History Module by entering N (for NET) when the source is requested. Some infrequently used software is not loaded onto the HM, but can be accessed from emulated disks, if needed.

After an operating personality is running, emulated disks are accessed by their logical-device numbers (for example, \$F1 for Drive 1 and \$F2 for Drive 2, and so on.).

Special Software—Two types of software are available for most node types. If a node contains an HMPU, K2LCN, or HPK2, you must load 68020 type software into that node. Also, in Task 10 you must specify the software used by these nodes as TYPE II (for example, HG II, USUNP II, and so on). If you do not know which nodes contain these boards, refer to Figure 4 for help. If a node contains a K4LCN, you must load 68040 type software into that node. In Task 10 you must specify the software used by these nodes as TYPE IV (for example, NIM IV, AM IV).

Organization—The organization of this section is based on the following hierarchy:

Procedure

 One or more Tasks

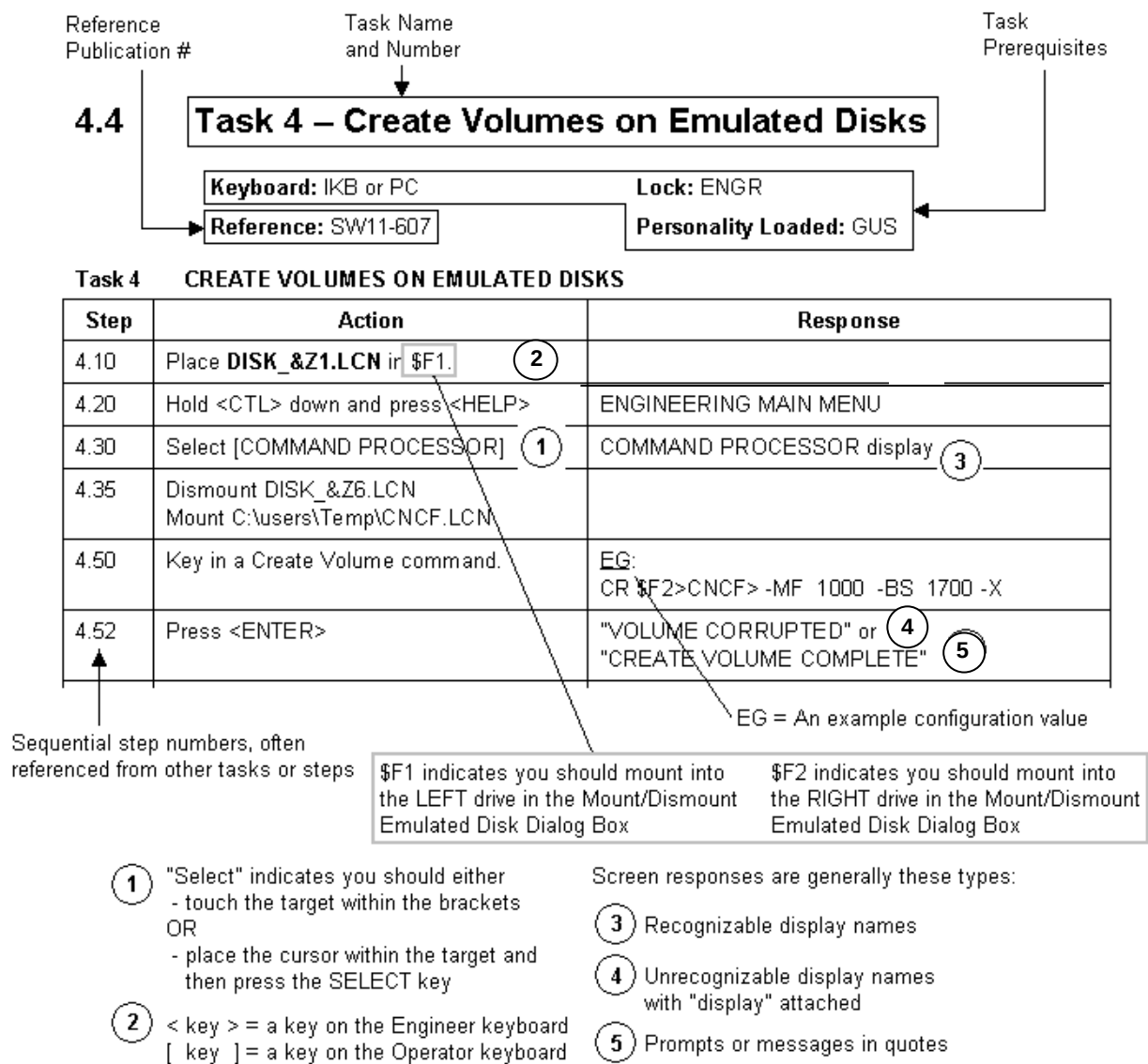
 Multiple Instructions

Eight procedures are made up of 32 individual tasks. The first seven procedures (Tasks 1-31) must be performed in sequence the first time you configure a system. Procedure 8 (Task 32) is a prerequisite for changing the database after your system is up and running on-process. Some tasks instruct you to perform a task previously performed; therefore, the instructions in a given task are generally written as free-standing or re-entrant.

To make it easier for you to find a particular task, a task number has been printed in the upper left-hand and right-hand corner of each page. You can use the Table of Contents to first find Task Numbers and then find instructions.

Instructions—Instructions are presented as Action/Response pairs. Figure 3 shows the standard format of a sample task within a procedure. You must understand the parts of the standard format before performing the instructions that follow.

Examples—Examples are used extensively. They are based upon the sample system and example values used for developing this guide. If you decide to use the examples, you must substitute your site-specific values for our example values in most cases, but be consistent. Use the same Area and Console IDs throughout the procedure.



16397-A

Figure 3 The Standard Format of System Startup Instructions

3.2 System Startup Task Record

As each task is performed and completed you should record the indicated information and check the associated task box. You will find the recorded information useful to have when performing subsequent tasks.

Task # Information to be recorded or checked off when configured. *As you reset Nodes, check them off; also record Node Nos.*

☐ 1 US ☐ AMs ☐ HMs ☐ HGs ☐ NIMs ☐ CGs ☐ PLCGs ☐

NGs ☐ UXs ☐ AXMs ☐ GUSs ☐ APP ☐

☐ 2 Physical Device Nos:

Zip/Cartridge Drive Nos: _____, _____, _____, _____, _____, _____

Printers: _____, _____, _____, _____

☐ 3 ☐ 4 ☐ 5 ☐ 6

☐ 7 Unit IDs: _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____; _____

Area Name: _____; Console Name: _____

☐ 8

☐ 9 Nodes: Type/Numbers : _____, _____, _____, _____, _____, _____

_____, _____, _____, _____, _____, _____

Hiway Numbers: _____, _____, _____, _____

UCN Numbers: _____, _____, _____, _____

* For configuration tasks in this document, CIU, CG and CM broadly refer to the same thing.

- ☐ 10 Area Names: _____; _____; _____
 User's Volume IDs: _____, _____, _____
 11 ☐ 12 ☐ 13 ☐ 14 ☐ 15 ☐
 16 ☐ 17 ☐ 18 ☐ 19 ☐
- ☐ 20 Hiway Box Types/Box Nos/Slot Nos:
 ____/____/____ ____/____/____ ____/____/____ ____/____/____ ____/____/____
 ____/____/____ ____/____/____ ____/____/____ ____/____/____ ____/____/____
- ☐ 21 Point-IDF Names: _____; _____; _____; _____
 Point IDs: _____

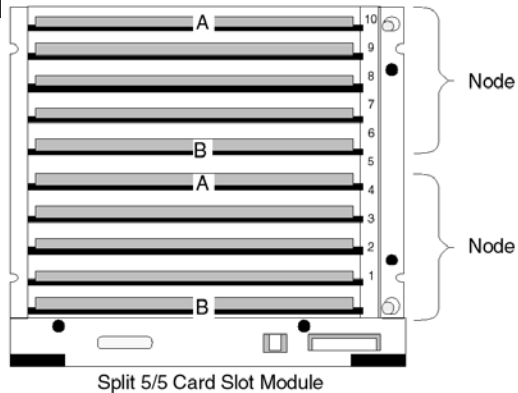
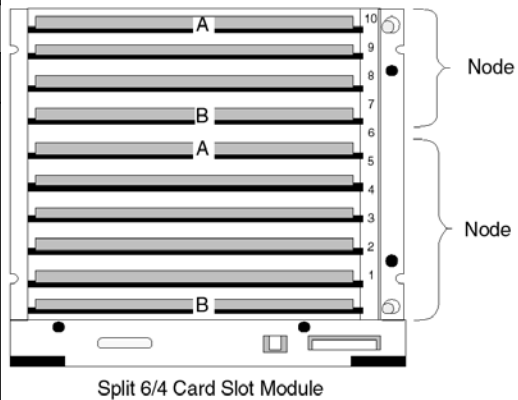
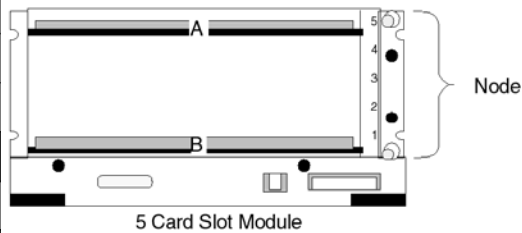
- ☐ 22 Logic Block Conf.
- ☐ 23 Node (PMM) No./Primary (P),Redundant (R):
 ____/____ ____/____ ____/____ ____/____ ____/____
- ☐ 24 Picture Filenames: _____
- ☐ 25 Button Filenames: _____
 Picture ____/Button # ____; Picture ____/Button # ____
 Picture ____/Button # ____; Picture ____/Button # ____
 Picture ____/Button # ____; Picture ____/Button # ____
- ☐ 26 HG Library Names: _____
- ☐ 27 Free Format Log Filenames
- ☐ 28 HM History Groups
- ☐ 29 CL Configuration
- ☐ 30 Area Database—IDFs: _____
 Areas Installed: _____
- ☐ 31 Node Numbers of USs Loaded: _____

3.3 Node Identification

For nodes other than GUS nodes, Figure 4 shows how to determine what type of nodes you have, their node numbers, and which ones require 68020 vs. 68040 software. Many modules have a tag to identify the module type. You will need this information when loading software, and for Task 10 configuration.

PRINTED WIRE BOARD A:

Board ID	Node Type
CNI	Plant Network CG
CLI	Computer Gateway
DHI	Hiway Gateway
NGI	Network Gateway
PNI/EPNI	Network Intf. Module
PLCI/EPLCI	PLC Gateway
WDC	History Module (HM)
AMR/ EAMR	Redundant AM
SPC MMEM EMEM PMEM or no board	If the node contains an EPDG, TPDG, or a VDG board in any slot, it is a US or U^XS. If not, it is an AM, A^XM, or HM. The A^XM also contains a WSI board. An HM also contains an SPC board.



PRINTED WIRE BOARD B:

If printed wire board B is an HPMU, HPK2 or K2LCN, the node uses Type II software.

If printed wire board B is a K4LCN, the node uses Type II software.

Examples:

Card Slot A = DHI
Card Slot B = HPMU
Node is Type II HG

Card Slot A = EPNI
Card Slot B = K4LCN
Node is a Type IV NIM

Node Number
The Node Number (or error code) is displayed on the board in Slot B.

Figure 4 Node Identification

3.4 CD-ROM Contents

GUS-TPN SYSTEM SOFTWARE on CD-ROM

The GUS-TPN System software is provided to you on CD-ROM. The following emulated disk files, and personality images can be found on the related CD-ROM for this release.

Table 1 CD-ROM Contents for GUS-TPN System Software

Emulated Disk File	Disk Name/ Area Affected	Directories/Contents
DISK_&Z1.LCN	System	&AMG (AM GDFs) &ARG (Area GDFs, initial AREA00 file) &CUS (External Load Module files) &DSY,&DS2 (Standard displays and display abstracts) &EC (Honeywell-supplied executable command files) &HGG (HG DDFs) &LDR (Boot loader files) &NMG, &NM2 (NIM GDFs) &OP1 (Universal Personality overlays) &OPR (US Operator personality files) &UNP (US Universal personality files) &OVG (Equipment list files) &CIO (Computing Module personality files) &HMO (HM Online personality files) &HMI (HM Offline personality files) &NMO (NIM personality files) &UCN (UCN node personality files) &NGO (NG personality files) &HGO (HG personality files) TLK1, DIA1 (PERFMENU and diagnostic displays) &KJF Kanji Font Files
DISK_&Z6.LCN	Customer Startup Files	&ASY (System startup NCF.CF and other files) &D01 (Null Area Database AREA01.DA and button object files) &I01-&I20 (Startup NIM and HG checkpoint files) &AMC (Startup AM checkpoint files) &CIC (Startup AM checkpoint files)

NOTE

Refer to your Software Change Notice for the latest information on CD-ROM contents.
Contents of CD-ROM may vary depending upon the release being used.

GUS-TPN APPLICATION MODULE on CD-ROM

Table 2 CD-ROM Contents for TPN Application Module

Emulated Disk File	Disk name/ Area Affected	Directories/Contents
DISK_&Z3.LCN	Application Module	&AMO (AM personality files) &LDR (Boot loader files) &CUS (Application Module External Load Module files) &CLX (Application Module Set Definition files) TLK2 (AM/CL Toolkit Files) &EC (Various Honeywell-supplied executable command files)

UCN SIMULATORS

Table 3 CD-ROM Contents for TPN Simulators

Emulated Disk File	Disk name/ Area Affected	Directories/Contents
DISK_&Z13.LCN	UCN Simulators	&UCN (TPS simulator personality files for APM and HPM) &EC (Honeywell-supplied executable command file to copy I/O UCN simulator images to the History Module)

TPS HARDWARE VERIFICATION TEST SUITE (HVTS) on CD-ROM

Table 4 CD-ROM Contents for TPN Hardware Verification

Emulated Disk File	Disk name/ Area Affected	Directories/Contents
DISK_&Z5.LCN	HVTS	&LDR Various hardware verification test files. &QLT &LCN &HVT &CMT &HIS &PMT &PM2 &PM3 &NMT &HMT &FAC &AMX

TPS Software on CD-ROM

The TPS software CD contains two emulated disk files that you will need for system startup.

Table 5 CD-ROM Contents for TPS System Startup

Emulated Disk File	Disk name/ Area Affected	Directories/Contents
DISK_&Z12.LCN	GUS Button File & Area Database GUS back-plane modules	&D01 (GUS Null Area Database Area01.DA and button source/object files) &CUS (GUSALA.LO, GUSALAG.LO, YGOCX.LO)
DISK_&Z14.LCN	GUS back-plane modules	&CUS (MSCHEM.LO, CSHEM.LO)

Section 4 – Procedure 1/Prepare For Network Configuration

4.1 Task 1 – Reset Nodes and Create Emulated Disks

Keyboard: N/A

Lock: N/A

Reference: This Document

Personality Loaded: N/A

NOTE

- All GUS node addressing and configuration should be done prior to the beginning of Task 1.
- Ensure that the GUS base software is installed prior to GUS personality installation.

The following procedure places the CD-ROM emulated disk file names in the Available (Dismounted) Emulated Disks list. To use these emulated disks, the proper CD-ROM must be mounted at the time of their use. Steps are saved later by creating all the necessary files now.

Task 1 RESET NODES AND CREATE EMULATED DISKS

Step	Action	Response
1.10	If not already running, execute the LCNP Status program from the "Honeywell GUS" program manager group and ensure that the LCNP is displaying the correct node number and is in the PWR_ON state.	
1.11	Start the Native Window program.	Native Window appears on the screen.
1.15	For all other LCN nodes: press the <RESET> button on the electronic chassis.	Nodes go to a reset state Note: After reset, the indicators near the reset button display the Node Number
1.20	Record all node numbers	
1.25	Turn power off to all HMs	
1.26	Mount the GUS-TPN System Software CD-ROM in the CD drive.	
1.27	From Native Window Access menu select "Mount/Dismount Emulated Disks"	Mount/Dismount emulated disk dialog box appears
1.28	Select Create	Select Directory and Filename Dialog Box appears
1.29	Select Drives (select CD drive letter)	Selected drive letter appears.
1.30	Select DISK_&Z1.LCN	
1.31	Select OK	Emulated disk file is added to Available (Dismounted) Emulated Disks list.
1.32	Repeat Steps 1.28 - 1.31 for the following: DISK_&Z6.LCN	
1.33	Replace the GUS-TPN System software CD-ROM with the TPN Application Module CD; repeat steps 1.28 - 1.31 selecting DISK_&Z3.LCN.	DISK_&Z3.LCN appears in the Available Dismounted Emulated Disk list.
1.34	Replace the TPN Application Module CD with the TPS System Software CD; repeat steps 1.28 - 1.31 selecting DISK_&Z12.LCN and again for DISK_&Z14.LCN	DISK_&Z12.LCN appears in the Available Dismounted Emulated Disk List. DISK_&Z14.LCN appears in the Available Dismounted Emulated Disk List.

Task 1 RESET NODES AND CREATE EMULATED DISKS

Step	Action	Response
1.35	Place the GUS-TPN System CD-ROM in the CD drive. Mount DISK_&Z1.LCN in \$F1 (Left); mount DISK_&Z6.LCN in \$F2 (Right)	
1.36	Place the TPS Keyfile Options floppy in the floppy disk drive; repeat steps 1.28 - 1.31 selecting the floppy disk drive and the file DISK_KF.LCN.	DISK_KF.LCN appears in the Available Dismounted Emulated Disk list.
1.37	Create several blank emulated disks on C:\USERS\TEMP\ for later use.	
1.40a	Select Create	Select Directory and Filename Dialog Box appears.
1.41b	Select the drive and directory where you want to store the emulated disk; type in the filename CNCF.LCN; press OK.	
1.42c	Enter a size and allocation type and press OK to add the file to the Available Disks list. Read/Write minimum size = 20.	Emulated Disk file is added to Available Dismounted Emulated Disk list.
1.43d	Repeat steps 1.40 - 1.42 using the following disk names: BKUPCNCF.LCN AREA_DB.LCN USER_VOL.LCN CHKPOINT.LCN	

NOTE

If the "Mount/Dismount Emulated Disks" dialog page is exited and then restarted, previously created disk files may be inadvertently removed from the Available Disks list.

If the disk you need is not in the Available Disks list, repeat steps 1.26 - 1.31 to add the disk to the list again.

NOTE

Mount the appropriate CD in the CD-ROM drive to use the following emulated disks:

Use the GUS-TPN System Software CD for:

DISK_&Z1.LCN

DISK_&Z6.LCN

Use the TPN Application Module CD for:

DISK_&Z3.LCN

Use the TPS Software CD for

Disk_&Z12.LCN

Disk_&Z14.LCN

Use the GUS-TPS Keyfile Options diskette for:

DISK_KF.LCN

Perform the following steps when a disk change is required during the execution of the PRS?VOLZ.EC file:

() _____ 1) DISMOUNT the currently mounted emulated disk and corresponding CD.

() _____ 2) MOUNT the required emulated disk and corresponding CD and continue as usual.

4.2 Task 2 – Load the GUS with the UNP Personality

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: None; Reset

CAUTION

In the tasks that follow, you must load software appropriate for the node type. If the node contains an HMPU, HPK2, or K2LCN printed wiring boards, you must use 68020 type software. If the node contains a K4LCN printed wiring board, you must use 68040 type software (see Table 1). The first GUS is started on Area 01. GUS processor type selection is automatic.

Task 2 LOAD THE GUS WITH THE UNP PERSONALITY

Step	Action	Response
2.05	Mount DISK_&Z1.LCN into the left drive.	
2.06	Mount DISK_&Z6.LCN into the right drive.	
2.07	Place the emulated key or keyswitch into the ENG position.	
2.10	Press (LOAD) on the IKB Keyboard (or <LOAD> on the Native Window Status Bar.	"W,N,1,2,3,4,X?"
2.20	Press (1) then press (ENTER)	"UNP? OPR?"
2.21	Press (U) then press (ENTER)	Series of messages, ending with "NODE STARTING UP" ... "NCF? N,1,2,3,4,X"
2.30	Press (2) and then press (ENTER)	"ABST N,1,2,3,4,X"
2.56	Press (1) and then press (ENTER)	"BUTT N, 1,2,3,4,X" (Will ask for button twice, but will not find any)
2.60	Press (1 or 2) and then press (ENTER)	SYSTEM STATUS Display appears
2.78	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
2.101	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
2.102	Select [MODIFY VOLUME PATHS]	MODIFY DEFAULT VOL PATHNAMES display
2.103	Select [SET DEVICE PATH TO REM. MEDIA]	Paths are changed from NET to \$Fn, \$Fn+1 (EG: \$F1 and \$F2)
2.104	Select [MAIN MENU]	ENGINEERING MAIN MENU
2.110	Select [CONSOLE STATUS]	CONSOLE STATUS AND ASSIGN display
2.120	Record device numbers for the printer and the drives.	PRINTER ----- DRIVES -----, ----- ACCESS ENGR Note: If the drive numbers are other than 1 and 2, remember to substitute the correct numbers in place of \$F1 and \$F2 wherever these are used in the tasks that follow.

4.3 Task 3 – Set Time and Date

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 3 SET TIME AND DATE

Step	Action	Response
3.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
3.20	Select [SYSTEM STATUS]	CONSOLE STATUS AND ASSIGN display
3.30	Select [TIME/DATE]	"ENTER TIME: HH MM" appears at top left
3.50	Key in hours-space-minutes	<u>EG</u> : 14 25
3.60	Press <ENTER>	<u>EG</u> : 14:25 in upper-right corner, "ENTER DATE: DD MMM YY" in upper left corner
3.80	Key in days-space-month-space-year	<u>EG</u> : 15 JUN 95
3.90	Press <ENTER>	<u>EG</u> : 15 Jun 95 in upper right corner

4.4 Task 4 – Create Volumes on Emulated Disks

Keyboard: IKB or PC

Lock: ENGR

Reference: Command Processor Operation

Personality Loaded: Universal

Task 4 CREATE VOLUMES ON EMULATED DISKS

Step	Action	Response
4.10	Place DISK_&Z1.LCN in \$F1.	
4.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
4.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
4.35	Dismount DISK_&Z6.LCN Mount C:\users\Temp\CNCF.LCN	
4.50	Key in a Create Volume command.	EG: CR \$F2>CNCF> -MF 1000 -BS 1700 -X
4.52	Press <ENTER>	"VOLUME CORRUPTED" or "CREATE VOLUME COMPLETE"
4.53	Key in a Create Directory command for the &CUS volume.	CD \$F2>CNCF> &CUS
4.54	Press <ENTER>	"CREATE DIRECTORY COMPLETE"
NOTE A 20 MB emulated disk supports up to 9995 files; however, some operations will be slowed as a function of the size of the directory; this example specifies 1000 files. Refer to the Command Processor manual for the maximum number of files supported by larger disks.		
4.55	Key in a Create Directory command for the &ASY volume.	EG: CD \$F2>CNCF> &ASY
4.56	Press <ENTER>	"CREATE DIRECTORY COMPLETE"
4.57	Repeat steps 4.55 and 4.56 to create &KFO volume.	EG: CD \$F2>CNCF> &KFO
4.65	Dismount the formatted emulated disk from \$F2	This will be the system-specific NCF.
4.66	Repeat steps 4.35 – 4.57 with emulated disk BKUPCNCF.LCN (using same commands)	These volumes are used for backing up customer databases as suggested in the note at the end of Task 29.
4.70	Repeat the Create Volume and Create Directory steps using the Volume/Directory names below to create additional volumes for future use.	
	1. Mount AREA_DB.LCN (Area database) in \$F2. for area button file for custom files Repeat step 4.70 if you have additional areas. 2. Mount USER_VOL.LCN (Volumes) in \$F2 for data points for custom graphics for CL	EG: CR \$F2>AR01> -MF 1000 -BS 1700 -X CD \$F2>AR01> &D01 CD \$F2>AR01> BUTT CD \$F2>AR01> &CUS EG: CD \$F2>AR02> &D02 CD \$F2>AR03> &D03 CR \$F2>USER> -MF 1000 -BS 1700 -X CD \$F2>USER> HMV1 CD \$F2>USER> PICT CD \$F2>USER> CL
4.80	Mount CHKPOINT.LCN in \$F2. Repeat the Create Volume steps to create a checkpoint volume.	CR \$F2>CKPT> -F -MF 1000 -BS 1700 -X

Task 4 CREATE VOLUMES ON EMULATED DISKS

Step	Action	Response
4.90	Mount Disk_&Z6.LCN in \$F1. Repeat the Create Directory steps to create checkpoint directories as needed	For HG or NIM: CD \$F2>CKPT> &Ixx where xx = Hiway or Process Network number
	Copy Null checkpoint files to the CKPT directories.	For HG or NIM: CP \$F1>&Ixx>*. * \$F2> &Ixx>= -D UNPT \$F2>&Ixx>*. * For CG: CD \$F2>CKPT> &Hxx where xx is the CG unit index number from Unit Names CP \$F1>&CIU>CIUxx MAS.CP \$F2>&Hxx>= -D For AM: CD \$F2>CKPT> &Fxx where xx is the AM unit index number from Unit Names CP \$F1> &AMC>AMxxxMAS.CP \$F2>&Fxx>= -D UNPT \$F2>&Fxx>*. * where xxx represents the AM Unit Name position number

4.5 Task 5 – Copy Files

Keyboard: IKB or PC

Lock: ENGR

Reference: Command Processor Operation

Personality Loaded: Universal

Task 5 COPY FILES

Step	Action	Response
5.10	Hold <CTL> down and press <HELP> and then select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
5.15	Dismount DISK_&Z1.LCN from the left drive.	
5.20	Mount DISK_&Z6.LCN into \$F1 or left drive.	
5.25	Dismount CHKPOINT.LCN from the right drive	
5.30	Mount the CNCF.LCN emulated disk created in steps 4.50 – 4.65 into \$F2 or right drive. Key in a copy command for the NCF.	<u>EG:</u> CPV \$F1>&ASY> \$F2>&ASY> -D where \$F1 and \$F2 are the drive numbers assigned in the DISK_&Z6.LCN disk (see step 2.120) or in Task 9. If the NCF has been installed (see step 9.95); these may differ
5.35	Press <ENTER>	Lists files copied, followed by: "COPY VOLUME COMPLETE"
5.36	Key in a copy volume command for the Key File Option files.	<u>EG:</u> CPV \$F1>&KFO> \$F2>&KFO> -D
5.40	Press <ENTER>	"COPY VOLUME COMPLETE"
5.42	Key in a copy volume for the &D01 File (&D01 contains null Area 1 and Button Files).	<u>EG:</u> CPV \$F1>&D01> \$F2>&D01> -D
5.44	Press <ENTER>	"COPY VOLUME COMPLETE"
5.60	If you intend to assign this GUS to any area 2 - 10, go perform steps 5.64 – 5.78. DISK_&Z6.LCN should be \$F1 and CNCF.LCN in \$F2 .	Note: This GUS started up on Area 1. The proper area database is needed to reload if intention is to configure this GUS on another area.
5.64	Create a &Dnn directory for whatever Area you will assign this GUS to (nn = 02 - 10).	<u>EG:</u> CD \$F2>CNCF> &D03 where nn = 03 area database for this example
5.66	Press <ENTER>	Create Directory Complete
5.70	Copy/Rename the &D01 volume for whatever Area (Dnn) you will assign this US to	<u>EG:</u> CPV \$F1>&D01> \$F2>&D03> -D where nn = 03 for this example
5.74	Press <ENTER>	"COPY VOLUME COMPLETE"
5.76	Rename the added area file within &Dnn	<u>EG:</u> RN \$F2>&D03>AREA01.DA AREA03
5.78	Press <ENTER>	RENAME COMPLETE

Section 5 – Procedure 2/Configure the Network

Figure 5 shows a graphic representation of the Tasks (6-11) that must be performed to Configure the Network.

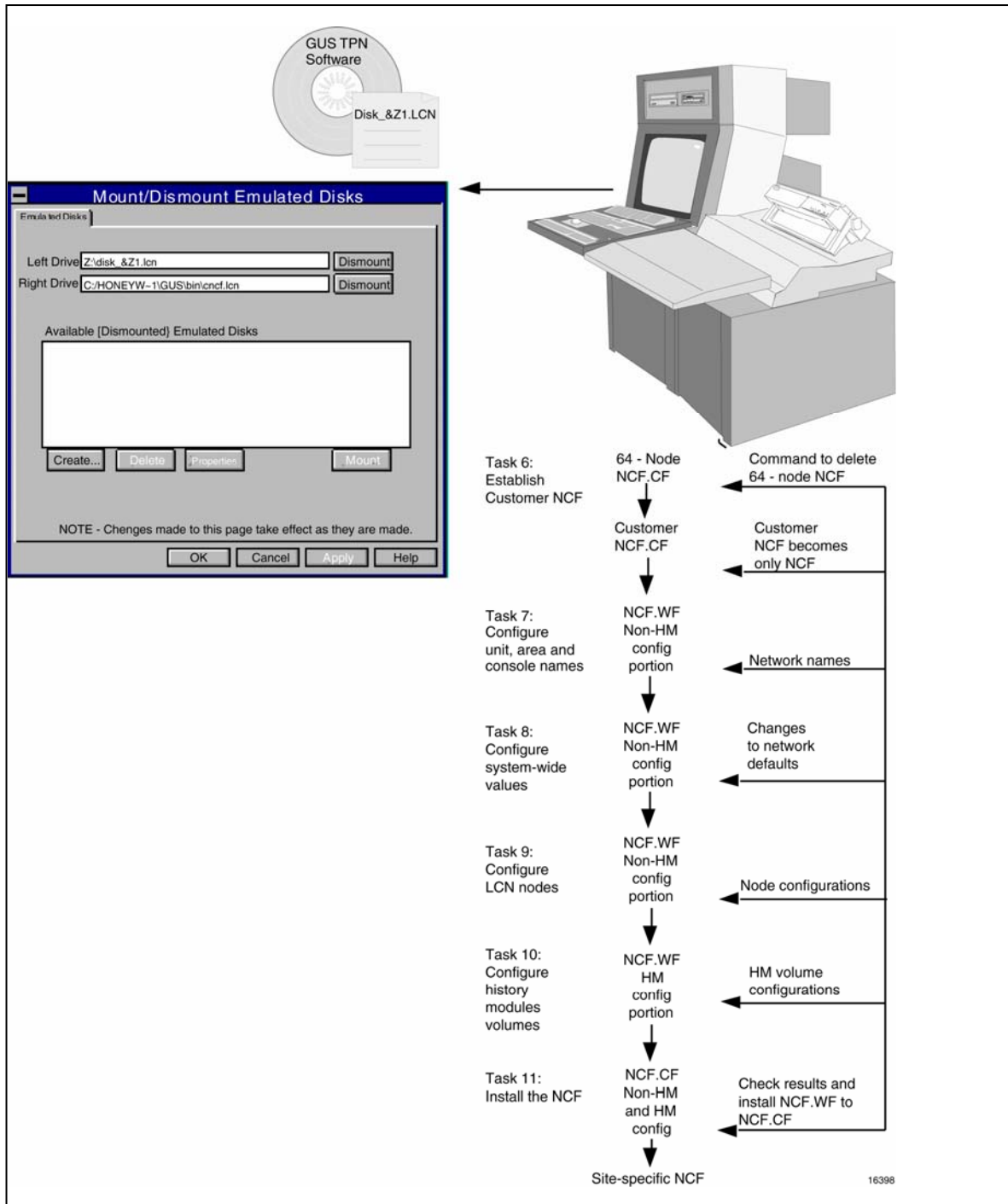


Figure 5 Overview of Configuring the Network

5.1 Task 6 – Establish Customer NCF

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

Task 6 ESTABLISH CUSTOMER NCF

Step	Action	Response
6.05	Dismount Disk_&Z6.LCN from the left drive.	
6.10	Mount Disk_&Z1.LCN into \$F1 (or left drive), and mount CNCF.LCN into \$F2.	CNCF is the primary copy made in Task 5.
6.20	If the COMMAND PROCESSOR display is on screen, go to step 6.50; if not, hold <CTL> down and press <HELP>.	ENGINEERING MAIN MENU
6.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
6.50	Use \$ F2 device number from step 2.120 and delete the NCF files.	EG: \$F2 EG: DL \$F2>&ASY>NCF.* -D
6.60	Press <ENTER>	"DELETE COMPLETE"
6.70	Key in a rename command (Rename customer NCF file).	<u>EG</u> : RN \$F2>&ASY>CSTMRNCF.CF NCF
6.80	Press <ENTER>	"RENAME COMPLETE"
6.90	Go to step 7.10. Be sure to read the warning below.	

WARNING

YOU SHOULD PERFORM TASKS 7 THROUGH 11 SEQUENTIALLY WITHOUT INTERRUPTION. DO NOT INTERRUPT THE SUGGESTED SEQUENCE AND DO NOT REMOVE THE CNCF DISK FROM THE DRIVE UNLESS YOU DO ONE OF THE FOLLOWING FIRST:

1. Return to Engineering Main Menu by holding <CTL> down and pressing <HELP>.
2. Perform the Install NCF function (Task 11).

5.2 Task 7 – Configure Unit, Area, Console Names

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

NOTE

If you are not entering data from filled in configuration forms or if you want a printout of your entries select File/Print Display to get a screen print after each configured page has been entered in Tasks 7-10.

With R681, NCF print option enables you to print NCF to a text file. Initiate a print using F4 function key from any NCF form to print the complete NCF as a text file. It will be stored in NET>&ASY by default with the file name as NCF.XX. If the default filename is not desired, specify a text file name using the extensions “.X?” or “.Y?” or “.Z?”.

Where,

? can be any alphabet.

If &ASY has memory shortage, you can print the NCF.XX to a different user volume.

You can use the existing functionality to abort printing – simultaneously press the keys CTL + Break. The printing is aborted and the partially printed text file is retained with a comment PRINT OPERATION ABORTED.

For an NCF parameter with multiple values, the values are printed continuously separated by a space. If the number of units characters exceeds 80 characters, the next line begins with &\, indicating it is a continuation of the previous line.

Note that in large networks having more than 64 nodes, the NCF text file takes approximately 30 minutes to print.

Task 7 CONFIGURE UNIT, AREA, CONSOLE NAMES

Step	Action	Response
7.10	Mount Disk &Z1.LCN into \$F1 (or left drive), and mount CNCF.LCN into \$F2.	
7.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.30	Select [UNIT NAMES]	EDIT UNIT NAMES—PAGE 1 display
7.32	If the words OFF LINE appears at the end of the 2nd line on the screen header, skip to step 7.40.	NCF.CF time stamp does not match network; Use F3 for OFF LINE to complete “Enter”
7.35	Set NCF to “ OFF LINE” Hold <CTL> down and press <3>	Warning Message: “OFF LINE” (Ignore backup reminder for now)
7.38	Press <ENTER>	EDIT UNIT NAMES display
7.40	Key in a UNIT ID and DESCRIPTION for each line item as required.	<u>EG</u> : 01 and UNIT 1 for line 1
7.50	When finished, press <ENTER>	Values and ports reappear in cyan
7.52	Page forward, if necessary, and continue entries. Press <ENTER> after each page.	
7.60	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.70	Select [AREA NAMES]	EDIT AREA NAMES display
7.80	Key in a AREA NAME and DESCRIPTOR for each line item as required. Do not leave blank lines between entries. Note: An Area Name cannot contain spaces.	<u>EG</u> : BOILER1 for line 1 Area Name <u>EG</u> : AREA 1 for Area 1 Description <u>EG</u> : REACTOR for line 2 Area Name <u>EG</u> : AREA 2 for Area 2 Description
7.90	When finished, press <ENTER>	Values and ports appear in cyan
7.100	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
7.110	Select [CONSOLE NAMES]	EDIT CONSOLES NAMES display
7.120	Key in a Console Description for each line item as required.	<u>EG</u> : WEST AREA CONSOLE for line item #1.

Task 7 CONFIGURE UNIT, AREA, CONSOLE NAMES

Step	Action	Response
7.130	When finished, press <ENTER>	Values and ports appear in cyan

5.3 Task 8A – Configure System Wide Values

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.10	Ensure DISK_&Z1.LCN is in \$F1; and CNCF.LCN is in \$F2.	
8.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
8.30	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU display
8.33	Select [SYSTEM ID]	SYSTEM ID display
8.36	Key into SYSTEM DESCRIPTOR port	<u>EG</u> : PLANT NUMBER 1
8.39	Key into SYSTEM ID port	
8.42	Press <ENTER>	Values and ports reappear in cyan
8.45	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.48	Select [CLOCK SOURCE]	CLOCK SOURCE display
8.51	Key into CLOCK LOCATIONS port (lowest number goes in left port).	<u>EG</u> : 24 and 38
8.54	Press <ENTER>	Values and ports reappear in cyan
8.57	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.60	Select [USER AVERAGE PERIOD]	USER AVERAGE PERIOD display
8.63	Select a User Average Period	<u>EG</u> : 5 MINUTES
8.66	Press <ENTER>	Values and ports reappear in cyan
8.69	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.72	Select [SHIFT DATA]	SHIFT DATA display
8.75	Key in changes as required	
8.78	Press <ENTER>	Values and ports reappear in cyan
8.81	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.84	Select [CONSOLE DATA]	CONSOLE DATA PAGE 1 display
8.87	Key in changes as desired for all pages (1-8).	<u>EG</u> : on page 3, Alarm Summary Freeze Timeout (seconds): 180
8.90	Press <ENTER>	Values and ports reappear in cyan
8.95	Press <PAGE FWD>	CONSOLE DATA—Page n display
8.110	Repeat steps 8.87 – 8.95 until pages 1 through 8 have been viewed and modified as required.	Note: Pages 9 and beyond are reserved. Do not change them without specific instructions.
8.111	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.114	Select [SOFTWARE OPTIONS]	SOFTWARE OPTIONS display
8.118	Ensure CNCF.LCN in \$F2	
8.119	Dismount DISK_&Z1.LCN from \$F1.	

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.120	Mount DISK_KF.LCN in \$F1 and ensure that the TPS Keyfile Options diskette is in the floppy drive.	
8.122	Return focus to Native Window. Hold down <CTL> and press <O>.	
8.136	Press <ENTER>	"BUSY"—SOFTWARE OPTIONS display
8.138	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.139	Dismount DISK_KF.LCN from \$F1 and mount DISK_&Z1.LCN again. If you do <u>not</u> have a Network Gateway, skip to step 8.172.	Note that NG Local System Ids must be configured before NG LCN Nodes.
8.140	Select [NG LOCAL SYSTEM ID]	NG LOCAL SYSTEM ID display
8.144	Key into LOCAL SYSTEM NUMBER port	<u>EG</u> : 1
8.148	Key into SYSTEM TITLE port (the first character must be an alpha)	<u>EG</u> : P1
8.152	Key into SYSTEM DESCRIPTION port	<u>EG</u> : Plant Number 1
8.156	Select a PIN ADDRESSING FORM option	<u>EG</u> : LOCAL
8.164	Press <ENTER>	Values and ports reappear in cyan Syst. Wide Value configuration for NGs continues in Task 9B after node configuration and HM volume configuration
8.172	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.176	Select [TAG NAME OPTIONS]	TAG NAMES display
8.180	Select a Tag Name option	<u>EG</u> : LONG
8.184	If you selected LONG, make a Formatting choice	<u>EG</u> : YES
8.188	Press <ENTER>	Values and ports reappear in cyan
8.192	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.196	Select [AM MULTIPLE PRIMMOD ALARMING OPTIONS]	AM MULTIPLE PRIMMOD ALARMING OPTIONS DISPLAY
8.200	Do either Step A or Step B: A. Select ENABLED EXCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters which have values. B. Select ENABLED INCLUSIVE MODE target to cause alarms to be reflected in alarm groups for those \$MPROD parameters which have values and also in the alarm group for PRIMMOD parameter.	
8.204	Press <ENTER>	Values reappear in cyan
8.208	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.212	Select [HG SPI/OP CLAMP OPTION]	HG SP/OP CLAMP OPTION DISPLAY

Task 8A CONFIGURE SYSTEM WIDE VALUES

Step	Action	Response
8.216	Select ENABLE OPTION target to enable the HG SP/OP Clamp Option functionality.	
8.220	Press <ENTER>	Values reappear in cyan
8.224	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
8.228	Hold <CTL> down and then press <PAGE BACK>	ENGINEERING MAIN MENU

5.4 Task 8B – Copy GUS Custom Backplane Software

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

Task COPY GUS CUSTOM BACKPLANE SOFTWARE

Step	Action	Response
8.300	Dismount DISK_&Z1.LCN	
8.305	Remove the GUS TPN Software CD and replace it with the TPS Software CD	
8.310	Ensure DISK_&Z14.LCN is in \$F1; and CNCF.LCN is in \$F2.	
8.320	Select [COMMAND PROCESSOR]	COMMAND PROCESS display
8.330	Key in a copy command for the GUS custom backplane files.	<u>EG</u> : CP \$F1>&CUS>*.LO \$F2>&CUS>= -D
8.350	Dismount DISK_&Z14.LCN from \$F1.	
8.360	Mount DISK_&Z12.LCN in \$F1.	DISK_&Z12.LCN is now mounted in \$F1.
8.370	From the [COMMAND PROCESSOR] key in a copy command for the GUS custom backplane files.	<u>EG</u> : CP \$F1>&CUS>*.LO \$F2>&CUS>= -D
8.380	Dismount DISK_&Z12.LCN	
8.90	Remove the TPS Software CD and replace it with the GUS TPN Software CD.	
8.400	Mount DISK_&Z1.LCN in \$F1.	DISK_&Z1.LCN is mounted.
8.410	Key in the Set Path command; type SP and press <ENTER>	Modify Volume Pathnames display.
8.420	Modify the volume path for EXT LOAD MODULES to now point to the logical device id of \$F2. (CNCF.LCN)	\$F2>&CUS>
8.430	Press <ENTER>	
8.440	Hold <CTL> down and press <HELP>	Engineering Main Menu appears on screen.

5.5 Task 9A – Configure LCN Nodes

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

WARNING

You must configure the Global Universal Station (GUS) you are using for system startup; otherwise, you will not be able to successfully start up your system. During Network Configuration, each display where you enter data must say OFF LINE at the end of line 2.

Task 9A CONFIGURE LCN NODES

Step	Action	Response
9.10	Ensure DISK &Z1.LCN is in \$F1; and CNCF.LCN is in \$F2.	
9.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
9.30	Select [LCN NODES] If necessary, <PAGE FWD> until you see the node number for one of your GUSs.	LCN NODE CONFIG—Page 1 display <u>EG</u> : NODE 1 LCN NODE CONFIG—Page n display
9.50	Select a GUS Node Number	<u>EG</u> : 1 SELECT NODE TYPE display
9.60	Select [UNIVERSAL STATION]	UNIVERSAL STATION NODE display
9.70	Key into CONSOLE # port	<u>EG</u> : 01 (for nodes 1 and 2)
9.80	Key into STATION # port	<u>EG</u> : 1 for node 1 (2 for node 2)
9.90	Key into DEFAULT AREA port	<u>EG</u> : BOILER1 (for nodes 1 and 2)
9.95	Key into CARTRIDGE DISK NUMBERS ports (ALWAYS)	<u>EG</u> : 1 and 2 for node 1 <u>EG</u> : on next pass: 3 and 4 for node 2 , etc.
9.110	Key into PRINTER NUMBER ports	<u>EG</u> : 1 for node 1, 2 for node 2, etc.
9.130	Select ACCESS LEVEL option	<u>EG</u> : [OPER] for nodes 1 and 2
9.134	Select DEFAULT LOAD PERS. option	<u>EG</u> : [UP] for nodes 1 and 2
9.140	Select OPERATORS KEYBOARD option (ALWAYS)	<u>EG</u> : [YES] for nodes 1 and 2
9.150	If this GUS supports touch screen, select TOUCH SCREEN option.	<u>EG</u> : [YES] for nodes 1 and 2
9.160	Select ENGINEERS KB option (ALWAYS).	<u>EG</u> : [YES] for nodes 1 and 2
9.161	<PAGE FWD> to enter values on the second page of the Universal Station display.	Universal Station Node page 2 of 2
9.162	If this is a GUS node, enter the following information for the External Load Module Names: Name Pers MSCHEM UP CSCHEM UP GUSALA UP GUSALAG UP YGOCX UP	
9.163	Press <ENTER>	Values and ports reappear in cyan.

Task 9A CONFIGURE LCN NODES

Step	Action	Response
9.164	Add 4000 to the Total (Modules plus additional memory) number displayed above the Maximum External Module Memory port; type the result in the Maximum External Module Memory port for "UP" personality.	
9.165	Press <ENTER>	Values and ports reappear in cyan.
9.180	If more GUSs, OPR, UNP, or UXs need to be configured, you can hold <CTL> down, press <PAGE BACK>, and go back to step 9.50.	<u>EG: 2</u> (configure as before, except Station 2, etc.)
9.190	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page 1 display.
9.195	<PAGE FWD> look for NIM node number (if no NIM on system, skip to 9.254)	LCN NODE CONFIG—Page n display.
9.200	Select a NIM Node Number. If no NIM on system, skip to 9.254.	<u>EG: 24</u> , SELECT NODE TYPE display
9.210	Select [NETWORK INTERFACE MODULE]	NETWORK INTERFACE MODULE Display
9.220	Key into REDUNDANT MEMBER ID port	<u>EG: 25</u> or 0 if none
9.230	Key into UNIV. CONT. NETWORK port	<u>EG: 3</u>
9.244	After last data is entered, press <ENTER>	Values and ports reappear in cyan.
9.245	Repeat 9.190 – 9.244 for each NIM node pair.	
9.250	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG - Page n display.
9.254	<PAGE FWD> (if necessary) to NG node number. If no NG on system, skip to 9.284.	LCN NODE CONFIG -- Page n display.
9.258	Select a Network Gateway Node Number	<u>EG: 30</u> , SELECT NODE TYPE display
9.260	Select [NETWORK GATEWAY]	NETWORK GATEWAY MODULE DISPLAY
9.264	Key into PIN LOCAL ADDRESS port	<u>EG: 8</u>
9.266	Key into PIN GLOBAL ADDRESS part (first half Honeywell provided global address, if any).	<u>EG:</u> No entry.
9.268	Select Configured Cables	<u>EG: A</u>
9.276	After last data is entered, press <ENTER>	Values and ports reappear in cyan.
9.278	Repeat 9.250 – 9.276 for each NG node.	
9.280	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG -- Page n display.
9.284	<PAGE FWD> (if necessary) to HG node number (if the system does not have an HG or a PLCG, skip to step 9.318)	LCN NODE CONFIG—Page n display
9.288	Select an HG or PLCG Node Number Note: HG and PLCG LCN nodes are configured exactly the same, but must be on different Data Hiways.	(<u>EG: 38</u>) SELECT NODE TYPE display
9.290	Select [HIWAY GATEWAY]	HIWAY GATEWAY NODE display
9.294	Key into REDUNDANT MEMBER ID port	<u>EG: 39</u> (using the next highest number for the REDUNDANT ID is recommended or 0 if none.)
9.298	Key into HIWAY NUMBER port	<u>EG: 2</u>
9.308	Press <ENTER> after last page.	
9.310	Repeat 9.280 - 9.308 for each HG node pair.	
9.314	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display

Task 9A CONFIGURE LCN NODES

Step	Action	Response
9.318	<PAGE FWD>; look for AM node No. (If the system does not have an AM, skip to step 9.370.)	LCN NODE CONFIG—Page n display
9.322	Select an AM or AXM Node Number	(EG: 40) SELECT NODE TYPE display
9.326	Select [APPLICATION MODULE]	APPLICATION MODULE NODE display
9.328	REDUNDANT MEMBER Node ID port	EG: 0 (Enter node number or enter 0 if no redundant node)
9.340	Select a STARTUP MODE option	EG: [Cold]
9.344	Key into ASSIGNED UNITS port (the units must have been previously configured) <u>Do not press Enter yet.</u>	EG: 01 02 03 04 (multiple units must be separated by spaces)
9.346	If AM/CL Background functions are used, or if this is an AXM, or if Custom AM Software will be used, Go to the next step—If not, skip to step 9.360.	
9.350	<PAGE FWD> to page 2. Type in Page 2 information. Obtain from: <i>AM Implementation Manual</i> (Background/CL) <i>AXM System Administration Manual</i> (AXM) or Custom AM Software instructions. These numbers may vary according to system requirements. <u>Do not press Enter yet.</u>	USER MEMORY ALLOCATION display. Note that Page 2 and 3 data can be entered or changed later with the On-Line Reconfiguration feature if you are uncertain. Typical entries for AXM/Background/CL are: EG: # BACKGROUND CL TASKS 10 # CONCURRENT DATA ACCESS 4 BACKGROUND TASK SIZE 15000 CVB SIZE (Default) 2000 INCLUDE INTERNETWORK PROC. YES USER MEMORY RESERVED 1
9.354	<PAGE FWD> to page 3. Enter information per same manuals as used in the previous steps. A combination of entries is allowable for multiple functions. Page 3—Further external directives must be set to “NO.” (This cannot be “YES” until the file NET>&CSD>APCFGxxx.x has been built.	EXTERNAL LOAD MODULE NAMES display. Typical entries for Background/CL are: EG: NAME PERSONALITY FILE AMO CONV AMO AMCL03 AMO Typical entries for AXM are: NAME PERSONALITY AMCL06 AMO XACCES AMO
9.360	Press <ENTER>	Page 1 and (if necessary Page 2/3) complete
9.365	Repeat 9.314- 9.360 for each AM or AXM	
9.366	Hold <CTL> down, press<PAGE BACK>	LCN NODE CONFIG—Page n display
9.370	<PAGE FWD>; look for HM node No. (If the system does not have an HM, skip to step 9.410.)	LCN NODE CONFIG—Page n display
9.372	Select an HM Node Number	EG: 43 SELECT NODE TYPE display
9.376	Select [HISTORY MODULE]	HISTORY MODULE NODE display
9.388	Key into NODE PAIR NUMBER port	EG: 1
9.402	Press <ENTER>	

Task 9A CONFIGURE LCN NODES

Step	Action	Response
9.406	Repeat 9.370 – 9.402 for each HM	
9.410	Hold down <CTL>, press<PAGE BACK>	LCN NODE CONFIGURATION display
9.412	<PAGE FWD> look for CM node number (if no CM on system, skip to step 9.454)	LCN NODE CONFIG—Page n display
9.416	Select a CM Node Number	<u>EG</u> : 51 , SELECT NODE TYPE display
9.420	Select [COMPUTING MODULE]	COMPUTING MODULE NODE display
9.422	Key in ASSIGNED UNITS port	<u>EG</u> : 01 02 03 04
9.430	If you have External Load Modules that affect the CM, press <PAGE FWD> (if External Load Modules not used, skip the next step).	EXTERNAL LOAD MODULE display. This can be done later with the On-Line Reconfiguration feature.
9.436	Key in data, select items as necessary. Press <ENTER> after last page.	Obtain entry information from the External Load Module documentation.
9.440	Repeat 9.410 – 9.436 for each CM	
9.446	Hold <CTL> down, press <PAGE BACK>	LCN NODE CONFIG—Page n display
9.454	To print the LCN Node Config. hold <CTL> down, then press <4>. Select printer target, then press <ENTER>.	(US must be connected to printer) Print instructions appear. LCN configuration is printed.

5.6 Task 9B – Configure System Wide Values For NGs

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

If you do not have a Network Gateway, skip to Task 10. Note that the following steps could not be completed until the NG nodes were configured (Task 9A). The following section completes NG configuration.

Task 9B CONFIGURE SYSTEM WIDE VALUES for NGs

Step	Action	Response
9.500	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
9.501	Select [SYSTEM WIDE VALUES]	SYSTEM WIDE VALUES MENU
9.502	Select [NG REMOTE SYSTEMS]	NG REMOTE SYSTEM display
9.503	Key into # port	<u>EG</u> : 2
9.504	Key into TITLE port	<u>EG</u> : P2
9.505	Key into DESCRIPTOR port	<u>EG</u> : Plant NO. 2
9.506	Select a CABLE FOR OUTPUT	<u>EG</u> : A
9.507	Key into RESP NG port	<u>EG</u> : 30
9.508	Key into REMOTE SERVERS RESP LOCAL port	<u>EG</u> : 16
9.509	Key in other ports as desired	
9.510	Press <ENTER>	Values and ports reappear in cyan.
9.520	Page forward and continue to make entries if necessary. Press <ENTER> after each page.	
9.530	Hold <CTL> down and then press <PAGE BACK>	SYSTEM WIDE VALUES MENU
9.540	Select [NG SECURITY DATABASE]	NG SECURITY DATABASE AUTH. display
9.541	Key in a REMOTE SYSTEM NUMBER or select ALL SYSTEMS	<u>EG</u> : ALL SYSTEMS
9.542	Select [DEFINE ACCESS RESTRICTIONS]	NG SECURITY DATABASE AUTHORIZ. display
9.543	Select a default access for System Vol/Dir.	<u>EG</u> : READ ONLY
9.544	Select a default access for USER Vol/Dir.	<u>EG</u> : READ /WRITE
9.545	Key in a SYSTEM VOL name under EXCEPTIONS if desired and select the access level.	<u>EG</u> : no entry
9.546	Key in a USER VOL name under EXCEPTIONS if desired.	<u>EG</u> : CSW1
9.547	Select default access for User Vol/Direc.	<u>EG</u> : READ ONLY
9.550	Page Forward to Parameter Access Defaults	<u>EG</u> : Page 2
9.551	Select a default access for Parameters	<u>EG</u> : READ ONLY
9.552	Select a NODE TYPE for EXCEPTIONS	<u>EG</u> : AM
9.553	Select a HIWAY/UNIT	<u>EG</u> : 01
9.554	Select Exceptions for Parameter Access	<u>EG</u> : READ/WRITE
9.560	Continue entries for other node types. When all entries are complete, press <ENTER>.	Values and ports reappear in cyan.
9.570	Hold <CTL> down and then press <PAGE BACK>	NG SECURITY DATABASE AUTHORIZ. display

Task 9B CONFIGURE SYSTEM WIDE VALUES for NGs

Step	Action	Response
9.580	Hold <CTL> down and then press <PAGE BACK> Unless Honeywell has requested that you change NG Modem parameters, skip the NG MODEM DEFINITION configuration target.	SYSTEM WIDE VALUES MENU

5.7 Task 10 – Configure HM Volumes

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry

Personality Loaded: Universal

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.10	Place DISK_&Z1.LCN into \$F1, ensure CNCF.LCN is in \$F2.	
10.13	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
10.16	Select [VOLUME CONFIGURATION]	HM PAIR SELECTION MENU
10.21	Select a Node Pair number	EG: 1 VOLUME CONFIGURATION display
10.24	Key in the number of Winchester disk drives in the History Module.	EG: 1 (each redundant pair counts as 1)
10.27	Key in the type or size of disks in the HM.	EG: 3 for Wren 3 EG: 4 or 215040 for 210 MB SCSI drive EG: 5 or 454130 for 445 MB SCSI drive EG: 6 or 896046 for 875 MB SCSI drive EG: 7 for 1.9 Gigabyte drive
10.30	Select configuration choices on the left side of the menu: HM INIT PERSONALITY SYSTEM VOLUME GDF FILES INCLUDED FILE MANAGER DESCRIPTORS	EG: [YES]—target changes in intensity [YES]—target changes in intensity [YES]—target changes in intensity [NO]
10.32	Key in AREA SOE JOURNAL size Numbers used in this section were for a Wren 3. Increase them for larger drives.	EG: 2000 (If this HM will collect SOE events). For SOE, Process Unit Journals must also be configured on the same HM in step 10.515.
10.34	Key in BURST BUFFER size	EG: 7500
10.38	Select SYS. UNIT JOURNALS option—YES if you want this HM assigned for journals	EG: YES —Added ports appear. If you select NO, skip to step 10.60
10.40	Key in STATUS CHANGE port	EG: 2048
10.46	Key in ERROR port	EG: 2048
10.50	Key in MAINT port	EG: 1024
10.60	Press <ENTER>	Values and ports reappear in cyan.
10.100	Select [PROGRAM IMAGE]	PROG IMAGE VOLUME CONFIG. display
10.110	Select NODE TYPE options. Select TYPE IV version for any node that has an K4LCN board, or TYPE II for others (see Figure 4). If both Type IV and II nodes exist for any function, select both versions.	EG: [HG II] [USOPR II] [USUNP II] [NG II] [AM IV] [NIM IV] The system must have the options you select. Select HG IV or HG II if the system has a PLCG.
10.112	Key in DUMP VOLUME SIZE	EG: 36000 or 0 for none
10.125	Press <ENTER>	Values and ports reappear in cyan.
10.130	Hold <CTL> down, press<PAGE BACK>	VOLUME CONFIGURATION display
10.140	Select [AREA]	AREA DATA VOLUME CONFIG. display
10.144	Select a File Manager Descriptor option	EG: NO

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.150	Key into ports for each Area line item as required.	EG: <u>A Name</u> <u>V Size</u> <u>Files</u> BOILER1 5000 150
10.160	Press <ENTER>	Values and ports reappear in cyan.
10.170	Hold <CTL> down, press<PAGE BACK>	VOLUME CONFIGURATION display
10.180	Select [CHECKPOINT]	P.N. FOR CHECKPOINT CONFIG display
Note: Refer to the <i>Engineer's Reference Manual</i> for volume size guidelines.		
10.190	Key into ports for each Node Pair Number line item as required. Note 1: Enter zero for HG or PLCG Vol. Size & No. of Files. Configure an AXM the same as an AM. Note 2: Redundant node members— enter sequentially with Size/Files = 0	EG: <u>NODE</u> <u>VOLUME</u> <u>NUMBER</u> <u>NO.</u> <u>SIZE</u> <u>FILES</u> NIM 24 40000 NIM 25 0 PLCG 31 0 HG 38 0 HG 39 0 AM 40 30000 Numbers used were for a Wren 3. Increase them for larger HM drives (refer to SW09-505).
10.200	Press <ENTER>	Values and ports appear in cyan.
10.210	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.310	Select [CONTINUOUS HISTORY]	CONT. HIST. UNIT AND GROUPS display
10.315	Key into UNIT ID port	EG: 01
10.320	Key into NBR GROUPS port	EG: 60
10.325	Continue entries as required	
10.330	Press <ENTER>	Values and ports reappear in cyan.
10.335	Press <PAGE FWD> and repeat steps 10.315 – 10.330 until all required entries are complete.	
10.350	Select [GROUP OPTIONS] (you can select the [HM/UNIT OPTIONS] display target if desired for a display that allows you to assign options for all).	CONT. HISTORY GROUP OPTIONS
10.360	Select a SAVE RATE (you must select a save rate to get <u>any</u> history)	EG: 20
10.365	Select either [SNAPSHOTS], [USER AVG] or both (you must select SNAPSHOTS to get trend data).	EG: Select both [SNAPSHOTS] and [USER AVG]
10.370	Key into PREARCHIVE HOURS port for SNAPSHOTS	EG: 48
10.373	Key into PREARCHIVE HOURS port for USER AVG	EG: 48
10.376	ARCHIVE [YES] [NO] Function not released	Leave [NO] options selected
10.380	Repeat steps 10.360 – 10.376 for all required entries.	
10.385	Press <ENTER>	Values and ports reappear in cyan.
10.390	<PAGE FWD> and repeat steps 10.354 – 10.385 for all required entries.	
10.400	Hold <CTL> down, press <PAGE BACK>	CONT. HISTORY UNIT AND GROUPS display
10.410	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display
10.510	Select [JOURNALS]	PROCESS UNIT JOURNAL CONFIG.

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response																
10.515	Key into UNIT ID port	<u>EG</u> : 01 Note that Unit numbers you specify here are also the units from which SOE events are collected (if you specified an SOE Journal for this HM node pair).																
10.535	Key into the PROCESS ALARM port	<u>EG</u> : 15000																
10.540	Key into the PROCESS CHANGE port	<u>EG</u> : 10000																
10.545	Key into the OPERATOR MESSG port	<u>EG</u> : 1000																
10.548	Repeat steps 10.515 – 10.545 for other applicable Units.																	
10.550	Press <ENTER>	Values and ports reappear in cyan.																
10.560	Press <PAGE FWD>	Page n of screen forms appears																
10.570	Repeat steps 10.515 – 10.560 for all applicable Units																	
10.575	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																
10.630	Select [CL STORAGE]	CL STORAGE VOLUME CONFIGURATION display																
10.634	Select a HIWAY FILE MGR Description choice	<u>EG</u> : NO																
10.638	Select a NETWORK FILE MGR Description choice	<u>EG</u> : NO																
10.640	Key into HIWAY/NETWORK NBR. port	<u>EG</u> : 02																
10.650	Key into NUMBER OF FILES port	<u>EG</u> : 30																
10.660	Key into VOLUME SIZE port	<u>EG</u> : 100																
10.670	Repeat steps 10.640-10.660 for other applicable process networks																	
10.680	Press <ENTER>	Values and ports reappear in cyan.																
10.690	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																
10.800	Select [USER FILE STORAGE]	USER FILE STORAGE CONFIG. display																
10.850	Key into Node Pair Number n ports for each line item as required (HMV1 is a general HM user volume, CL is for CL programs, CSW1 is for custom programs). Numbers used were for a Wren 3. Increase them for larger drives.	File Mgr. <u>EG</u> : <table><tr><td><u>Vol ID</u></td><td><u># of Files</u></td><td><u>Vol Size</u></td><td><u>Descriptor</u></td></tr><tr><td>HMV1</td><td>1000</td><td>15000</td><td>FAST</td></tr><tr><td>CL</td><td>200</td><td>3000</td><td>NO</td></tr><tr><td>CSW1</td><td>200</td><td>3000</td><td>YES</td></tr></table>	<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>Descriptor</u>	HMV1	1000	15000	FAST	CL	200	3000	NO	CSW1	200	3000	YES
<u>Vol ID</u>	<u># of Files</u>	<u>Vol Size</u>	<u>Descriptor</u>															
HMV1	1000	15000	FAST															
CL	200	3000	NO															
CSW1	200	3000	YES															
10.860	Press <ENTER>	Values and ports reappear in cyan.																
10.870	Hold <CTL> down, press <PAGE BACK>	VOLUME CONFIGURATION display																
CHECK HM VOLUME CONFIGURATION																		
10.880	Hold <CTL> down and press <1>. This performs the CHECK function.	Messages: "NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED....HM CHECKER RUNNING."																
10.881	To configure volumes for another HM, repeat all steps in Task 10.																	
10.885	If errors are indicated, go to step 10.890; otherwise, skip ahead to Task 8C (next page).																	
10.890	Hold <CTL> down and press <8>	NCF.ER file displayed																
10.920	If you want a printed copy, hold <CTL> down and press <4>. Make sure the printer is on.	Print instructions appear																
10.930	Select printer target, press <ENTER>	NCF.ER file is printed																

Task 10 CONFIGURE HM VOLUMES

Step	Action	Response
10.940	If total HM size exceeds 100%, If the number of virtual volumes is exceeded	Go to step 10.850 and reduce the size of the User Volume. Go to step 10.850 and delete one or more User volumes, or delete the dump volume (10.112) or call a Honeywell support engineer.
10.944	Hold <CTL> down and press <PAGE BACK>	EDIT VOLUME CONFIG display
10.948	Go repeat steps 10.800 – 10.880 until errors are cleared.	
10.950	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

5.8 Task 11 – Install the NCF

Keyboard: IKB or PC

Reference: Network Data Entry

Lock: ENGR

Personality Loaded: Universal

Task 11 INSTALL THE NCF

Step	Action	Response
11.01	Ensure DISK_&Z1.LCN is in \$F1, and CNCF.LCN in \$F2.	
11.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
11.30	Select [UNIT NAMES]	Units—Page 1 display
11.40	Press <ENTER>	
11.50	Hold <CTL> down and press <1>	"NCF FILE IS BEING PACKED INTO MEMORY...NEW NCF.WF IS BEING COPIED HM CHECKER RUNNING"
11.52	Hold <CTL> down and press <2>	"WARNING:...PRESS ENTER"
11.60	Press <ENTER>	"OFFLINE INSTALL COMPLETE—DATE/TIME)"
11.65	The "INSTALL COMPLETE" message appears.	
11.70	Dismount DISK_&Z1.LCN from the left drive.	
11.75	Make a backup copy of CNCF.LCN emulated disk. Mount BKUP_CNCF.LCN disk from step 4.66 into \$F1.	
11.80	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
11.85	Key in a disk copy command.	<u>EG</u> : FCP \$F2 \$F1
11.90	Press <ENTER>	"Media Copy Complete" If your primary CNCF disk becomes corrupted, you now have a backup copy.
11.95	Dismount the emulated disks from the left and right drives.	

Section 6 – Procedure 3/Prepare the History Module(s)

Figure 6 shows a graphic representation of the Tasks (12-17) that must be performed to prepare the History Module.

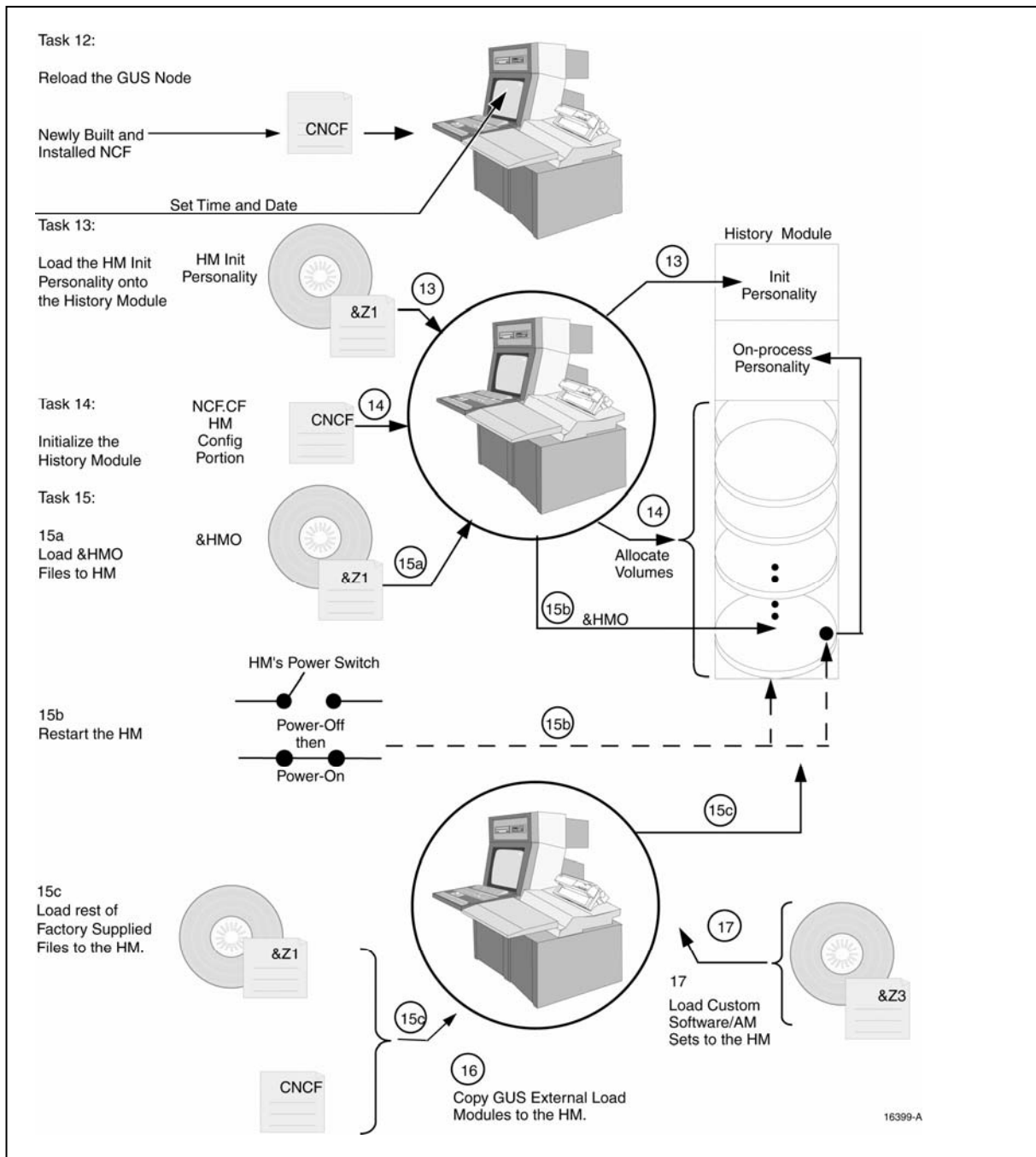


Figure 6 Prepare the History Module(s)

6.1 Task 12 – Reload the GUS Personality and Set Time and Date

Keyboard: N/A

Lock: N/A

Reference: This Document

Personality Loaded: N/A

Task 12 RELOAD THE GUS PERSONALITY AND SET AND TIME AND DATE

Step	Action	Response
12.05	Go perform Task 1 (Reset all nodes)	Note: Not required for nodes previously reset.
12.10	Go perform Task 2 using CNCF.LCN from Task 11 in the right drive and return here.	You must reload the GUS using your site-specific CNCF (from Task 11).
12.20	Go perform Task 3 and return here.	Current time/date in "this US"
12.30	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

6.2 Task 13 – Load the History Module with the HM Initialization Personality

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 13 LOAD THE HISTORY MODULE WITH THE HM INITIALIZATION PERSONALITY

Step	Action	Response
13.05	Turn power ON at all HMs. RECORD THE TIME _____	HM Power was turned off in Task 1
13.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
13.15	Select [SYSTEM STATUS]	SYSTEM STATUS display
13.18	Select [DE-SELECT]	All nodes de-selected
13.20	Select the HISTORY MODULE Node	HM Node box selected
13.24	Select [NODE STATUS]	HISTORY MODULES display (Ignore HM status)
CAUTION The HMs must have had power applied for at least 5 minutes before you press an HM's Reset button <u>in the next step</u>. Make sure you press it only once.		
13.30	Reset each HM by pressing white RESET button on HM module.	Wait until the HM Status changes from FAIL or PWR-OFF to POWER_ON.
13.35	If any HM does not show POWER_ON status, turn its power off for a few seconds; then turn its power on and repeat step 14.30.	
13.40	Mount DISK_&Z1.LCN in \$F1, CNCF.LCN in \$F2.	
13.50	Select Number under NODE heading on the HISTORY MODULE display.	<u>EG</u> : 43 (Number changes color)
13.60	Select [LOAD/DUMP]	LOAD/DUMP targets appear
13.64	Select [MANUAL LOAD]	PGM targets appear
13.70	Select [INIT PROGRAM]	PGM SOURCE FOR NODE NN display
13.80	Select [ALTERNATE SOURCE]	<u>EG</u> : Boxes [1] and [2] appear
13.90	Select [1] (if this is the number for the first emulated disk drive entered in step 9.95).	"VOL ID: &Z1" and targets appear
13.100	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
13.110	Select [ALTERNATE SOURCE]	<u>EG</u> : Boxes [1] and [2] appear
13.120	Select [2]	"VOL ID: CNCF" and targets appear
13.130	Select [EXECUTE COMMAND]	HISTORY MODULES display
13.140	Press <ENTER>	Messages appear at the top of the screen "SINGLE LOAD; BUSY: HM NN, STATUS: LOAD, ...TYPE: HMOF, . . . OK" This can take several minutes to complete.
13.150	Repeat 14.50 - 14.140 to load the HM Init Personality to other HMs	

Task 13 LOAD THE HISTORY MODULE WITH THE HM INITIALIZATION PERSONALITY

Step	Action	Response
NOTE 1. If you get an error message, refer to the <i>TPS Messages Directory</i> for possible error recovery procedures. 2. From this point on, the emulated disk drive numbers will be as entered in Step 9.95 and may differ from the examples.		

6.3 Task 14 – Initialize the History Module

Keyboard: IKB

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 14 INITIALIZE THE HISTORY MODULE

Step	Action	Response
14.10	Mount DISK_&Z1 in \$F1, and CNCF.LCN in \$F2.	
14.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
14.30	Select [VOLUME CONFIGURATION]	HISTORY MODULE PAIR SEL. Menu
14.40	Select Node Pair Number	<u>EG</u> : 1 (for Primary Node 43), VOLUME CONFIGURATION display
14.50	Hold <CTL> down and press <6> Note: Requires several minutes to complete.	"HM INITIALIZATION RUNNING... DISC SURFACE HAS BEEN INITIALIZED AND A NEW LOCAL-NVCF FILE HAS BEEN WRITTEN"
NOTE If you get an error message, refer to the <i>TPS Messages Directory</i> publication, for possible error recovery procedures.		
14.60	Hold <CTL> down, press <PAGE BACK>	HISTORY MODULE PAIR SELECTION Menu
14.70	Repeat 15.40 – 15.60 for each HM	
14.80	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU

6.4 Task 15 – Copy Software from Emulated Disks to the History Module

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 15 COPY SOFTWARE FROM EMULATED DISKS TO THE HISTORY MODULE

Step	Action	Response
HISTORY MODULE PREPARATION		
15.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
15.30	Check that the HM Local Vol. file is on the HM by keying in a Catalog command.	EG: CAT PN:NN>!9np>*.* where NN = HM Node Number and np = node pair (CAT PN:43>!901 for this sample system)
15.40	Press <ENTER>	"CATALOG COMPLETE"
15.50	If the L01_NVCF MM file is listed on the display, continue with step 16.70; else call for help.	Your Honeywell engineer
15.60	Assure printer is powered on.	To obtain a record of the files transferred in the upcoming steps
15.70	Key in " PD ON "	
15.80	Press <ENTER>	"Prompt To Dataout Turned On"
15.90	Key in " DO (Print Device ID)"	EG: DO \$P1
15.100	Press <ENTER>	"DATA OUT COMPLETE" Note: Printer must be connected to the station.
15.110	Insert DISK_&Z1.LCN in \$F1.	
15.120	Key in an EC command for the Local Volume	EG: EC \$F1>&EC>LOC_VOLZ.EC \$F1 43 01
15.130	Press <ENTER>	"THIS EC IS USED TO COPY HM PERSONALITIES TO ITS LOCAL VOLUME."
At this point you have initiated an "Execute Command" program to simplify the task of loading a series of files from cartridge to the HM. You will be asked a series of questions to which you must respond by keying in a "Y" for yes or an "N" for no, followed by pressing <ENTER>. Notes: - The &HMI files can be copied to more than one HM. - The &HMO files must be copied to <u>all</u> HMs. DISK_&Z1.LCN contains both 68020 and 68040 HMO & HMI software. Answers to the EC questions will determine which type is loaded (HMs with K4LCN boards require 68040 software).		
15.140	Shown here are the files that should be transferred during the execution of the above EC file. Check your printout to determine that all required files have been loaded on the HM(s). ___ &LDR ___ &HMI (HMOFxxx) files ___ &HMO (HMONxxx) files When the EC program is complete, the message "Write Boot ...EC Complete" appears.	
15.150	If you have other HMs, return to Step 15.120 to copy the &HMO (&HMI if desired) directory to the additional HM(s).	
NOTE The &ASY directory in the following steps must be copied to <u>only one</u> HM, the system HM.		
15.160	Ensure CNCF.LCN in \$F2.	
15.170	Copy the custom NCF files to the HM	EG: CP \$F2>&ASY>*. * PN:43>&ASY>= -D
15.180	Press <ENTER>	"COPY COMPLETE"

Task 15 COPY SOFTWARE FROM EMULATED DISKS TO THE HISTORY MODULE

Step	Action	Response
RESTART THE HISTORY MODULE		
15.190	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.200	Select [SYSTEM STATUS]	SYSTEM STATUS display
15.210	Select the HISTORY MODULE node	Node is selected
15.230	Select [SHUT DOWN]	Target changes color
15.240	Press <ENTER>	Status changes to QUALIF
15.250	At the HM module, turn power OFF , wait at least 5 seconds, and then turn power ON .	Status of the HM changes to: LOC LOAD, then READY, and finally OK . HMON On-process personality is now running in the HM This process can take several minutes.
NOTE If you get an error message, check the <i>TPS Messages Directory</i> publication (see References), for possible error recovery procedures.		
COPY REMAINING SOFTWARE TO THE HISTORY MODULE		
15.260	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
15.270	Select [COMMAND PROCESSOR]	Command Processor display
15.280	Assure printer is on.	To obtain a record of the files transferred in the upcoming steps.
15.290	Key in " PD ON "	
15.300	Press <ENTER>	"Prompt To Dataout Turned On"
15.310	Key in " DO (Printer Device ID)"	<u>EG:</u> DO \$P1
15.320	Press <ENTER>	"DATA OUT COMPLETE"
15.330	If you want to enter files for modules with 68020 microprocessors, ensure DISK_&Z1.LCN in \$F1 and key in an EC command for the 68020 personality volume For 68040 processor files, ensure DISK_&Z1.LCN in \$F1 and key in an EC command	<u>EG:</u> EC \$F1>&EC>PRS2VOLZ.EC \$F1 <u>EG:</u> EC \$F1>&EC>PRS4VOLZ.EC \$F1
15.340	Press <ENTER>	"THIS EC IS USED TO COPY PERSONALITY FILES TO THE SYSTEM HM."
You should answer "Y" for each of the following files that is needed by a node that you have on your system. If you have inserted DISK_&Z1.LCN into \$F1, answer "Y" only for nodes that use 68020 or 68040 microprocessors. In either case, save HM space, decline the copying of files for nodes that you do not have. As part of this procedure, you will be requested to place other cartridges into the drive. The EC program you invoke will select the correct software from DISK_&Z3 for a Yes or No answer.		

Task 15 COPY SOFTWARE FROM EMULATED DISKS TO THE HISTORY MODULE

Step	Action	Response
NOTE Mount the appropriate CD in the CD-ROM drive to use the following emulated disks: Use the GUS-TPN System Software CD for: DISK_&Z1.LCN DISK_&Z6.LCN Use the TPN Application Module CD for: DISK_&Z3.LCN Use the TPS Software CD for: DISK_&Z12.LCN DISK_&Z14.LCN Perform the following steps when a disk change is required during the execution of the PRS?VOLZ.EC file: (●)_____1) DISMOUNT the currently mounted emulated disk and corresponding CD. (●)_____2) MOUNT the required emulated disk and corresponding CD and continue as usual.		
NOTE If the "Mount/Dismount Emulated Disks" dialog page was exited and then restarted, previously created disk files may have been inadvertently removed from the Available Disks list. If the disk you need is not in the Available Disks list, repeat steps 1.26 - 1.31 to add the disk to the list again.		

Task 15 COPY SOFTWARE FROM EMULATED DISKS TO THE HISTORY MODULE

Step	Action	Response																																																			
15.350	This step provides a list of the files that can be transferred during the execution of the above EC file(s), and the module with which each is associated. The "PRS2VOLZ" EC loads files to nodes with 68020 microprocessors and the "PRS4VOLZ" EC loads files to nodes with 68040 microprocessors. Most of the files on &Z1 are for 68040 microprocessors and 68020 microprocessors (in some case there is no distinction). This software is normally copied to <u>only</u> the system HM. Note that file &LDR is loaded along with the other files. <table> <tr> <td>Module</td><td>DISK_&Z1</td><td>DISK_&Z3</td></tr> <tr> <td>OPR II</td><td>___ &OPR</td><td></td></tr> <tr> <td>OPR IV</td><td>___ &OPRk</td><td></td></tr> <tr> <td>UNP II</td><td>___ &UNP (Also requires overlay files &OP1 on DISK_&Z1.LCN)</td><td></td></tr> <tr> <td>UNP IV</td><td>___ &UNPk (Also requires overlay files &OP1 on DISK_&Z1.LCN).</td><td></td></tr> <tr> <td>UNP</td><td>___ &OVG</td><td></td></tr> <tr> <td>HG II</td><td>___ &HGO</td><td></td></tr> <tr> <td>HG IV</td><td>___ &HGOk</td><td></td></tr> <tr> <td>NIM II</td><td>___ &NMO</td><td></td></tr> <tr> <td>NIM IV</td><td>___ &NMOk</td><td></td></tr> <tr> <td>PM, APM, HPM, LM, SM</td><td>___ &UCN</td><td></td></tr> <tr> <td>AM II</td><td>___ &AMO</td><td></td></tr> <tr> <td>AM IV</td><td>___ &AMOk</td><td></td></tr> <tr> <td>CG II</td><td>___ &CIO</td><td></td></tr> <tr> <td>CG IV</td><td>___ &CIOk</td><td></td></tr> <tr> <td>NG II</td><td>___ &NGO</td><td></td></tr> <tr> <td>NG IV</td><td>___ &NGOk</td><td></td></tr> </table> k means type 68040 software (required when the node contains a K4LCN board). When the EC asks if you want to create/copy &OVG, answer Yes if you want the Equipment List functions. When the EC asks where, specify a user volume such as HVM1 or CSW1. When the EC copies the AM files it asks if you want to create/copy &CUS and CLX. This is optional software, but you should probably copy it in unless you know that it is not needed. If you copy the Computer Gateway files, the EC asks if you want to copy &CUS. You probably should. If the EC also asks where you want to create the directories, Key in the name of one of your User volumes such as CSW1 or HVM1. When the EC program is complete, the message "EC COMPLETE" appears. Check the printout to assure that all required files have been loaded.	Module	DISK_&Z1	DISK_&Z3	OPR II	___ &OPR		OPR IV	___ &OPRk		UNP II	___ &UNP (Also requires overlay files &OP1 on DISK_&Z1.LCN)		UNP IV	___ &UNPk (Also requires overlay files &OP1 on DISK_&Z1.LCN).		UNP	___ &OVG		HG II	___ &HGO		HG IV	___ &HGOk		NIM II	___ &NMO		NIM IV	___ &NMOk		PM, APM, HPM, LM, SM	___ &UCN		AM II	___ &AMO		AM IV	___ &AMOk		CG II	___ &CIO		CG IV	___ &CIOk		NG II	___ &NGO		NG IV	___ &NGOk		
Module	DISK_&Z1	DISK_&Z3																																																			
OPR II	___ &OPR																																																				
OPR IV	___ &OPRk																																																				
UNP II	___ &UNP (Also requires overlay files &OP1 on DISK_&Z1.LCN)																																																				
UNP IV	___ &UNPk (Also requires overlay files &OP1 on DISK_&Z1.LCN).																																																				
UNP	___ &OVG																																																				
HG II	___ &HGO																																																				
HG IV	___ &HGOk																																																				
NIM II	___ &NMO																																																				
NIM IV	___ &NMOk																																																				
PM, APM, HPM, LM, SM	___ &UCN																																																				
AM II	___ &AMO																																																				
AM IV	___ &AMOk																																																				
CG II	___ &CIO																																																				
CG IV	___ &CIOk																																																				
NG II	___ &NGO																																																				
NG IV	___ &NGOk																																																				
15.360	If necessary, return to Step 16.330 and invoke the other ECs.																																																				
15.370	Ensure DISK_&Z1.LCN is in \$F1.																																																				
15.380	Key in an EC command for the System Volume.	<u>EG:</u> EC \$F1>&EC>SYS_VOLZ.EC \$F1 43 (the last parameter is the system HM node number)																																																			
15.390	Press <ENTER>																																																				
15.400	The files that can be transferred using the above EC command are listed here for comparison with your printout to determine that all required files have been loaded on the HM.																																																				

Task 15 COPY SOFTWARE FROM EMULATED DISKS TO THE HISTORY MODULE

Step	Action	Response
NOTE To the question "COPY VOLUME &ASY (YES/NO)" answer No ("N"). To the question "DO YOU WISH TO CHANGE TO ANOTHER DRIVE" usually answer No ("N"). To the question "DO YOU WISH TO SKIP &DSY...." usually* answer No ("N"). *This EC loads several small files. If you answer "No," the EC will load all of them. If you answer "Yes," you can choose to load or not load any or all of the following files: ___ &DSY ___ &HGG ___ &AMG ___ &ARG ___ &NMG & NM2 The system does not have to have the node type to load any of these files.		
15.410	Ensure CNCF.LCN in \$F2.	
15.420	Copy the customer key file to the HM	<u>EG</u> : CPV \$F2>&KFO> NET>&KFO> -D
15.424	Copy the null Area Database file to the HM	<u>EG</u> : CPV \$F2>&D01> NET>&D01> -D
15.430	Press <ENTER>	"COMMAND COMPLETE"
15.443	You can copy certain standard displays (NCF Status, and so on.) to a user volume on an HM other than that which contains the &DSY volume. The systems hunts for these displays in directory &DS2 on another HM If the system HM fails. To load the displays, ensure emulated DISK_&Z1.LCN is in \$F1. Key in the EC command EC \$F1>&EC>DS2 BKUP.EC \$F1 . Press <ENTER>. When asked "Do you wish to create &DS2...?", answer, Yes. Key in the user volume name when asked.	
15.440	Return to Task 16.330 if any of the above software must be copied to a different HM (as determined earlier during Volume Configuration in Task 10).	
15.450	Key in " PD OFF "	
15.460	Press <ENTER>	"PROMPT TO DATAOUT TURNED OFF"
15.470	Key in " DO "	"DATA OUT COMPLETE"

6.5 Task 16 – Copy GUS External Load Modules to the History Module

Keyboard: IKB or PC

Lock: ENGR

Reference:

Personality Loaded: Universal

Task 16 COPY GLOBAL USER STATION EXTERNAL LOAD MODULES TO THE HISTORY MODULE

Step	Action	Response
16.50	Mount the TPS Software CD ROM in the CD Drive	
16.51	From Available Disks, select and mount DISK_&Z14.LCN	DISK_&Z14.LCN is mounted in \$F1
16.52	From the Native Window COMMAND PROCESSOR, key in the command: EC \$F1>&EC>MSCHEMZ.EC	Follow the prompts
16.53	Dismount DISK_&Z12.LCN	

6.6 Task 17 – Copy Optional Software to the History Module

Keyboard: IKB or PC

Lock: ENGR

References: CL/AM Reference Manual
Engineer's Reference Manual
Command Processor Operation Manual

Personality Loaded: Universal

If you purchased optional custom software (load modules) that were not loaded in Task 16, perform the following steps. If not, skip to Task 18.

These steps assume the COMMAND PROCESSOR is still selected.

Task 17 COPY OPTIONAL SOFTWARE TO THE HISTORY MODULE

Step	Action	Response								
17.10	Create directories for the software you intend to load. List the contents of the cartridges, if necessary, to find the directories. <u>EG</u> : Key in a Create Directory Command. Press <ENTER>	In Task 10 you may have created a user volume for custom software (CSW1 in the example). You can also create a suitable directory under another volume, such as the example user volume HMV1. <u>EG</u> : CD NET>CSW1 &nnn* (where &nnn is custom software)								
17.80	Copy the optional software into the History Module: <table><tr><td><u>n \$F1</u></td><td><u>Cmd Type</u></td><td><u>Example of command to be keyed</u></td><td><u>Press</u></td></tr><tr><td>DISK_&Z__</td><td>Copy</td><td><u>EG</u>: CP \$F1>&nnn> ** NET>&nnn>= -D</td><td><ENTER></td></tr></table>	<u>n \$F1</u>	<u>Cmd Type</u>	<u>Example of command to be keyed</u>	<u>Press</u>	DISK_&Z__	Copy	<u>EG</u> : CP \$F1>&nnn> ** NET>&nnn>= -D	<ENTER>	
<u>n \$F1</u>	<u>Cmd Type</u>	<u>Example of command to be keyed</u>	<u>Press</u>							
DISK_&Z__	Copy	<u>EG</u> : CP \$F1>&nnn> ** NET>&nnn>= -D	<ENTER>							
*Substitute the volume name of the custom software cartridge(s) for &nnn.										
17.90	Repeat Task 17 for each HM where you wish custom software copied.									
NOTE If the system has redundant disk drives, they can be synchronized now (or later). Refer to the <i>Engineer's Reference Manual</i>, Section 7.5 and Section 5 in the <i>Command Processor Operation</i> manual for the procedure. The sample system did not have this feature.										

Section 7 – Procedure 4/Configure Process Networks and Hiways

Figure 7 shows a graphic representation of the Tasks (18-23) that must be performed to Configure Process Networks and Hiways.

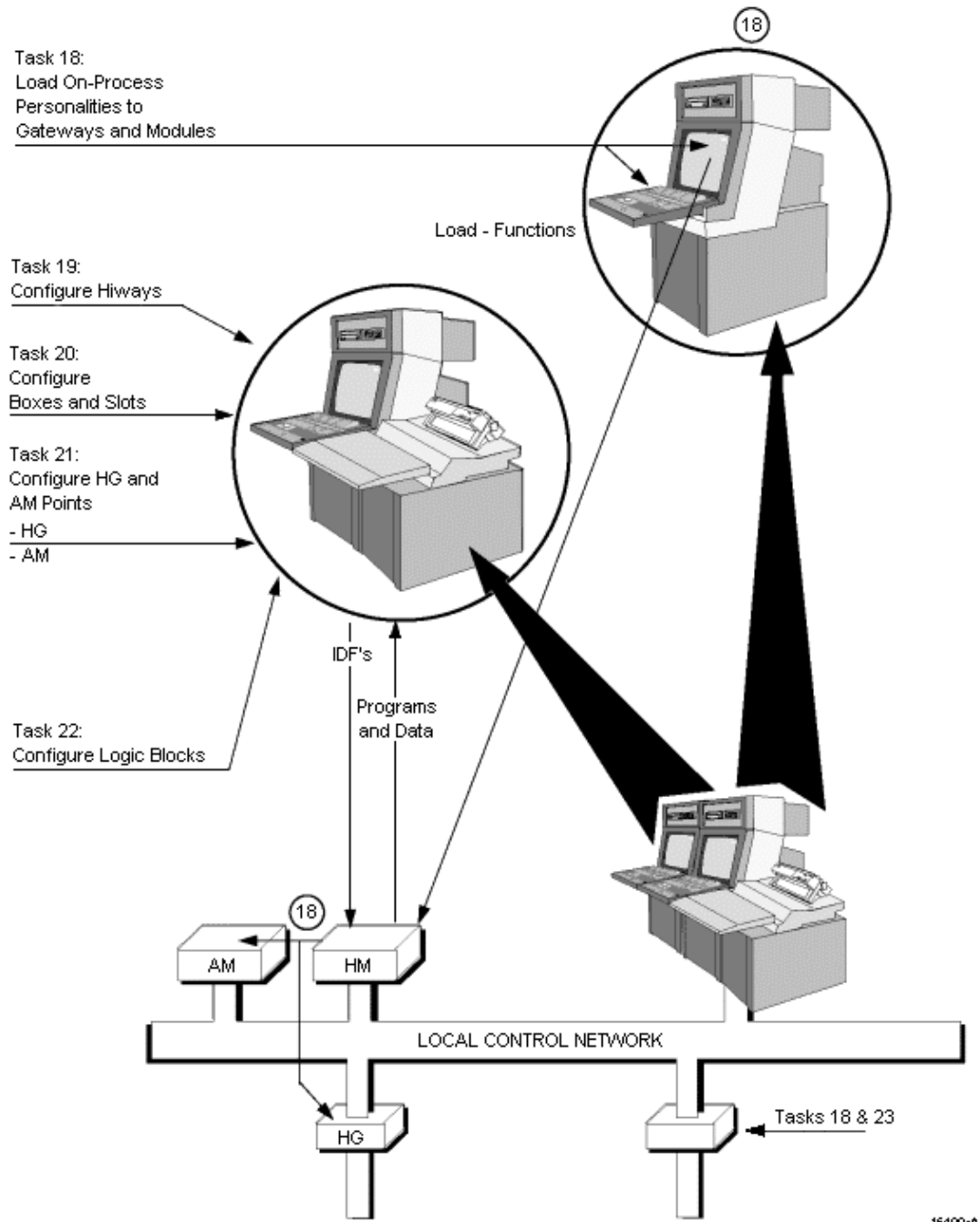


Figure 7 Configure Process Networks and Hiways

7.1 Task 18 – Load On-Process Personalities to Gateways and Modules

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 18 LOAD ON-PROCESS PERSONALITIES TO GATEWAYS AND MODULES

Step	Action	Response
18.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
18.03	Select [SUPPORT UTILITIES]	SUPPORT UTILITY MENU
18.04	Select [MODIFY VOLUME PATHS]	MODIFY DEFAULT VOL PATHNAMES display
18.05	Select [SET DEVICE PATH TO "NET"]	LDIDs (Logical Device IDs) are all filled in with NET .
18.06	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
See Appendix A if you wish to LOAD another Global User Station (GUS) at this point.		
18.10	Select [SYSTEM STATUS]	SYSTEM STATUS display
18.20	Select an AM, NIM, or Gateway Node NIM example: Redundant NIM: NG example: HG/PLCG example, before Tasks 19-20: HG/PLCG example, after Tasks 19-20: AM example: CM example:	Only one node should be selected at a time <u>EG</u> : 24 for Network Interface Mod (NIM) <u>EG</u> : 25 for Redundant NIM <u>EG</u> : 30 for Network Gateway <u>EG</u> : 38 for Primary HG or PLCG Module <u>EG</u> : 39 for Redundant HG or PLCG Module <u>EG</u> : 40 for Application Modules or AXMs <u>EG</u> : 46 for Computing Modules
WARNING - Do not select a redundant (secondary) HG or PLCG node unless Hiway and Box/Slot Configuration has been done in Tasks 19 and 20. - The primary HG or PLCG of a redundant pair must be loaded first, using the CHKPOINT.LCN checkpoint disk created in Task 4. After Hiway & Box/Slot configuration, the primary HG or PLCG is checkpointed. The secondary HG or PLCG is loaded later and must be started with a checkpoint volume from the primary node. Refer to the Engineer's Reference Manual, Section 8. - Do not attempt to load a Primary HG or PLCG from one station and a Secondary node from a different station at the same time. The nodes will both fail. - Always load the primary AM before loading the redundant AM (if any).		
18.30	Select [NODE STATUS]	Process Networks Node Status Display or Application Modules Node Status Display
18.40	Select [MANUAL LOAD]	NODE LOAD targets appear.
18.50	Select a LOAD option	<u>EG</u> : [COLD LOAD] for AM <u>EG</u> : [OPERATOR PROGRAM] for HG, NG, PLCG, or NIM PGM SOURCE FOR NODE NN display
18.60	Select [DEFAULT SOURCE]	[EXECUTE COMMAND] target appears.
18.70	Select [EXECUTE COMMAND]	DATA SOURCE FOR NODE NN display
18.80	Mount CHKPOINT.LCN in \$F1.	CHKPOINT contains null checkpoints for all checkpoint nodes (HG, NIM, AM, AXM, CG, and PLCG). You created this in Task 4.
18.90	Select [ALTERNATE SOURCE]	Boxes [1], [2], etc. appear.
18.100	Select <u>EG</u> : [1]	Volume Identifier = CKPT and [EXECUTE COMMAND]
18.110	Select [EXECUTE COMMAND]	GATEWAYS or AM display reappears.

Task 18 LOAD ON-PROCESS PERSONALITIES TO GATEWAYS AND MODULES

Step	Action	Response
18.120	Press <ENTER>	"BUSY: AM or GWY nn or CM" Status goes to LOAD.
18.130	Wait a few minutes.	Status goes to OK when loaded.
18.134	Repeat 18.06-18.130 including other redundant modules.	Note that PMs, APMs, and HPMs are loaded in a later task.
18.136	After loading an AM, Select the Node number under NODE.	AM node number color changes to white Status should still be OK
18.138	Select [SAVE DATA]	CHKPT DEST display appears
18.140	Select [DEFAULT SOURCE]	Additional targets appear
18.142	Select [EXECUTE COMMAND]	BUSY AM nn. Node status =SAVE... then OK
18.144	Select [SHUTDOWN]	Node should still be selected
18.145	Press <ENTER>	Node status changes to READY...QUALIF
18.146	Select [LOAD/DUMP]	Additional targets appear
18.147	Select [AUTOLOAD NET]	
18.148	Press <ENTER>	Node status = LOAD...LOC LOAD...OK Node Type changes to AM
18.149	Repeat steps 18.136 - 18.148 for all.	
18.150	Remove the emulated disk from the disk drive.	

7.2 Task 19 – Configure Hiways

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry
Data Entity Builder

Personality Loaded: Universal

Task 19 CONFIGURE HIWAYS

Step	Action	Response
19.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
NOTE The primary HG or PLCG must be loaded and running with the startup data base (loaded in Task 18). If the system does not have an HG or PLCG, skip to task 21.		
\$HIWAYnn Where nn is hiway number		
19.70	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
19.75	Select [HIWAY CONFIG]	HG-HIWAY POINT ASSIGN. PED display
19.80	Key into HIWAY NUMBER port	<u>EG</u> : 02
19.90	Select HTD ASSIGNMENT option	<u>EG</u> : [THISHG] (HTD is mandatory if a Basic OP Station is on the Hiway)
19.100	Fill in the remaining ports as required and press <ENTER>	PED reappears with values in cyan
19.110	Press <COMND>	COMMAND DISPLAY
NOTE 5. You should make an IDF with a different name for each HG pair and each PLCG pair. 6. When building a database create the IDFs in this order: a) In this task, create the Hiway Configuration database for the HG pair or PLCG pair. b) In Task 20, create the box/slot database for the HG pair, or PLCG pair and then create the database for all the boxes and devices on the Hiway.		
19.120	If this is the <u>first</u> write to this IDF, go to step 19.130; If <u>not</u> , go to step 19.160.	IDF for this example is HIWAY2
19.130	Select [WRITE TO IDF]	Pathname ports and target appear.
19.140	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HMOV1>
19.150	Key into pathname for IDF port	<u>EG</u> : HIWAY2
19.160	Press <ENTER>	Hiway Config data written to IDF. HG-HIWAY POINT ASSIGNMENT display
19.170	Press <COMND>	COMMAND DISPLAY
19.180	Select [LOAD]	
19.190	Press <ENTER>	Hiway Config data loaded to HG or PLCG. PARAM ENTRY display reappears
19.200	Repeat 19.10 – 19.190 for each other Hiway to be configured.	<u>EG</u> : HIWAY4 for a PLCG

7.3 Task 20 – Configure Boxes and Slots

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry
Data Entity Builder

Personality Loaded: Universal

Task 20 CONFIGURE BOXES AND SLOTS

Step	Action	Response
20.05	Hold <CTL> down and press <9>. If the HG BUILD TYPE MENU appears go to step 20.20; if not go to 20:10.	HG-BUILD TYPE AND CONFIG. MENU
20.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
\$HYxxBnn Where xx is the Hiway number and nn is the box number		
20.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
20.20	Select [BOX/SLOT CONFIG]	HG-BOX POINT ASSIGNMENT PED display
20.30	Key into HIWAY NUMBER port	<u>EG</u> : 02
20.40	Key into BOX NUMBER port	<u>EG</u> : 02 for an HG (<u>Configure HG or PLCG before other Hiway boxes</u>) 03 if you have an HG or PLCG pair 04 if you have a Basic Station 05 for a MC, 10 for CB, etc. (configure 1 box/pass)
NOTE If this box is the HG or PLCG, the box number of the first node of a redundant pair must be an even number and the second node must be the next odd number.		
20.50	Select a BOX TYPE option [CB] if box is CB [MC] if box is MC	<u>EG</u> : [HG]
20.55	Select a BOX ASSIGNMENT option	<u>EG</u> : [THISHG]
NOTE If an HG or PLCG pair is not your master HG pair for this Hiway, the BOX ASSIGNMENT for that HG or PLCG pair must be either [REMOTEHG] or [ADDEDHG].		
20.90	Select a LOAD DESTINATION (if appl.)	<u>EG</u> : [HG_HIWAY]
20.95	Press <ENTER>	PED reappears with values in cyan. (If PLCG, perform steps in Note at end of Sec. 20 to clear address error).
NOTE If you selected any options outlined in yellow or if you keyed data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.		
20.100	Press <PAGE FWD> to next PED page key in data, select options, and press <ENTER> after each page.	HG-BOX CONFIGURATION PED
CHECK THE DATA YOU HAVE ENTERED		
20.110	When you have paged forward and entered all PED pages hold <CTL> down and press <8>.	PARAM ENTRY DISP SET STATUS display

Task 20 CONFIGURE BOXES AND SLOTS

Step	Action	Response
20.120	If "Display Set Entered?" is Yes, and no errors are indicated, go to step 20.150; else go to 20.130.	Display Set Entered? Yes Errors? NO
20.130	Select a Display Title option, correct data and press <ENTER>	Applicable PED appears.
20.135	Repeat step 20.120	Go check that data is all OK.
WRITE TO IDF		
20.150	Press <COMND>	COMMAND DISPLAY
20.160	If this is the <u>first</u> write to this IDF, go to step 20.165; if not, go to step 20.190.	IDF for this example is HY2BXSL .
20.165	Select [WRITE TO IDF]	Pathname ports and target appear.
20.170	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HVM1>
20.180	Key into pathname for IDF port	<u>EG</u> : HY2BXSL
20.190	Press <ENTER>	Hiway Config data written to IDF. HG-BOX POINT ASSIGNMENT PED display
20.195	Repeat steps 20.30 – 20.190 for the redundant HG or PLCG.	
20.200	Repeat steps 20.30 – 20.190 for each box/slot.	All configured boxes/slots on IDF
20.210	Press <COMND>	COMMAND DISPLAY
20.220	Select [LOAD MULTIPLE]	Additional pathname ports appear
20.230	Key into REFERENCE PATH NAME port	<u>EG</u> : Net>HVM1>
20.240	Key into pathname for IDF port	<u>EG</u> : HY2BXSL
20.245	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" (See note below if an error results)
20.250	Press <COMND>	COMMAND DISPLAY
20.260	Select [PRINT ENTITIES]	Path name ports/targets appear. REF. PATH NAME = NET>HVM1>
20.270	Select [PRINT IDF ENTITIES]	
20.280	Use default Print Device ID	<u>EG</u> : \$P1
20.290	Press <ENTER>	Box Slot configuration parameters are printed

NOTE

Entity names in the printout identify Hiway and Box numbers as follows:

"\$HIWAYnn" where nn is the Hiway number.

"\$HYnnBnn" where nn after HY is Hiway number and nn after B is Box number.

Address Error—If an address error results when loading Box/Slot data to a Hiway Box, make sure you saved the data in an IDF. Then proceed as follows (doing this on another station is preferable):

Call up the System Status Display.

Select the appropriate HG Node.

Select [**NTWK/HWY STATUS**] near the bottom of the display.

Select [**HIWAY CMND**].

Select [**INIT ADDR**].

Select [**EXECUTE COMMAND**]. The address errors should be cleared.

Return to the BOX/SLOT configuration display and repeat the LOAD MULTIPLE steps.

This time the result should be "OPERATION COMPLETE."

PLCG Interface—Certain PLCG interface functions have to be programmed with a Local Loader Station, refer to PLCG documentation for details.

7.4 Task 21 – Configure Process Networks and Hiways

Keyboard: IKB or PC

Lock: ENGR

Reference: Network Data Entry
Data Entity Builder

Personality Loaded: Universal

Task 21 CONFIGURE PROCESS NETWORKS AND HIWAYS

Step	Action	Response
21.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
21.20	Select [HIWAY GATEWAY] , or [APPLICATION MODULE] , or [COMPUTING MODULE] [NTWK INTERFACE MODULE] (if the system has a PLCG, select [HIWAY GATEWAY]. If the system has none of the above, skip to Task 22)	EG: [HIWAY GATEWAY] HG/AM/CM BUILD TYPE MENU
21.21	Select a Point Build option	EG: [REGULATORY]
NOTE - For detailed information on point building, refer to the <i>Data Entity Builder Manual</i>, (see References). For point building tips and precautions, refer to section 9 in the <i>Engineer's Reference Manual</i>. - The general scenario of point building is to key into ports, select options on each PED page, press <ENTER> and then <PAGE FWD> to the next PED page. The pages in a particular PED set may change depending upon what type of box, algorithm, or point you are configuring. - If you select any options outlined in yellow or if you key data into any port outlined in yellow, the content of the PED set or PED changes when you press the <ENTER> key. You should press <ENTER> after each page. - Steps 21.30 through 21.420 describe the process of building a typical PID Normal point for a Basic Controller. For other types of points see Tables 3-2 through 3-4. When building any of the example points, use default values and options for items not on the list.		
POINT BUILD		
21.30	Key into TAG NAME port	EG: HG1001
21.40	Key into UNIT ID port	EG: 01
21.50	Key into POINT DESCRIPTOR port	EG: CB REG POINT
21.60	Key into POINT KEYWORD port	EG: CBpid
21.64	PRIMARY MODULE POINT ID	-----
21.66	Key into POINT ASSOCIATED DISPLAY	EG: DISP1 (R510 and later systems)
21.70	Key into HIWAY NUMBER port	EG: 02
21.75	Select BOX TYPE option	EG: [CB]
21.80	Key into BOX NUMBER port	EG: 10
21.90	Select POINT DISCLOSURE option	EG: [FULL]
21.100	Select LOAD DESTINATION option	EG: [HG_HIWAY]
21.110	Press <ENTER>	PED reappears with values in cyan
21.120	Press <PAGE FWD>	ALGORITHM Selection display
21.125	Select ALGORITHM option	EG: [PIDNORM]
21.130	Press <ENTER>	PED reappears with values in cyan
21.135	Press <PAGE FWD>	HG-PV CONFIGURATION display
21.140	Key into E.U. DESCRIPTOR port	EG: GAL/HOUR
21.150	Select CHARACTERIZATION option	EG: [SQRROOT]

Task 21 CONFIGURE PROCESS NETWORKS AND HIWAYS

Step	Action	Response
21.160	Select INPUT CONDITIONING option	<u>EG</u> : [SQRROOT]
21.170	Select PV DECIMAL FORMAT option	<u>EG</u> : [D2] (places point at 999.99)
21.180	Key into OVERVIEW VALUE port	<u>EG</u> : 20
21.190	Select PV RANGE OPTION	<u>EG</u> : [CLMPZERO]
21.200	Select CLAMP VALUE OPTION	<u>EG</u> : [NOCLAMP]
21.210	Press <ENTER>	PED reappears with values in cyan
21.220	Press <PAGE FWD>	HG—ALARMING display
21.225	Select ALARM FORMAT option	<u>EG</u> : [ALFMT00]
21.230	Key into PVLOTP port	<u>EG</u> : 10
21.240	Key into PVHITP port (far right)	<u>EG</u> : 100
21.250	Select ALARM PRIORITY option	<u>EG</u> : [HIGH]
21.260	Press <ENTER>	PED reappears with values in cyan
21.270	Press <PAGE FWD>	HG-OPERATING CONFIGURATION display
21.290	Select PV SIGNAL Type option	<u>EG</u> : [PV]
21.310	Select INITIALIZATION option	<u>EG</u> : [INIT]
21.320	Key into SETPOINT port	<u>EG</u> : 43
21.330	Press <PAGE FWD>	
21.350	Key into OUTPUT HIGH LIMIT port	<u>EG</u> : 100
21.360	Key into INTEGRAL HIGH LIMIT port	<u>EG</u> : 100
21.370	Key into FILTER CONSTANT port	<u>EG</u> : 1
21.380	Press <ENTER>	PED reappears with values in cyan
21.390	Press <PAGE FWD>	HG-MODE CONFIGURATION display
21.400	Select NORMAL MODE option	<u>EG</u> : [CAS]
21.410	Select NRML MODE ATTRIBUTE option	<u>EG</u> : [OPERATOR]
21.420	Press <ENTER>	PED reappears with values in cyan
CHECK THE DATA YOU ENTERED		
21.430	When you have paged forward and entered all PED pages, hold <CTL> down and press <8>.	PARAM ENTRY DISPLAY SET STATUS display
21.440	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to step 21.460; or else to to step 21.450.	
21.450	Select a Display Title option, correct data and press <ENTER>.	Applicable PED appears. PED set is corrected.
21.455	Repeat step 21.440.	
NOTE You should make IDFs with different names for the points that you build in each box on the hiway. (HG1, CB1, and so on.)		
WRITE TO IDF		
21.460	Press <COMND>	COMMAND DISPLAY
21.470	If this is the <u>first</u> write to this IDF, go to step 21.475. If not, go to step 21.510.	In this example, the IDF is HGPNTS for HG, AMPNTS for AM, etc.
21.475	Select [WRITE TO IDF]	Additional port/target appear.
21.480	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HMOV1>

Task 21 CONFIGURE PROCESS NETWORKS AND HIWAYS

Step	Action	Response
21.500	Key into pathname for IDF port	<u>EG</u> : HGPNTS for HG, AMPNTS for AM
21.510	Press <ENTER>	Point Config data written to IDF PARAMETER ENTRY DISPLAY
21.520	Repeat steps 21.30 – 21.510 for each other point.	All configured points are on IDF for HG, or IDF for AM, etc.
21.530	Press <COMND>	COMMAND DISPLAY
21.540	Select [LOAD MULTIPLE]	Additional ports/target appear
21.550	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
21.555	Select [WITH OVERWRITE]	
21.560	Key into Pathname For IDF port	<u>EG</u> : HGPNTS for HG; AMPNTS for AM
21.570	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES" Point Config data loaded to HG. "OPERATION COMPLETE"
21.572	If errors occur, select the point on the error display. Make corrections; then repeat steps 21.530 – 21.570.	
21.580	If the Command Display is not on the screen, press <COMND>	COMMAND DISPLAY
21.590	Select [LIST ENTITIES IN IDF]	Additional ports/target appear.
21.600	Key into REFERENCE PATH NAME	<u>EG</u> : NET>HNV1>
21.605	Key into pathname for IDF port	<u>EG</u> : HGPNTS for HG, AMPNTS for AM
21.610	Key into pathname For SEL LIST	<u>EG</u> : BXSL.EL
21.614	Select BUILD TYPES of entities to list	<u>EG</u> : HG or AM or CIU as applies. Additional targets appear.
21.618	Select types of entities	<u>EG</u> : ALL
21.620	Press <ENTER>	"BUSY", DISPLAY OF LIST FILE display
21.630	For a printout, press <COMND>	COMMAND DISPLAY
21.640	Select [PRINT FILE]	Ports/std. suffixes appear
21.650	Key into pathname for FILE port	<u>EG</u> : BXSL.EL
21.655	Key into Printer Print Device ID port	<u>EG</u> : \$P1 (default used)
21.660	Press <ENTER>	Point IDs are printed. (For more details, select PRINT ENTITIES)
21.670	Repeat Task 21 to build "master" points for other Data Hiway boxes.	<u>EG</u> : MC point (see Table 3-3)
21.680	Go to Appendix D of this document to EXCEPTION BUILD other Data Hiway box points as required.	
21.690	Repeat Task 21 to build other "master" points for AM and CL.	<u>EG</u> : AM points (vratio is used in task 28 for CL). See Table 3-2.
21.695	Go to Appendix D of this document to EXCEPTION BUILD other Data Hiway box points as required.	

NOTE

You should save HG and AM checkpoints to emulated disk at this time. To do so, go perform instructions in Appendix B (HG) or C (AM) of this document.

Table 5 AM/CL Point VLRATIO (Regulatory) Example

TAG NAME	VLRATIO
UNIT ID	01
POINT DESCRIPTOR	CL Point
ENGR UNITS DESCRIP.	Fur/Ftnt
POINT KEYWORD	CLPOINT
POINT ASSOCIATED DISPLAY	(blank)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM [CL]	
CONTROL ALGORITHM	[NULL]
OVERVIEW VALUE	25
I/O DISPLAY SUPPR.	[NOSUPPR]
POINT EXEC. PERIOD	[NOPERIOD]
PV DECIMAL FORMAT	[D1]
PV RANGE HIGH 100.0	
PV RANGE LOW 0.0	
PV RANGE EXT. HIGH	102.9
PV RANGE EXT. LOW	-2.9
NUMBER ASSC. CL BLKS	1
All others—default values.	

Table 6 AM Point (Regulatory) Example

TAG NAME	AM101
UNIT ID	01
POINT DESCRIPTOR	TEMPERATURE
ENGR UNITS DESCRIP.	KBTU/MCF
POINT KEYWORD	TEMPA
POINT ASSOCIATED DISPLAY	(blank)
PRIMARY MODULE POINT ID	-----
POINT DISCLOSURE	[FULL]
PV ALGORITHM [SUMMER]	
CONTROL ALGORITHM	[NULL]
POINT EXEC. PERIOD	5SEC
PV DECIMAL FORMAT	[D2]
PV RANGE HIGH 1.0	
PV RANGE LOW 0.0	
PV RANGE EX. HI	1.0
PV RANGE EX. LOW	-1.0
ALGORITHM EQN TYPE	[EQA]
OVERALL GAIN 0.01	
NUMBER INPUT CONN.	1
SOURCE POINT. PARAM.	HG1001.OP
ALM ENB STATE DISABLE	
All others—default values.	

Table 7 Example Parameters for (Regulatory) MC Point HG0501

TAG NAME	HG0501
UNIT ID	01
POINT DESCRIPTOR	MC REGULATORY POINT NO 1
POINT KEYWORD	MCPID
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank)
POINT DISCLOSURE	[FULL]
HIWAY NUMBER 02	
BOX TYPE	[MC]
BOX NUMBER	05
POINT DISCLOSURE	[FULL]
SLOT NUMBER 01	
LOAD DESTINATION	[HG]
ALGORITHM	[PIDNORM]
E. U. DESCRIPTOR	GAL/HR
OVERVIEW VALUE	20
PV RANGE OPTION	[CLMPZER0]
ALARM FORMAT [ALFMT00]	
ALARM PRIORITY	[LOW]
DISPLAY SUPPRESSION	[NOSUPPR]
OPERATOR MODE CHANGE	[PERMIT]
All others—default values.	

Table 8 Example Parameters for PLCG Digital Input Point PLC0801

TAG NAME	PLC0801
UNIT ID	01
POINT DESCRIPTOR	PL DIGITAL POINT NO 1
POINT KEYWORD	PL DIGIN1
PRIMARY MODULE POINT ID	-----
POINT ASSOCIATED DISPLAY	(blank)
HIWAY NUMBER 04	
BOX TYPE	[DHP]
BOX NUMBER	8
POINT DISCLOSURE	[BRIEF]
LOAD DESTINATION	[HG_HIWAY]
INPUT SLOT NUMBER	1
INPUT SUBSLOT NUMBER	1
ALARM FORMAT [STATE1]	
ALARM PRIORITY	[HIGH]
UPPER BOX TEXT	ON
LOWER BOX TEXT	OFF
PC TYPE	[MODICON]
INPUT 1 SPECIFIER CODE	0
All others—default values.	

7.5 Task 22 – Configure Logic Blocks

Keyboard: IKB or PC

Lock: ENGR

Reference: MC11-500

Personality Loaded: Universal

NOTE

If the system has no Multifunction Controllers, skip to Task 23 to configure NIMs, PMs, APMs, and HPMS. If neither Multifunction Controllers nor NIMs, go to Task 24.

Task 22 CONFIGURE LOGIC BLOCKS

Step	Action	Response
22.01	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.03	Select [SYSTEM STATUS] display	
22.04	Select the Hiway Gateway Node	Node box is selected
22.05	Select [NTWK/HWYSTATUS]	Multifunction Controller Status display
22.07	Select [BOX STATUS]	
22.09	Enter Box Number	<u>EG</u> : 5
22.11	Press <ENTER>	BOX STATUS DISPLAY
22.14	Select [IDLE] Target changes color	
22.16	Select [EXECUTE COMMAND]	MF CTLR STATUS: IDLE
22.18	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
CONFIGURE LOGIC BLOCKS		
22.20	Select [LOGIC BLOCKS]	LBC-BOX SELECTION display appears
22.22	Key into PATHNAME port	<u>EG</u> : NET>HMV1>LB020520 (02 = Hiway 2; 05 = Box 5, 20 = revision level 20)
22.24	Press <ENTER>	Message: "SELECT FUNCTION"
22.25	Hold <CTL> down and press <2> (function key = F2)	Message: "TEXT BUILDER IS SELECTED"
22.26	Select [CREATE]	LBC-TEXT BUILDER display
22.27	Key into BLOCK NUMBER	<u>EG</u> : 1
22.28	Press <ENTER>	Additional ports displayed
22.29	Key into ENTER ALGORITHM port	<u>EG</u> : 1
22.30	Press <ENTER>	Algorithm number changes to AND; additional ports appear
22.31	Key into INPUT PARAMETER #1	<u>EG</u> : LP(01).SOA
22.32	Key into INPUT PARAMETER #2	<u>EG</u> : LP(11).SOA
22.33	Key into INPUT PARAMETER #3	<u>EG</u> : LP(12).SOA
22.34	Tab to NEXT BLOCK, [YES] , press <SELECT>	Block 1 information appears in lower half of display. At top left, Block Number changes to 2
22.35	Tab to ENTER ALGORITHM; key into port	<u>EG</u> : 3
22.36	Press <ENTER>	Algorithm number changes to XOR; Additional ports appear
22.37	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
22.38	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
22.39	Key into INPUT PARAMETER #3	<u>EG</u> : LB(104)
22.40	Tab to NEXT BLOCK, [YES] , press <SELECT>	Block 2 information is added to lower half of display. At top left, Block Number changes to 3

Task 22 CONFIGURE LOGIC BLOCKS

Step	Action	Response
22.41	Tab to ENTER ALGORITHM; key in	<u>EG</u> : 7
22.42	Press <ENTER>	Algorithm number changes to FLIP-FLOP; additional ports appear
22.43	Key into INPUT PARAMETER #1	<u>EG</u> : LB(102)
22.44	Key into INPUT PARAMETER #2	<u>EG</u> : LB(103)
22.45	Key into OUTPUT PARAMETER	<u>EG</u> : FL(001)
22.46	Tab to NEXT BLOCK [YES] , press <SELECT>	Block 3 information is added to lower half of display.
22.47	Hold <CTL> down and press <^> (function key = F12)	Message: "WRITING FILE NET> HMV1>LB020520.JS"
22.48	Hold <CTL> down and press <4> (function key = F4) Wait	LBC Compile Display appears; Message: "LBC COMPILE STARTED" Message: "WRITING FILE NET> HMV1>LB020520.JL/.JO" Message: "COMPILE COMPLETED" when compile is finished.
22.49	Hold <CTL> down and press <5> (function key = F5)	Message: "LBC DATA LOAD STARTED"
22.50	Hold <CTL> down and press <6> (function key = F6)	Message: "READING DATA BASE"- then LBC Compare Display appears with last line "NO DIFFERENCES FOUND"
22.51	For other Multifunction Controllers, hold <CTL> down and press <1>. Then, repeat 22.22-22.50 The following steps add comments to the file. If you do not want to add comments, skip to 22.68	Comments are used to document the logic block configuration.
22.52	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
22.53	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
22.54	Type in EDIT command (.JS = Source File)	<u>EG</u> : EDIT NET>H MV1>LB020520.JS
22.55	Press <ENTER>	Logic Block data from previous steps appears
22.56	Move the cursor to the end-of-file: Hold <CTL> down and press <3>, Hold <CTL> down and press <2>	Jump Menu appears Cursor moves to end-of-file following existing text
22.58	Press <RETURN>	
22.60	Type in the comments, for example: — <RETURN> — LOGIC BLOCKS 001 THROUGH 003 ARE OVERTEMP/CONTROL ALARMS.<RETURN> — THE OUTPUT OF THIS CIRCUIT CONTROLS A SECONDARY CIRCUIT<RETURN> — CONSISTING OF LOGIC BLOCKS 010 THROUGH 017.<RETURN>	
22.62	Move the cursor to the end-of-file: Hold <CTL> down and press <3>, Hold <CTL> down and press <1>	Jump Menu appears Cursor moves to beginning-of-file
22.64	Update and exit from the file: Hold <CTL> down, press <1>, Hold <CTL> down, press <2>	Quit Menu appears Message: "FILE WAS UPDATED- X LINES WRITTEN"
22.66	Repeat 22.54-22.64 for other Logic Block Files.	

Task 22 CONFIGURE LOGIC BLOCKS

Step	Action	Response
22.68	Hold down <CTL> and press <HELP>	ENGINEERING MAIN MENU

NOTE

If the system has **Network Interface Module/UCN devices**, go to Task 23 near the back of this manual. When the NIM and UCN nodes have been configured, continue on to Task 24.

7.6 Task 23 – Configure UCN Nodes and Subsystems

Keyboard: IKB or PC

Lock: ENGR

Reference: PM09-640, AP09-640, HP09-640, Personality Loaded: Universal and SW09-605

CAUTION

The NIM(s) must be loaded with its operating personality and have a status of **OK** on the System Status Display before starting this procedure. The NIM was loaded in Task 18.

If you are configuring a PM or HPM, the procedure is nearly identical to the APM configuration described below except where noted. The sample system did not have an SM or LM. Refer to the manuals in Section 1, References, if necessary, to configure these.

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.110	Hold <CTL> down and press <HELP> If you do not have a NIM, skip to Task 24.	ENGINEERING MAIN MENU
\$NMhhNnn Where hh is the UCN number and nn is the NIM Node number		
CONFIGURE THE NIM UCN ADDRESSES		
23.115	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
23.120	Select [UCN NODE CONFIGURATION]	UCN-NODE CONFIGURATION display
23.130	Key into NETWORK NUMBER port	<u>EG</u> : 03
23.140	Key into UCN NODE NUMBER port (NIM node number)	<u>EG</u> : 01 (primary NIM node # on 1st pass) <u>EG</u> : 02 (redundant NIM node number, if any, on 2nd pass)
23.150	Select an NODE TYPE option	<u>EG</u> : [NIM]
23.160	Select a NODE ASSIGNMENT option	<u>EG</u> : [THISNIM]
23.165	Press <ENTER>	PED reappears
23.169	Select a LOAD SCOPE option	<u>EG</u> : [NIMANDPM]
23.173	Select a NUMBER OF MESSAGE TEXTS	<u>EG</u> : [3]
23.180	Press <ENTER>	Message Text ports appear
23.184	Key in message text	<u>EG</u> : [OUT/SERV] [RED TAG] [IN USE]
23.188	Select an SOE SYNC option	<u>EG</u> : [ENABLE] (if NIM has EPNI board)
23.190	Press <ENTER>	PED reappears with values in cyan
NOTE		
If you select any options outlined in yellow or if you key data into any port outlined in yellow, the content of the PED-set or PED changes when you press the <ENTER> key. You should press <ENTER> after each PED page.		
23.210	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page.	UCN NODE CONFIGURATION Ped
WRITE TO IDF		
23.320	Press <COMND>	COMMAND DISPLAY
23.400	If this is the <u>first</u> write to this IDF, go to step 32.460; if not, go to step 32.485	IDF for this example is NIMIDF

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.460	Select [WRITE TO IDF]	Pathname ports and target appear.
23.470	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.480	Key into pathname for IDF port	<u>EG</u> : NIMIDF
23.485	Press <ENTER>	NIM Config data written to IDF. UCN NODE CONFIGURATION display
23.488	Repeat steps 32.130 – 32.485 for redundant NIM (if any)	
23.490	Repeat steps 32.130 – 32.488 for each other NIM pair.	All configured NIMs on IDF
23.500	Press <COMND>	COMMAND DISPLAY
23.510	Select [LOAD MULTIPLE]	Additional pathname ports appear
23.515	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.520	Key into pathname for IDF port	<u>EG</u> : NIMIDF
23.525	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" COMMAND DISPLAY
23.530	Hold <CTL> down and press <1>	UCN NODE CONFIGURATION display
\$NMhhNnn Where hh is the UCN number and nn is the APM Node number		
(PM/APM/HPM/LM/SM) NODE CONFIGURATION		
The UCN NODE CONFIGURATION display should be on the screen.		
23.550	Key into NETWORK NUMBER port	<u>EG</u> : 03
23.560	Key into UCN NODE NUMBER port (APMM node number)	<u>EG</u> : 03 (Primary node # on first <u>EG</u> : 04 pass, redundant node #, if any, on second pass)
23.570	Select [UCN NODE TYPE]	<u>EG</u> : [APM]
23.580	Select a NODE ASSIGNMENT option	<u>EG</u> : [THISNIM]
23.590	Press <ENTER>	PED reappears with values in cyan
23.595	Key in any other data/select options press <ENTER>	
WRITE TO IDF		
23.610	Press <COMND>	COMMAND DISPLAY
23.620	If this is the <u>first</u> write to this IDF, go to step 32.630; if not, go to step 32.660	IDF for this example is APMMIDF
23.630	Select [WRITE TO IDF]	Pathname ports and target appear.
23.640	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.650	Key into pathname for IDF port	<u>EG</u> : APMIDF
23.660	Press <ENTER>	APM Config data written to IDF. UCN NODE CONFIG. PED display
23.670	Repeat steps 32.550 – 32.660 for the redundant APM (if any).	
23.680	Repeat steps 32.550 – 32.670 for each other APM	All configured APMs in IDF
23.700	Press <COMND>	COMMAND DISPLAY
23.710	Select [LOAD MULTIPLE]	Additional pathname ports appear
23.720	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.730	Key into pathname for IDF port	<u>EG</u> : APMMIDF
23.740	Press <ENTER> Repeat steps 32.550 - 32.740 for other PMs, or HPMS (if any).	"WAIT" in yellow;"...OPERATION COMPLETE" Use different IDF names for PMs and HPMS
LOAD THE APM		
23.742	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
23.744	DISK_&C6.LCN copy into \$F1	
23.746	Select [SYSTEM STATUS]	SYSTEM STATUS display
23.748	Select the NIM Node	Node Box is selected
23.750	Select [NTWK/HWY STATUS]	UCN nn STATUS display
23.754	Select [nn APM nn] The status must be <i>ALIVE</i> (Select target 2nd time if loading the redundant node)	Target fills with color; selected node is highlighted
23.758	Select [LOAD/SAVE RESTORE]	Additional targets appear
23.762	Select [PROGRAM LOAD]	Target turns white
23.766	Select [ENTER]	PGM SOURCE FOR NODE nn display
23.770	Select [DEFAULT SOURCE]	EXECUTE target appears
23.774	Select [EXECUTE]	DATA SOURCE FOR NODE nn display
23.778	Select [ALTERNATE SOURCE]	Targets [1], [2], and so on appear
23.782	Select <u>EG</u> : [1]	&C6 and [EXECUTE COMMAND]
23.786	Select [EXECUTE COMMAND]	" BUSY UCN nn " [nn APM nn LOADING] " * " "MOUNTAPM nn DATA BASE"
23.790	Select [MOUNT APMnn DATA BASE] (Primary node only)	Additional targets appear
23.794	Select [ABORT REQUEST] (Primary node only)	<u>EG</u> : [nn APM nn] [IDLE] [ALIVE/IDLE] [LOADED]
23.796	Repeat Tasks 32.754 – 32.794 for redundant APM (if any)	<u>EG</u> : [nn APM nn] [IDLE] [BACKUP] [LOADED]
23.798	Repeat Tasks 32.754 – 32.796 for other PMs, APMs, or HPMS.	
\$NMhhBuu Where hh is the UCN number and uu is the APMM Node number		
PM/APM/HPM NODE SPECIFIC CONFIGURATION		
23.800	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
23.802	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
23.806	Select [NODE SPECIFIC CONFIG.]	NODE SPECIFIC CONFIGURATION MENU
23.808	Key into NETWORK NUMBER	<u>EG</u> : 03
23.814	Key into NODE NUMBER	<u>EG</u> : 03 (Primary APMM)
23.816	Key into NODE TYPE	<u>EG</u> : APM

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.818	Press <ENTER>	
23.820	Press <PAGE FWD> to next PED page, key in data, select options, and press <ENTER> after each page or gold port	NODE SPECIFIC CONFIG. PED
23.822	Select a CONTROL PROC. OPTION (Not Applicable for HPMs)	EG: [ON]
23.826	Key into NUM OF REG CONTROL SLOTS	EG: 25
23.830	Key into NUMBER OF FAST REG	EG: 5
23.834	Key into NUMBER OF REG PV SLOTS	EG: 20
23.840	Key into NUM OF FAST REG PV SLOTS	EG: 5
23.844	Key into NUMBER OF LOGIC SLOTS	EG: 10
23.846	Key into NUM OF FAST LOGIC SLOTS	EG: 3
23.850	Key into NUM OF DIG. COMP. SLOTS	EG: 15
23.852	Key into NMBER OF FAST DIG. COMP.	EG: 2
23.854	Key into NUM OF DEV. CTL SLOTS	EG: 4
23.856	Key into NMBER OF FAST DEV. CTL	EG: 2
23.858	Key into SEQUENCE PROCESSING	EG: 1_PU
32.860	Key into NUM OF PROC MODULE SLOTS	EG: 10
23.862	Key into NUM OF NUMERICS SLOTS	EG: 48 (Multiples of 16 only)
23.864	Key into NUM OF STRINGS SLOTS*	EG: 16 (Multiples of 16 only)
23.866	Key into NUM OF TIMES SLOTS*	EG: 32 (Multiples of 32 only)
23.868	Key into NUM OF ARRAY SLOTS*	EG: 10
23.870	Select an SI DATA... SCAN Period*	EG: 1.0
23.874	Select a SCAN RATE	EG: [REG1LOG1]
23.880	Select a PACKAGING OPTION (Select REDUN if redundant nodes)	EG: [REDUN]
23.883	Select a Simulation Indicator option*	EG: ON
23.885	Press <PAGE FWD> to next PED page, key data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED
23.900	Choose an I/O Module number, press <PAGE FWD> if necessary	EG: I/O MODULE NUMBER 1
23.910	Key into FILE NUMBER	EG: 1
23.920	Key into CARD NUMBER (Example is 1st card Number if REDUN is selected)	EG: 11 for APM ; (9 for PM, 03 for HPM)
23.930	Select a MODULE TYPE	EG: [HLAI]
23.932	If the system has redundant IOPs, select REDUN. If not, skip to 32.940	EG: IOMB File and Card number ports
23.934	Enter redundant IOP File and Card No.	EG: File 2, Card No. 11
23.940	Choose another I/O Module number, press <PAGE FWD> if necessary	EG: MODULE NUMBER 2
23.950	Key into IOM-A FILE NUMBER	EG: 1
23.960	Key into IOM-A CARD NUMBER	EG: 12 for APM

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.970	Select MODULE TYPE	<u>EG</u> : [AO]
23.975	Select a FAILURE OPTION	<u>EG</u> : [HOLD]
23.980	Key in any other data/select options, press <ENTER>	
23.990	Press <PAGE FWD> to next PED page (if any) key in data, select options, and press <ENTER> after each page	I/O MODULE CONFIGURATION PED
23.995	Repeat steps 32.940 through 32.990 to configure other I/O Modules	
* Not applicable for PMs		
WRITE TO IDF		
23.1000	Press <COMND>	COMMAND DISPLAY
23.1020	If this is the <u>first</u> write to this IDF, go to step 32.1030; if not, go to step 32.1060	IDF for this example is APMCONFIG
23.1030	Select [WRITE TO IDF]	Pathname ports and target appear.
23.1040	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.1050	Key into pathname for IDF port	<u>EG</u> : APMCONFIG or HPMCONFIG
23.1060	Press <ENTER>	I/O Module Config data written to IDF. IO MODULE CONFIG. display
23.1070	Repeat steps 32.800 – 32.1060 for other APMs on this UCN (if any)	
23.1080	Repeat steps 32.800 – 32.1070 for each other PM, APM, or HPM primary (if any)	All configured PMs, APMs, HPMs on IDFs (use different IDF names for PMs and HPMs)
23.1100	Press <COMND>	COMMAND DISPLAY
23.1110	Select [LOAD MULTIPLE]	Additional pathname ports appear
23.1115	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
23.1120	Key into pathname for IDF port	<u>EG</u> : APMCONFIG
23.1125	Press <ENTER>	"WAIT" in yellow;.."OPERATION COMPLETE"
23.1150	Repeat steps 32.1100 – 32.1150 for PMs or HPMs (if any)	PMs, APMs, HPMs loaded with Node Specific Configuration Data
PM/APM/HPM POINT BUILDING		
23.1200	Hold <CTL> down and press <9>	NIM BUILD TYPE SELECT MENU
23.1220	Select [PROCESS POINT BUILDING]	NIM PROC. POINT BUILD TYPE display
23.1230	Select a Point Build Type	<u>EG</u> : [REGULATORY CONTROL] NIM-POINT ASSIGNMENT display
23.1240	Key into TAG NAME port	<u>EG</u> : FC2401
23.1244	Select a NODE TYPE	<u>EG</u> : APM
23.1250	Select a POINT FORM option	<u>EG</u> : [FULL]
23.1260	Key into the POINT DESCRIPTOR port	<u>EG</u> : TOWER FLOW CONTROL
23.1265	Key into the E. U. DESCRIPTOR port	<u>EG</u> : KB/D
23.1275	Key into the POINT KEYWORD port	<u>EG</u> : FLOW
23.1285	Key into the UNIT ID port	<u>EG</u> : 01
23.1290	Key into the NETWORK NUMBER port	<u>EG</u> : 03

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.1295	Key into the NODE NUMBER port	<u>EG</u> : 03
23.1300	Key into the SLOT NUMBER port	<u>EG</u> : 01
23.1310	Key in other data/select options press <ENTER>	
23.1315	Press <PAGE FWD>	Next PED page
23.1320	Select a Control Algorithm; press <ENTER>	<u>EG</u> : [PID]
23.1325	Press <PAGE FWD>	Next PED page
23.1330	Key into PV RANGE HIGH port	<u>EG</u> : 25.00000
23.1335	Select a PV SOURCE OPTION	<u>EG</u> : [ALL]
23.1340	Press <ENTER> after each page and after any entry boxed in yellow	
23.1345	Press <PAGE FWD>	Next PED page
23.1350	Select a NORMAL MODE	<u>EG</u> : [AUTO]
23.1355	Select a NORMAL MODE ATTRIBUTE	<u>EG</u> : [OPERATOR]
23.1360	Key in any other data/select options; press <ENTER> after each page	
23.1365	Press <PAGE FWD>	Next PED page
23.1370	Key into SET POINT HIGH LIMIT port	<u>EG</u> : 25.00000
23.1380	Key into SET POINT LOW LIMIT port	<u>EG</u> : 0.000000
23.1390	Key into SET POINT port	<u>EG</u> : 15.00000
23.1395	Key in any other data/select options; press <ENTER> after each page	
23.1400	Press <PAGE FWD>	Next PED page
23.1420	Select an ALGORITHM FORM option	<u>EG</u> : [IDEAL]
23.1430	Key into OVERALL GAIN VALUE port	<u>EG</u> : 0.699200
23.1440	Key into INTEG. TIME IN MINUTES port	<u>EG</u> : 0.100000
23.1445	Key in any other data/select options; press <ENTER> after each page	
23.1450	Press <PAGE FWD>	Next PED page
23.1460	Key into INPUT CONN. SOURCE port	<u>EG</u> : - - . - -
23.1462	Key into OUTPUT DEST. SOURCE port	<u>EG</u> : !AO02S02.OP
23.1465	Key in any other data/select options; press <ENTER> after each page	
23.1470	Press <PAGE FWD>	Next PED page
23.1485	Key in any other data/select options; press <ENTER> after each page	
23.1490	Press <PAGE FWD>	Next PED page
23.1500	Key into PV HIGH TRIP POINT port	<u>EG</u> : 25.00000
23.1502	Key into PV LOW TRIP POINT port	<u>EG</u> : 0.000000
23.1504	Key into POSITIVE PV ROC TRIP port	<u>EG</u> : 32.00000
23.1508	Press <ENTER> after each page and after each entry boxed in yellow	
23.1510	Press <PAGE FWD>	Next PED page

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.1512	Key into NEG. PVROC TRIP POINT port	<u>EG</u> : 20.00000
23.1514	Key into DEV HIGH TRIP POINT port	<u>EG</u> : 5.000000
23.1516	Key into DEV LOW TRIP POINT port	<u>EG</u> : 5.000000
23.1518	Press <ENTER>	
23.1520	Press <PAGE FWD> to next PED page. Enter any other data/select options and press <ENTER> after each page until all pages are configured.	
23.1522	Hold down <CTL> and press <8>	PED STATUS display
23.1524	Check for proper status	"Display Set Entered?" Yes Errors? No
23.1526	If status is OK, go to step 32.1530; else select faulty item; enter corrections.	
WRITE TO IDF		
23.1530	Press <COMND>	COMMAND DISPLAY
23.1532	If this is the <u>first</u> write to this IDF, go to step 32.1534; if not, go to step 32.1560	IDF for this example is APMPNTS
23.1534	Select [WRITE TO IDF]	Pathname ports and target appear.
23.1540	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.1550	Key into pathname for IDF port	<u>EG</u> : APMPNTS
23.1560	Press <ENTER>	NIM Point Config. data written to IDF. NIM Point PED display
23.1570	Repeat steps 32.1240 – 32.1560 to build other points of this type, or hold <CTL> down and press <9>. Then repeat steps 32.1230 – 32.1570 to build other types of NIM Points until all NIM points are built. If you want to clone points from a master point, go to Appendix D.	
23.1600	Press <COMND>	COMMAND DISPLAY
23.1610	Select [LOAD MULTIPLE]	Additional pathname ports appear
23.1620	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
23.1630	Key into pathname for IDF port	<u>EG</u> : APMPNTS
23.1640	Press <ENTER> Build/load points for PMs and HPMS (if any) by repeating steps 32.1200 - 32.1640	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE" Use different IDF names for PM or HPM points
NIM LIBRARY CONFIGURATION		
\$NhhLIBn Where hh is the UCN number and n is the library number		
Note that the NIM Library is usually automatically loaded while compiling an APM Sequence program. The following steps illustrate how to access, manually load, or modify it.		
23.1710	If the COMND or Point CONF. display is on screen, hold <CTL> down and press <9> <u>twice</u> . Then go to step 32.1720. Else hold <CTL> down and press <HELP>	POINT TYPE BUILD MENU, then NIM BUILD TYPE SELECT MENU ENGINEERING MAIN MENU

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.1715	Select [NETWORK INTF. MODULE]	NIM BUILD TYPE SELECT MENU
23.1720	Select [LIBRARY CONFIGURATION]	NIM LIBRARY CONFIGURATION display
23.1722	Key into the NETWORK NUMBER port	<u>EG</u> : 03
23.1724	Key into the Library Number port	<u>EG</u> : 1
23.1728	Press <ENTER>	
23.1730	<PAGE FWD>	
23.1732	Key into the INDEX ENTRY ports	<u>EG</u> : TRUE for 002, FALSE for 003, etc.
23.1734	Press <ENTER>	
23.1736	<PAGE FWD>. Enter any other data, press <ENTER> after each page.	
WRITE TO IDF		
23.1740	Press <COMND>	COMMAND DISPLAY
23.1745	If this is the <u>first</u> write to this IDF, go to step 32.1755; if not, go to step 32.1765	IDF for this example is LIBCNFG
23.1750	Select [WRITE TO IDF]	Pathname ports and target appear.
23.1755	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HNV1>
23.1760	Key into pathname for IDF port	<u>EG</u> : LIBCNFG
23.1765	Press <ENTER>	NIM Library Config data written to IDF. NIM LIBRARY CONFIGURATION display
23.1767	Repeat steps 32.1724 – 32.1765 to build Libraries 2/3 (if desired) for this UCN	
23.1770	Repeat steps 32.1722 – 32.1767 to build Libraries for other UCNs.	
23.1800	Press <COMND>	COMMAND DISPLAY
23.1810	Select [LOAD MULTIPLE]	Additional pathname ports appear
23.1820	Key into Reference Pathname port	<u>EG</u> : NET>HNV1>
23.1830	Key into pathname for IDF port	<u>EG</u> : LIBCNFG
23.1840	Press <ENTER>	"WAIT" in yellow; "PROCESSING ..." "OPERATION COMPLETE"
NOTE In the following steps you will not be able to save data for any IO Processor for which you have not previously built and loaded at least one data point. When saving data, all IOPs that are configured for a given PM, APM, or HPM will assume a "DB Invalid" status (and should not be validated until a point has been built and loaded).		
SAVE CHECKPOINT DATA FOR A NIM AND UCN device (PM, APM/HPM/LM/SM)		
23.1910	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
23.1930	Select [SYSTEM STATUS]	SYSTEM STATUS display
23.1934	Select the NIM Node	Node Box is selected
23.1938	Select [NTWK/HWY STATUS]	UCN nn STATUS display
23.1945	Select the appropriate APM target	nn APM nn target fills with color
23.1950	Select [LOAD/SAVE RESTORE]	Additional targets appear.
23.1955	Select [SAVE DATA]	Target changes color; additional targets
23.1960	Select [ENTER]	SAVE UCN NODE display

Task 23 CONFIGURE UCN NODES AND SUBSYSTEMS

Step	Action	Response
23.1965	If you wish to save checkpoints on emulated disk, go to step 32.1970. If you wish to save checkpoints on an HM, then you must copy NULL METADATA file to HM checkpoint volume. Mount copy &Z6.LCN Select [DEFAULT SOURCE] and then go to step 32.1980.	Note: Both NIM data and the associated APM data is saved by this procedure. CP \$F5>&Ixx>METADATA.FF NET>&Ixx>= -d
23.1970	CHKPOINT.LCN emulated disk into \$F1	
23.1975	Select [ALTERNATE SOURCE]	Targets [1] and [2] etc. appear
23.1978	Select [1]	<u>EG</u> : Volume ID = CKPT
23.1980	Select [EXECUTE COMMAND]	"BUSY UCN nn display and " * "
23.1999	Wait for status to change	"SAVED"
23.2000	Repeat steps 32.1945 – 32.1999 for other PMs, APMs, and HPMS.	

Section 8 – Procedure 5/Configure Custom Functions

Figure 8 shows a graphic representation of the Tasks (24-29) that must be performed to Configure Custom Functions.

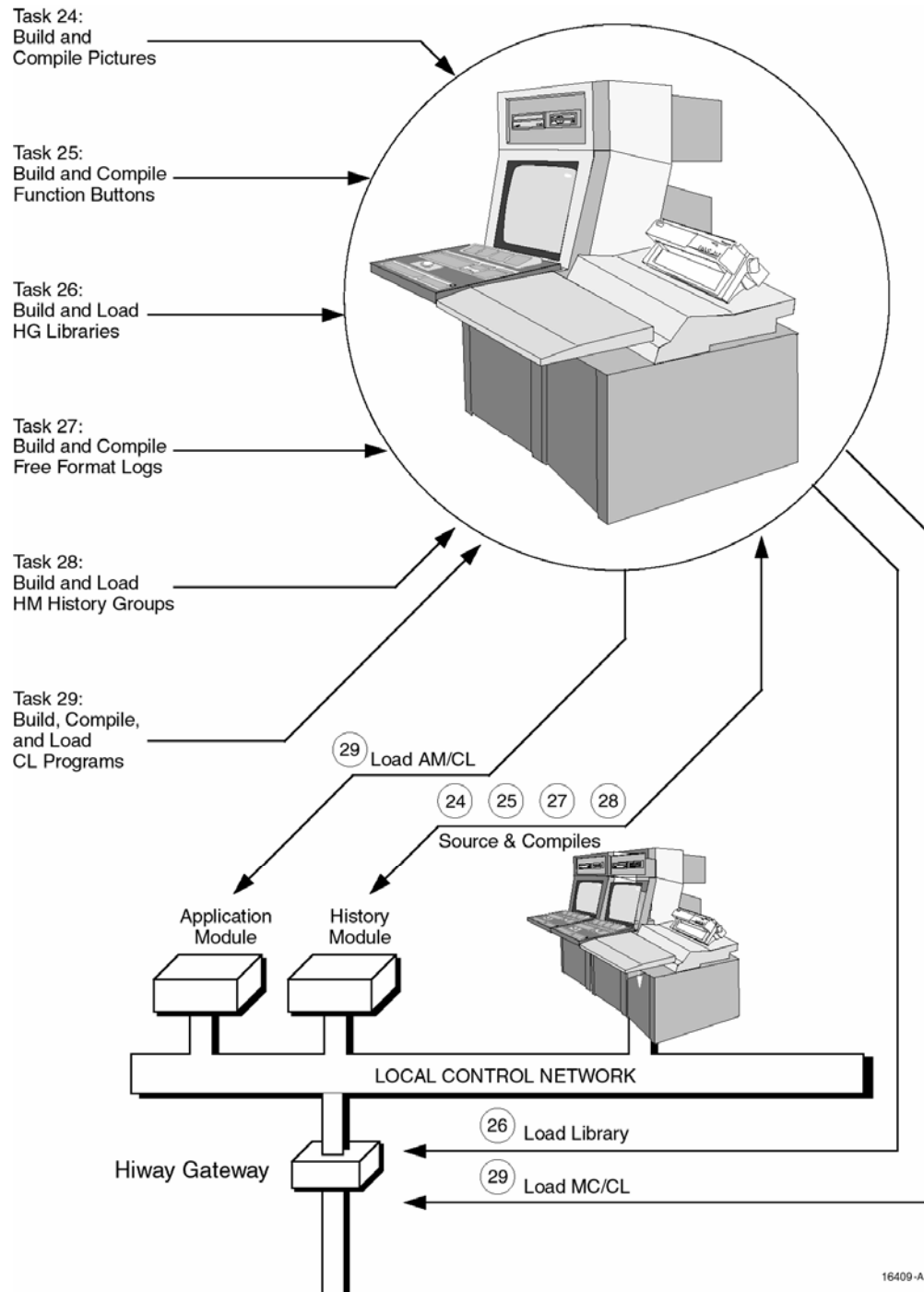


Figure 8 Configure Custom Functions

8.1 Task 24 – Build and Compile Pictures

Keyboard: IKB or PC

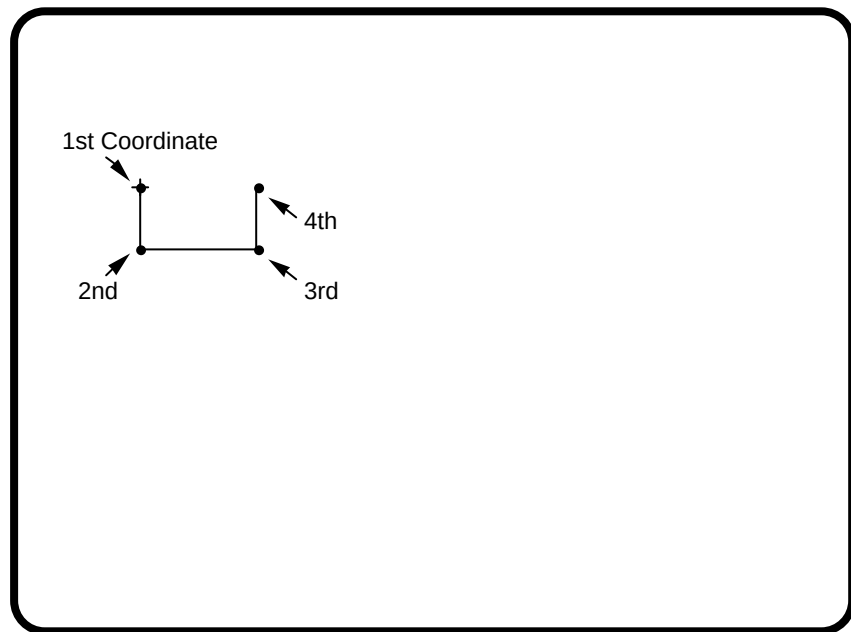
Lock: ENGR

Reference: SW09-650 and SW11-650

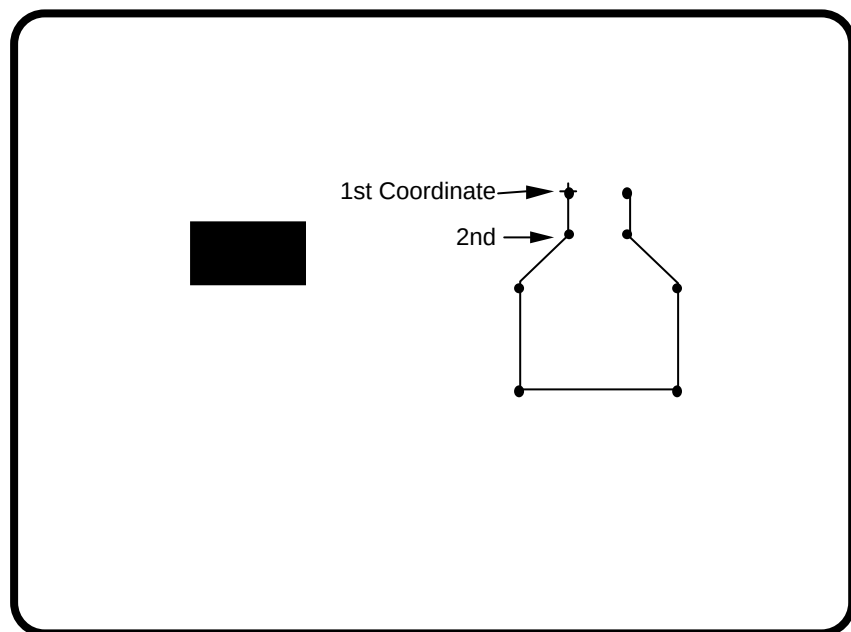
Personality Loaded: Universal

Task 24 BUILD AND CONFIGURE PICTURES

Step	Action	Response
24.10	Make sure points referenced by pictures have been built and loaded.	
24.14	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
24.18	Select [PICTURE EDITOR]	PICTURE EDITOR display
The following steps and Figure 9 show how to build a simple Custom Graphic Display. If pictures were prebuilt, go to step 24.160 to read them in and then compile them on the HM.		
24.22	Key in a Picture Editor command	<u>EG</u> : ADD SOLID
24.28	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
24.32	Move cursor to desired location	Hold <CTL> down, press <TAB> keys
24.40	Press <SELECT>	Cross marks the coordinate (see Figure 9a)
24.44	Repeat steps 24.32 – 24.40 until all of the coordinates are selected.	Note that deletes the last coordinate selected if you need to make a correction.
24.48	Press <ENTER>	Command Completed Note: The NEW command clears the screen if you want to redraw.
24.56	Repeat steps 23.22 – 23.48 to build additional objects.	<u>EG</u> : Another Solid. (Figure 9b)
24.60	Hold <CTL> down and press a color key (color keys are on the keypad).	<u>EG</u> : <CTL> and <4> Status indicator at top center of screen changes to white.
24.65	Key in a Picture Editor command	<u>EG</u> : ADD LINE
24.70	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
24.75	Move cursor to desired location.	<u>EG</u> : Hold <CTL> down; press TAB
24.80	Press <SELECT>	Cross marks the Coordinate
24.85	Repeat steps 24.75 – 24.80 until all of the coordinates are selected.	<u>EG</u> : Lines connect objects (See Figure 10c).
24.90	Press <ENTER>	Command Completed



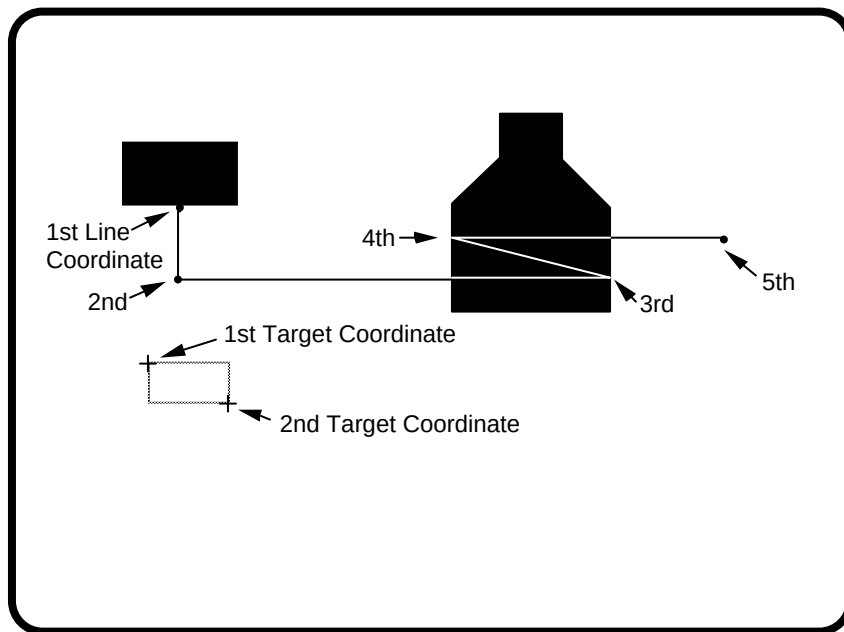
(a)



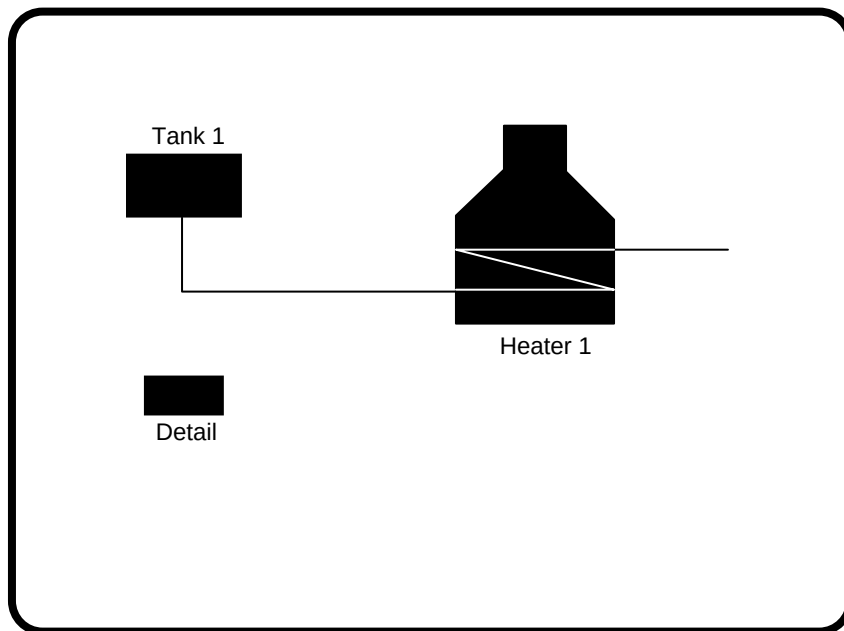
(b)

2600

Figure 9 Picture Building



(c)



(d)

2661

Figure 10 Picture Building Continued

Task 24 BUILD AND CONFIGURE PICTURES

Step	Action	Response
24.94	Key in Picture Editor commands to add conditional behavior, and so on.	<u>EG</u> : SEL SOL;ADD COND
24.98	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
24.102	Move cursor to desired location	<u>EG</u> : Place cursor on solid rectangular object
24.106	Press <SELECT>, then press <ENTER>	Object turns white and blinks; Screen Form appears
24.110	Key in screen form information (points you refer to must have been built and loaded in Task 21)	<u>EG</u> : (in condition block): IF HG1001.PV > 75 THEN SET RED ELSE SET CYAN
24.114	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
24.118	Key in commands to add targets	<u>EG</u> : ADD TARG
24.124	Press <ENTER>	Prompt: "ENTER ... COORDINATES"
24.128	Move cursor to desired location	<u>EG</u> : Place cursor under first solid rectangular object (see Figure 10c)
24.132	Press <SELECT>	Cross marks 1st target coordinate
24.136	Move cursor; press <SELECT>	Rectangle drawn around coordinates
24.140	Press <ENTER>	Target screen form appears
24.145	Key into Action area	<u>EG</u> : DETAIL("HG1001")
24.150	Press <ENTER>	Picture returns (or point Type/Format requested if point not found)
24.154	Enter any other commands to complete desired custom display.	<u>EG</u> : ADD TEXT Label objects (see Figure 10d)
24.156	Press <ENTER> to complete command	
24.159	Go to step 23.210 to compile the display	
Follow steps 23.160 – 23.190 to read in prebuilt pictures from emulated disks.		
24.160	Insert emulated disk with picture files in \$F1; key in the SET PATHNAME command.	<u>EG</u> : S P \$F1>VOL1>DISP1 (where VOL1 is the volume name and DISP1 is the file name)
24.170	Press <ENTER>	Pathname appears at top of display
24.180	Key the READ command onto the command line.	<u>EG</u> : READ DISP1 (Assumes DISP1 is filename)
24.190	Press <ENTER>	Picture appears on US screen
24.210	Key in a COMPILE command	<u>EG</u> : COMPILE NET>HVM1>DISP1
24.220	Press <ENTER>. Picture on the screen is compiled onto the HM. To have a redundant schematic, you should compile it to other HMs, cartridge disks, or floppies now.	Message: "WRITING SOURCE FILE" Message: "WRITING OBJECT FILE" No redundant schematics were configured on the example system.
24.230	Repeat steps 23.19 – 23.220 until each of your pictures is compiled.	
24.240	If you want to save picture source files insert cartridge in \$F1.	<u>EG</u> : USER (volume created in task 4)
24.241	Press <ESC>	"ESCAPED FROM PICTURE EDITOR"

Task 24 BUILD AND CONFIGURE PICTURES

Step	Action	Response
24.244	Key in a copy command	<u>EG</u> : CP NET>HMOV1>DISP1.DS \$F1>PICT>DISP1 -D
24.248	Press <ENTER>	Message: "COPY COMPLETE" Source file is copied
24.249	Hold <CTL> down and press <HELP>	Return to Picture Editor

8.2 Task 25 – Build and Compile Buttons

Keyboard: IKB or PC

Lock: ENGR

Reference: SW11-570

Personality Loaded: Universal

Task 25 BUILD AND CONFIGURE BUTTONS

Step	Action	Response
Make sure all the points and pictures you need were created in previous tasks		
25.20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
25.30	Select [BUTTON CONFIGURATION]	BUTTON CONFIGURATION MENU
25.35	Select a button (EG: 7)	ACTION & LAMP DATA regions appear
25.40	Key into the Action region	EG: DETAIL("HG1001")
25.45	Press <ENTER>	BUTTON CONFIGURATION MENU appears
25.50	Select another button (EG: 8)	ACTION & LAMP DATA regions appear
25.55	Key into the Action region	EG: SCHEM("DISP1")
25.60	Press <ENTER>	BUTTON CONFIGURATION Menu
25.65	Select another button (EG: 9)	Action & Lamp Data regions appear
25.70	Key into the Action region	EG: SCHEM(R_STR)
25.75	Press <ENTER>	PARAMETER ENTRY display
25.80	Key into ports exposed by Actor.	EG: 10 into "Tip Length" EG: "ENTER SCHEMATIC NAME" into "Prompt 1–30 Char"
25.85	Press <ENTER>	BUTTON CONFIGURATION MENU
CONFIGURE A DISPLAY SET		
25.90	Select another button	EG: 10 ACTION & LAMP DATA regions appear
25.100	Key into the Action region	EG: CROSSCRN(1);SCHEM("DISP1"); CROSSCRN(2);GROUP(1,0)
25.140	Press <ENTER>	BUTTON CONFIGURATION MENU
25.150	Repeat steps 25.35 – 25.140 to configure rest of buttons in set.	Buttons for all stations in an Area are configured by this task.
NOTE If you want a button on the Operator Keyboard to call up either the Overview Display or the Area Trend Display, configure them here. For the Universal Work Station's Engineering Keyboard, <PAGE FWD> from the main Button Menu to the PF Button Menu and configure as desired.		
25.160	Key COMPILE xxx into the cmd line.	EG: COMPILE NET>&D01>BUTTON
25.270	Press <ENTER>	"WRITING SOURCE/OBJECT FILE"
25.180	Key in a PRINT command	EG: PRINT 7..10 \$P1
25.190	Press <ENTER>	Button Config. record is printed

8.3 Task 26 – Build and Load HG Libraries

Keyboard: IKB or PC

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

Task 26 BUILD AND LOAD HG LIBRARIES

Step	Action	Response
Note: If you do not have a Hiway Gateway, skip to Task 27.		
HG LIBRARY \$HnnLIBx where nn = Hiway No. and x = Library No.		
26.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
26.15	Select [HIWAY GATEWAY]	HG-BUILD TYPE AND CONFIG. MENU
26.20	Select [HG LIBRARY]	HG LIBRARY CONFIGURATION display
26.30	Key into HIWAY NUMBER port	<u>EG</u> : 02 (configured in Task 9)
26.40	Key into LIBRARY NUMBER port	<u>EG</u> : 1
26.50	Key into INDEX ports	<u>EG</u> : ON (into "Index 001" port) <u>EG</u> : OFF (into "Index 002" port) <u>EG</u> : TRUE (into "Index 003" port) <u>EG</u> : FALSE (into "Index 004" port)
26.60	Key into other Index ports as required. Press <ENTER> after page is done.	PED reappears with values in cyan
26.70	Press <PAGE FWD> and key into Index ports as required.	New PED page appears.
26.80	When a given library is complete, press <COMND>	COMMAND DISPLAY
26.82	Select [WRITE TO IDF]	Pathname Ports and target appear.
26.83	Key in a REFERENCE PATHNAME	<u>EG</u> : NET>HNV1>
26.84	Key in an IDF Pathname Press <ENTER>	<u>EG</u> : HGLIB Library is saved in IDF. HG-LIB. CONFIG display
26.85	Press <COMND>	COMMAND DISPLAY
26.90	Select [LOAD]	
26.100	Press <ENTER>	"BUSY" HG-LIBR. CONFIGURATION display
NOTE You should save the HG checkpoint(s) on HM and perhaps emulated disk at this time. See Appendix B of this document for instructions.		
26.110	Repeat Task 25 for each other HG Library to be configured.	<u>EG</u> : Library numbers 2, 3, and 4

8.4 Task 27 – Build and Compile Free Format Logs

Keyboard: IKB or PC

Lock: ENGR

Reference: HM11-660

Personality Loaded: Universal

Task 27 BUILD AND COMPILE FREE FORMAT LOGS

Step	Action	Response
27.10	Make sure points referenced in the logs have been built and loaded in Task 21	Prebuilt logs can be read in from removable media and compiled on the HM per step 26.90
27.14	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
27.18	Select [FREE FORMAT LOGS]	FFL BUILDER display. At top of screen, R- 0, 42 indicates the Edit region is now at top left of drawing area.
27.22	On Cmd line, type: SET GRID ON	
27.26	Press <ENTER>	Grid appears
27.34	Enter Command: ADD TEXT , press <ENTER>	"ENTER TEXT" System ready for text input
27.38	Move cursor to locations where you want text and type it in. Figure 11 shows an example of text in a Free Format Log.	EG: WEST AREA LOG (title) FURNACE INLET TEMPERATURE = COIL OUTLET TEMPERATURE = etc. See Figure 11a.
27.42	Press <ENTER>	Text is added to display
27.46	Enter Command: ADD VALUE Press <ENTER>	"ENTER VALUE COORDINATES" System waits for you to specify Value coordinates
27.50	Move cursor to each point where you want a value; press <SELECT>. Note cursor location (for example, C-224, 944) on 2nd line at that time.	+ Marks Value locations. (see Figure 11b)
27.54	Press <ENTER>	Screen form appears requesting EXPRESSION for Value at XXX, YYY. (for example, VALUE AT 224, 944)

DD MMM YY HH MM 01									
R 0, 42 C 0, 1040									
WEST AREA LOG									
60	FURNACE INLET TEMPERATURE = COIL OUTLET TEMPERATURE = PUMP MOTOR STATUS = AVERAGE FLOW = OIL FLOW =								
50									
	10	20	30	40	50	60	70		
ADD TEXT									
Enter Text									

a. Text Locations

DD MMM YY HH MM 01									
R 0, 42 C 0, 1040									
WEST AREA LOG									
60	FURNACE INLET TEMPERATURE = + COIL OUTLET TEMPERATURE = + PUMP MOTOR STATUS = + AVERAGE FLOW = + OIL FLOW = +								
50									
	10	20	30	40	50	60	70		
ADD VALUE									
Enter Value Coordinates									

b. Value Locations. Selected locations are marked by +.

2663

Figure 11 Free Format Log Example

Task 27 BUILD AND COMPILE FREE FORMAT LOGS

Step	Action	Response
27.58	Enter EXPRESSION information for each value (one per pass).	<u>EG</u> : SYS_TIME (1st Value) SYS_TIME (2nd Value) HG1001.PV HG1002.PV etc. for others (points must exist)
27.62	Press <ENTER>	Screen form requests FORMAT
27.74	Enter Format information (one per pass)	<u>EG</u> : DATEMM-DD-YYENDDATE (1st Value) TIMEHH:MM:SSENDTIME (2nd Value) R-ZZZZ9.99 (other Values)
27.78	Press <ENTER>	String of letters replaces cross at value location (for example, DTDTDT, R-ZZZZ9.99, etc.). EXPRESSION for the next value appears.
27.79	Repeat steps 27.58 – 27.78 until all Value information is entered.	
27.80	If TYPE is requested, Enter TYPE information for that Value.	EG: DATE, TYPE, REAL, etc. (If type is known to the system, this screen form is omitted.)
27.82	Enter command: SET GRID OFF , press <ENTER>	Grid disappears
27.90	If FFL was prebuilt you can read it in from cartridge disk at this time.	<u>EG</u> : READ \$F1>VOL2>FFL001 (reads file FFL001 from cartridge named VOL2)
27.94	Enter Command to compile log.	<u>EG</u> : COMPILE NET>HMV1>FFL001
27.96	Press <ENTER>	"WRITING SOURCE/OBJECT FILE" Log named FFL001 is compiled on HM
27.100	Repeat steps 26.22 – 26.96 for each FFL.	

8.5 Task 28 – Build and Load HM History Groups

Keyboard: IKB or PC

Lock: ENGR

Reference: HM12-500 and SW11-511

Personality Loaded: Universal

Task 28 BUILD AND LOAD HM HISTORY GROUPS

Step	Action	Response
HM HISTORY \$CHuu(n) uu = Unit No., n = Hist. Group Note: The HM Online Personality must be running for Task 28. Check the System Status display for "HISTORY MODULES-STATUS OK"		
28.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
28.20	Select [HM HISTORY GROUPS]	CH HEADER DISPLAY
28.30	Key into UNIT port	<u>EG</u> : 01
28.40	Key into HISTORIZATION GROUP	<u>EG</u> : 1
28.50	Key into ENTITY.PARAMETER table Suggestion: Move cursor to the port; then, press <CLR-ENT>.	<u>EG</u> : HG1001.PV
28.60	When all applicable entries to the ENTITY.PARAM table are complete, press <ENTER>	Values and ports turn cyan
28.70	Press <COMND>	COMMAND DISPLAY
28.80	Select [WRITE TO IDF]	Pathname ports and target appear.
28.90	Key in a REFERENCE PATHNAME	<u>EG</u> : NET>H MV1>
28.100	Key in an IDF Pathname Press <ENTER>	<u>EG</u> : H MHIS Library is saved in IDF. PED display.
28.105	Press <COMND>	COMMAND DISPLAY
28.110	Select [LOAD]	
28.115	Press <ENTER>	PARAMETER ENTRY DISPLAY

8.6 Task 29 – Build, Compile and Load CL Programs

Keyboard: IKB or PC

Lock: ENGR

Reference: HM12-500 and SW11-511

Personality Loaded: Universal

Task 29 BUILD, COMPILE AND LOAD CL PROGRAMS

Step	Action	Response
If you are not using CL, skip to Task 30.		
CREATE CL SOURCE FILE		
29.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
29.20	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
29.30	Key in an EDIT command	<u>EG</u> : EDIT NET>CL>VLR_CALC.CL
29.40	Press <ENTER>	Message: "OPEN FILE - FILE NOT FOUND"
29.50	Key in CL source file statements	<u>EG</u> : See Figure 12. This example is similar to Fig. 3-4 in the <i>CL/AM Overview Manual</i> . The point named in the BLOCK header must be a point built in task 21 (for example, VLRATIO). All points referenced in the statement must exist (for example, HG1001, HG1002). CLSLOTS must equal 1 or be greater than 1. In this example, PVALGID = CL.
29.60	Move cursor to 1st letter on line 1: Hold down <CTL>, then press <3>, Hold down <CTL>, then press <1>	JUMP MENU Cursor moves to start of file.
29.70	Update and exit: Hold down <CTL>, then press <1>, Hold down <CTL>, then press <2>	QUIT MENU Source file is transferred from temporary file to user volume. Message: "FILE WAS UPDATED xx Lines...."
COMPILE THE CL/AM BLOCK		
29.90	Key in a Compile command	<u>EG</u> : CL VLR_CALC
29.100	Press <ENTER> If errors were detected, examine the .LE file. A caret (^) marks the error(s). Make corrections, then recompile the file.	Elapsed time for Pass 1, Pass 2, Pass 3, time for compilation = nn.n sec. "Errors detected:None" <u>EG</u> : PRINT NET>CL>VLR_CALC.LE
LINK (AND LOAD) THE CL BLOCK		
29.110	Key in a Link Command	<u>EG</u> : LK VLR_CALC VLRATIO
29.120	Press <ENTER>	Elapsed time for linker = nn.n seconds. "Errors Detected None"

```

BLOCK VLR_CALC  (POINT VLRATIO; AT PV_ALG)
--
-- CALCULATE INTERNAL VAPOR/LIQUID RATIO FOR COLUMN NO. 2.
--
EXTERNAL  HG1001,  HG1002
LOCAL WL, WV, RATIO
LOCAL CP = 0.806    --LIQUID HEAT CAPACITY
LOCAL LAMBDA = 653.0    --HEAT OF VAPORIZATION
--
-- CALCULATE INTERNAL LIQUID FLOW RATE FROM HEAT AND MATERIAL BALANCE
SET WL = ((HG1001.PV - HG1002.PV)
&        * CP/LAMBDA)
--
-- CALCULATE INTERNAL VAPOR FLOW RATE FROM MATERIAL BALANCE
SET WV = HG1001.PV + WL
--
-- CALCULATE VAPOR/LIQUID RATIO AND STORE AS PV
SET RATIO = WV/WL
--

IF BADVAL (RATIO) THEN (CALL SET_BAD (PVCALC);
&                      SET PVAUTOST = BAD)
&                      ELSE (SET PVCALC = RATIO;
&                             SET PVAUTOST = NORMAL)
--
END VLR_CALC

```

Figure 12 CL Source File Example

Section 9 – Procedure 6/Configure Area Database

Figure 13 shows a graphic representation of the Task (30) that must be performed to Build and Install the Area Database.

Task 30:

- 1 Build the area data base

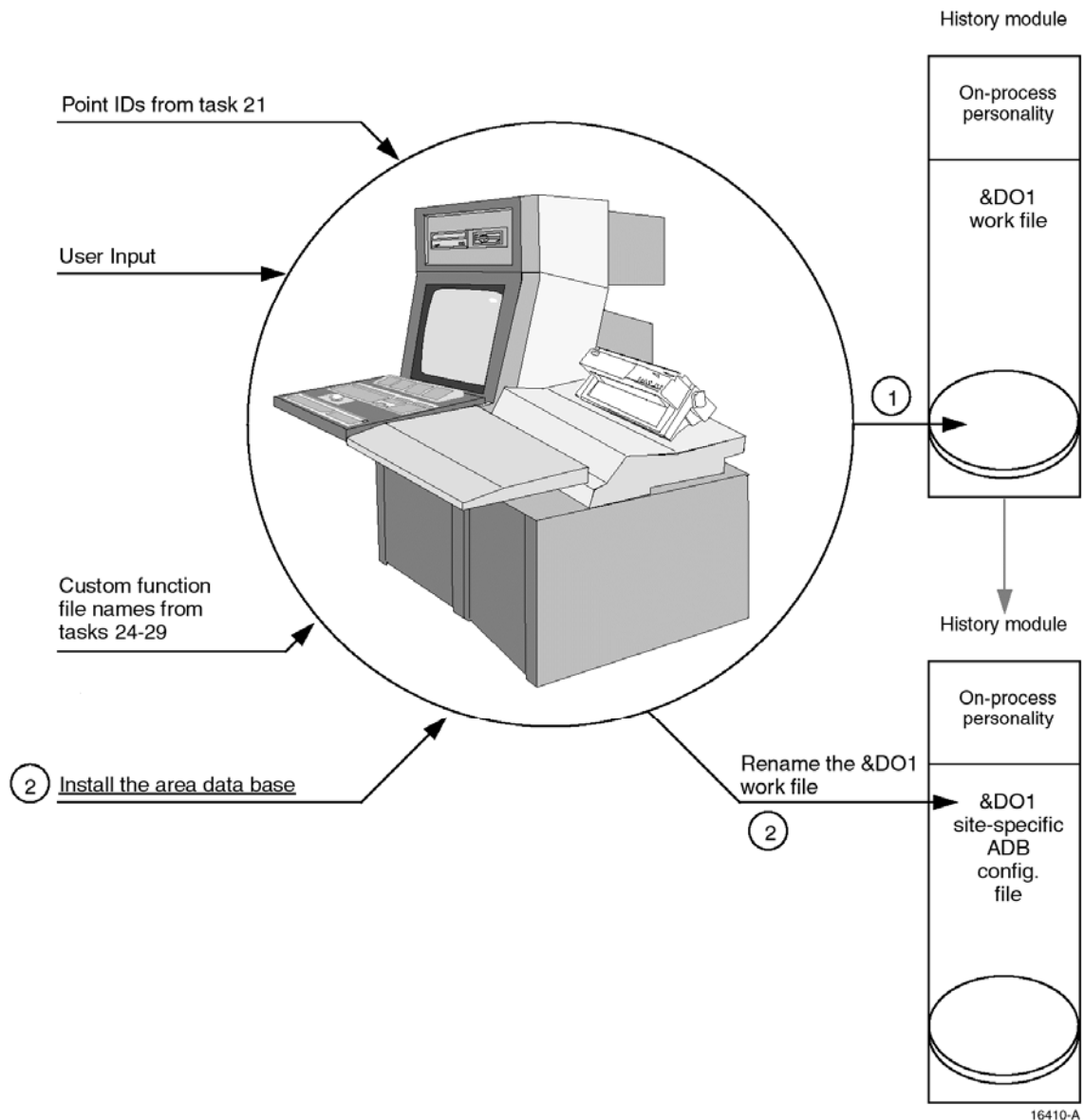


Figure 13 Build and Install the Area Database

9.1 Task 30 – Build, Select, and Install Area Database

Keyboard: IKB or PC

Lock: ENGR

Reference: SW11-511

Personality Loaded: Universal

NOTE

Points to be referenced in the Area Database must already be configured in Task 21. Custom functions to be referenced in the Area Database must already be configured in Tasks 24-29.

In the steps that follow, the Area Database entity IDs required for reconstituting, loading, or reading a given part of the Area Database is listed just to the right of the bold heading for each part.

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
COPY BLANK AREA DATA BASE		
30.06	Key in an Unprotect command to the Area Data Base; then press <ENTER>.	<u>EG</u> : UNPT NET>&D01>AREA01.DA "FILE UNPROTECT COMPLETE"
UNIT ASSIGN \$RUNTSTS		
30.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30.20	Select [AREA DATA BASE]	AREA CONFIGURATION Menu
30.30	Select [UNIT ASSIGNMENT]	UNIT ASSIGNMENT DISPLAY CONFIG.
30.40	Select Unit ports that you want to be in the area.	<u>EG</u> : 01 Selected port(s) turn green
30.45	Press <ENTER>	UNIT ID targets appear
30.47	Enter Unit ID(s)	<u>EG</u> : 01
30.50	Select INITIALLY ASSIGNED? option	Select [YES] for each unit initially assigned to this area.
30.70	Press <ENTER>	PED reappears with values in cyan
SET A PATHNAME AND WRITE TO IDF		
30.80	Press <COMND>	COMMAND DISPLAY
30.90	Select [WRITE TO IDF]	Pathname ports and target appear
30.100	Key into Reference Path Name port	<u>EG</u> : NET>HVM1>
30.110	Key into pathname for IDF port	<u>EG</u> : BOILER1
30.120	Press <ENTER>	Unit Assignment data written to IDF
GROUP DISPLAY \$OGROUP(n) where n can be 1 to 400		
30.130	If UNIT ASSIGN display is on screen, hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.140	Select [GROUP]	GROUP DISP. CONFIGURATION display
30.150	Key into GROUP NUMBER port	<u>EG</u> : 1
30.160	Key into GROUP TITLE port	<u>EG</u> : WEST GROUP
30.170	Key into the POINT IDS port	<u>EG</u> : HG1001
30.180	Select IN THE TREND SET? option	<u>EG</u> : [YES]
30.190	Repeat 30.170 – 30.180 for each other point in this group.	
30.200	Select a TREND TIMEBASE option	<u>EG</u> : The default option
30.201	Press <ENTER>	PED reappears with values in cyan

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.203	<PAGE FWD> to the Associated and Help display definitions. Key into ports as desired.	
30.210	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
30.220	Press <COMND>	COMMAND DISPLAY with [WRITE TO IDF] selected
30.230	Press <ENTER>	GROUP DISPLAY CONFIG data to IDF
30.240	To configure more groups, hold <CTL> down and press <PAGE BACK>. Go repeat 30.140 – 30.230.	
UNIT TREND DISPLAY \$OUNTRND(n) where n can be 1 to 36		
30.250	If GROUP DISP CONFIG is on screen hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.255	Select [UNIT TREND]	UNIT TREND DISPLAY CONFIG display
30.260	Key into TREND DISP NUMBER port	<u>EG:</u> 1
30.265	Key into TREND TITLE port	<u>EG:</u> UNIT 1 TRENDS
30.270	Select TREND TIMEBASE option	<u>EG:</u> [2 HOURS]
30.275	Key in names of point(s) to be trended in cyan and/or yellow.	<u>EG:</u> HG1001 (CYAN)
30.280	Press <ENTER>	PED reappears with values in cyan
30.285	Press <PAGE FWD>	RELATED DISPLAY 001 DEF display
30.290	Key into SCREEN NUMBER port(s) and DISPLAY TYPE port(s)	<u>EG:</u> 1 <u>EG:</u> ALARMSUM
30.295	Press <ENTER>	PED reappears with values in cyan or SCHEMATIC NAME port appears if DISPLAY TYPE entered was SCHEM
30.300	Key into SCHEM. NAME port (if any)	<u>EG:</u> DISP1
30.305	Press <ENTER>	PED reappears with values in cyan
30.310	<PAGE FWD>	Next page appears
30.315	Repeat 30.290 – 30.310 to configure remaining Related Displays, and on Page 6, Assoc. Display and Help Display	Related, Associated, and Help displays configured

WRITE TO IDF WHEN PATHNAME IS ALREADY SET

30.320	Press <COMND>	COMMAND DISPLAY with correct pathnames entered.
30.325	Press <ENTER>	UNIT TREND CONFIG. data to IDF
30.330	Repeat 30.260 – 30.330 to configure other Unit Trend displays. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Unit Trend displays configured and stored in IDF

SYSTEM STATUS DISPLAY \$OSYSTAT

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.350	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.360	Select [SYSTEM STATUS]	SYSTEM STATUS DISPLAY CONFIG display
30.364	If the screen is blank, press <PAGE FWD>	Next page appears
30.370	Key into Associated Display ports: SCREEN NO. port DISPLAY TYPE port Press <ENTER> CONSOLE NUMBER	<u>EG</u> : 0 <u>EG</u> : CONSTAT CONSOLE # port appears <u>EG</u> : 1
30.380	Key into HELP DISPLAY DEF ports: SCREEN NO. port DISPLAY TYPE port	<u>EG</u> : 1 <u>EG</u> : SYSSTAT
30.390	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
30.400	If this is not the first write to IDF, press <COMND> and go to 30.410. If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY with [WRITE TO IDF] selected. Example IDF is BOILER1
30.410	Press <ENTER>	SYS STAT DISP CONFIG-data to IDF
FREE FORMAT LOGS \$OFFLOG(n) where n can be 1 to 100		
30.420	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.430	Select [FREE FORMAT LOG]	FREE FORMAT LOG CONFIG. display
30.440	Key into LOG NUMBER	<u>EG</u> : 1
30.450	Key into LOG NAME	<u>EG</u> : FFL001
30.470	Key into LOG DESCRIPTION	<u>EG</u> : WEST AREA LOG
30.480	Accept default for INVOC. TYPE	<u>EG</u> : DEMAND
30.490	Accept default for PRINTER ID	<u>EG</u> : 1
30.500	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
30.510	If this is not the first write to IDF, press <COMND> and go to 30.515 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with path name set)
30.515	Press <ENTER>	FFL CONFIG DATA to IDF
30.520	Repeat 30.440 – 30.515 for other Free Format Logs	FFL CONFIG DATA to IDF
HIWAY BOX ANNUN POLICY \$DMONBLK		
30.610	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.620	Select [PROC NTKW ANNUN POLICY]	PROCESS NETWORK DEV. ANNC. CONFIG display
30.630	Key into PROCESS NTKW NBR port	<u>EG</u> : 2
30.640	Select ADDRESS ANNUNC. options	<u>EG</u> : [YES] for all addresses
30.650	Press <ENTER>	PED reappears with values in cyan
30.660	Check PED set status (<CTL> <8>)	Display Set Entered?-Yes If errors, see: 30.1060 – 30.1080
30.670	If this is not the first write to IDF, press <COMND> and go to 30.680. If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.680	Press <ENTER>	HIWAY ANN PLCY CONFIG-data to IDF
PATHNAME CATALOG \$OABSTR		
30.690	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.700	Select [PATHNAME CATALOG]	PATHNAME CATALOG CONFIG. display
30.710	Key into BUTN CONF VOLUME ID port	<u>EG</u> : &D01
30.720	Key into BUTN CONF FILE NAME port	<u>EG</u> : BUTTON
30.724	Key into SCHEMATIC or FFL PATHNAMES—most critical pathname in top left-hand port, then left to right, then second line left to right, etc.	<u>EG</u> : NET>HNV1 (your user volume), NET>&DSY, NET>&D01, NET>DIA1, etc.
30.750	Key into MEM. RES SCHEMATIC OR FFL FILE NAME port and select [FFL] if you entered a FFL filename.	<u>EG</u> : FFL001 [FFL], DISP1, DISP2, DISP3, etc
30.760	Press <ENTER>	PED reappears with values in cyan
30.764	Use <PAGE FWD> to add more schematic/FFL names	All schematic/FFL file names entered
WRITE TO IDF		
30.770	If this is not the first write to IDF, press <COMND> and go to 30.780 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.780	Press <ENTER>	PATHNAME CAT CONFIG-DATA to IDF
PROCESS MODULE GROUP DISPLAY \$OMODGRP(n) where n can be 1-50		
NOTE If a MC process module point was not configured in Task 21, go to 30.890.		
30.790	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.795	Select [PROCESS MODULE GROUP]	MODULE GROUP DISP CONFIG display
30.800	Key into GROUP NUMBER port	<u>EG</u> : 1
30.810	Key into GROUP TITLE	<u>EG</u> : MIX AND SOAK GROUP

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.815	Key into PROCESS MODULE ID ports	<u>EG</u> : SCALE1
30.820	Press <ENTER>	PED reappears with values in cyan
30.825	Press <PAGE FWD>	RELATED DISPLAY 001 DEF display
30.830	Key into Screen Number port(s) and Display Type port(s)	<u>EG</u> : 1 <u>EG</u> : SCHEM
30.835	Press <ENTER>	PED reappears with values in cyan and additional port(s) displayed.
30.840	Key into added port(s)	<u>EG</u> : DISP1
30.845	Press <ENTER>	PED reappears with values in cyan
30.850	<PAGE FWD> Next page appears	
30.855	Repeat 30.830 – 30.850 to configure ASSOCIATED DISPLAY and HELP DISPLAY	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF		
30.865	If this is not the first write to IDF, press <COMND> and go to 30.870 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] (with pathname BOILER1 set)
30.870	Press <ENTER>	PED display
30.880	Repeat 30.800 – 30.870 to configure other Process Module Groups. Use <PAGE FWD> and <PAGE BACK> as necessary.	All Process Module Groups configured and stored in IDF

MODULE SUMMARY DISPLAY \$OUNITSM(n) where n can be 1 to 36

30.890	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.900	Select [MODULE SUMMARY]	UNIT SUMMARY DISP CONFIG display
30.910	Select an option for each Line	<u>EG</u> : [HEADER] for Line 01, page 1
30.920	Press <ENTER>	<u>EG</u> : TITLE port appears for Line 1
30.930	Key into each Title port	<u>EG</u> : WEST UNIT 1
30.940	Press <ENTER>	PED reappears with values in cyan
30.950	Use <PAGE FWD> to get to more pages to configure and repeat steps 30.910 – 30.940 for each.	UNIT SUMMARY DISP CONFIG display – n
30.960	<PAGE FWD> to RELATED DISP DEF	RELATED DISP DEFINITION display – 1
30.970	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports	<u>EG</u> : 0 <u>EG</u> : ALARMSUM
30.980	Press <ENTER>	PED reappears with values in cyan
30.990	If desired, <PAGE FWD> to next page and repeat steps 30.970 – 30.980.	RELATED DISP DEFINITION display – n
30.1000	<PAGE FWD> to ASSOCIATED DISPLAY	ASSOCIATED & HELP DISPLAY DEFINITION DEFINITION
30.1010	Key into each pair of SCREEN NUMBER and DISPLAY TYPE ports.	<u>EG</u> : 0 <u>EG</u> : UNITALRM

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.1020	Press <ENTER>	Unit ID port appears.
30.1030	Key into Unit ID port	<u>EG</u> : 01
30.1040	Press <ENTER>	PED reappears with values in cyan

CHECK PED SET STATUS

30.1050	Hold <CTL> down and press <8>	PED Set Stat display
30.1060	If "Display Set Entered?" is Yes, and no missing data or errors are indicated, go to 30.1090; else 30.1070.	If "No," check "Missing Data?" and "Errors?"
30.1070	Select a Display Title option, correct data and press <ENTER>.	Applicable PED appears.
30.1080	Repeat step 30.1060.	

WRITE TO IDF

30.1090	If this is not the first write to IDF, press <COMND> and go to 30.1100 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1100	Press <ENTER>	MOD SUM DISP CONFIG data to IDF

ANNUNCIATOR DISPLAY \$ANNDATA

30.1110	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1115	Select <ANNUNCIATOR>	ANNUNCIATOR DISPLAY CONFIG. display
30.1120	Key into BOX NUMBER port	<u>EG</u> : 1
30.1125	Key into BOX TITLE port Note: Annunciator display boxes have two lines of four characters each	<u>EG</u> : AIR FLOW
30.1128	Select a Point Usage	<u>EG</u> : [ENTITY ID]
30.1130	Key into NUMBER OF POINTS port	<u>EG</u> : 1
30.1131	Press <ENTER>	
30.1132	Key into POINT NAMES ports	<u>EG</u> : HG1001
30.1135	Key into SCREEN NUMBER port	<u>EG</u> : 1
30.1140	Key into DISPLAY TYPE	<u>EG</u> : SCHEM
30.1145	Press <ENTER>	Additional port(s) appear
30.1155	Key into added port(s)	<u>EG</u> : DISP1
30.1160	Press <ENTER>	PED reappears with values in cyan

WRITE TO IDF

30.1165	If this is not the first write to IDF, press <COMND> and go to 30.1170 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1170	Press <ENTER>	ANNUNC. CONFIG DATA to IDF

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.1175	Repeat 30.1110 – 30.1170 for other boxes on Annunciator	All Annunciator Box configuration data in IDF
PROCESS JOURNALS \$OPRCJOR(n)		
30.1200	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1202	Select [PROCESS JOURNAL]	PROCESS JOURNAL CONFIG. display
30.1204	Key into JOURNAL NUMBER port	<u>EG</u> : 1
30.1206	Key into JOURNAL NAME port	<u>EG</u> : PRJNL001
30.1208	Key into JOURNAL TITLE port	<u>EG</u> : WEST AREA PROCESS ALARMS
30.1212	Select JOURNAL TYPE	<u>EG</u> : [ALARM]
30.1214	Select INVOCATION TYPE	<u>EG</u> : [PERIODIC]
30.1216	Select INITIALLY ACTIVE option	<u>EG</u> : [NO]
30.1220	Select DAYS OF WEEK option	<u>EG</u> : ALL (default)
30.1222	Key into START TIME port	<u>EG</u> : 0:10
30.1224	Key into PERIOD port	<u>EG</u> : 12:0
30.1226	Key into PRINTER ID port	<u>EG</u> : 1
30.1228	Press <ENTER>	PED reappears with values in cyan
30.1230	Press <PAGE FWD>	Next page of PED appears
30.1232	Key into TIME SPAN port	<u>EG</u> : 12
30.1234	Key into NUMBER OF UNITS port	<u>EG</u> : 1
30.1236	Press <ENTER>	PED reappears with Unit ID Box(s)
30.1240	Key into (Unit ID) box	<u>EG</u> : 01
30.1243	Press <ENTER>	PED reappears with values in cyan
30.1246	If this is not the first write to IDF, press <COMND> and go to 30.1248 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1248	Press <ENTER>	PROC. JOURNAL CONFIG data to IDF
30.1250	Repeat 30.1204 – 30.1248 to configure remaining journals.	
SYSTEM JOURNALS \$OSYSJOR		
30.1260	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1300	Select [SYSTEM JOURNAL]	SYSTEM JOURNAL CONFIG. display
30.1310	Key into JOURNAL NUMBER port	<u>EG</u> : 1
30.1320	Key into JOURNAL NAME port	<u>EG</u> : SYJNL001
30.1330	Key into JOURNAL TITLE port	<u>EG</u> : SYSTEM JOURNAL NO. 1
30.1340	Select JOURNAL TYPE	<u>EG</u> : [SYSTAT]
30.1350	Select INVOCATION TYPE	<u>EG</u> : [DEMAND]

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
30.1360	Key into PRINTER ID port	EG: 1
30.1370	Key into TIME SPAN port	EG: 24
30.1390	Press <ENTER>	PED reappears with values in cyan
30.1400	If this is not the first write to IDF, press <COMND> and go to 30.1410 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1410	Press <ENTER>	SYS. JOURNAL CONFIG-data to IDF
30.1420	Repeat 30.1310 – 30.1410 to configure remaining journals.	

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
REAL-TIME JOURNALS \$RTJNL		
30.1430	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1440	Select [R-T JOURNAL ASSIGNMENT]	R-T JOURNAL ASSIGNMENTS DISPLAY CONFIGURATION
30.1450	Key into PRINTER ID ports	EG: 1 (into all ports)
30.1460	Select JRN TYPE-INIT ACT? options	EG: [No] for SYS STATUS CHANGES
30.1470	Press <ENTER>	PED reappears with values in cyan
30.1490	If this is not the first write to IDF, press <COMND> and go to 30.1500 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1500	Press <ENTER>	REAL-TIME JRNL config-data to IDF
OVERVIEW DISPLAY \$OOVERVW		
30.1510	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1520	Select [OVERVIEW]	AREA OVERVIEW DISP CONFIG display
30.1530	Key into GROUP NUMBERS ports	EG: 1 (into Index 01)
30.1540	Key into NUMBER OF FIELDS port	EG: 1
30.1550	Press <ENTER>	FIELD DEFINITION ports appear
30.1560	Key into FIELD DEF ports: FLD TITLE TOP LEFT ELEMENT WIDTH HEIGHT	EG: WEST GROUP EG: 1 EG: 1 EG: 1
30.1570	Press <ENTER>	PED reappear with values in cyan

30.1573	<PAGE FWD> to the Related/Assoc. and Help Display definitions. Key into ports as desired.	
30.1578	Press <ENTER>	PED reappears with values in cyan
30.1580	Check PED-set status: Hold <CTL> down and press <8>	"Display Set Entered?"-Yes If errors, see: 30.1060 – 30.1080
30.1590	If this is not the first write to IDF, press <COMND> and go to 30.1600 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.1600	Press <ENTER>	OVERVIEW DISP CONFIG data to IDF

Task 30 BUILD, SELECT, AND INSTALL AREA DATABASE

Step	Action	Response
AREA TREND DISPLAY \$OTREND		
30.1610	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1615	Select [AREA TREND]	AREA TREND OV. DISP CONFIG display
30.1620	Key into TREND TITLE port	<u>EG</u> : WEST AREA TRENDS
30.1625	Select TREND TIMEBASE option	<u>EG</u> : [8 HOURS]
30.1630	Key in names of points to be trended in cyan and yellow.	<u>EG</u> : HG1001 (cyan)
30.1635	Press <ENTER>	PED reappears with values in cyan
30.1640	Press <PAGE FWD>	RELATED DISP 001 DEF display
30.1645	Key into each SCREEN NUMBER port and each Display Type port	<u>EG</u> : 1 <u>EG</u> : SCHEM
30.1650	Press <ENTER>	PED reappears with values in cyan and SCHEMATIC NAME appears (or other port depending on the DISPLAY TYPE entered).
30.1655	Key into each additional name port	<u>EG</u> : DISP1
30.1660	Press <ENTER>	PED reappears with values in cyan
30.1665	Press <PAGE FWD>	Next page appears
30.1670	Repeat 30.1650 – 30.1665 to configure remaining Related Displays, and on page 7, Associated and Help display	RELATED, ASSOCIATED, and HELP displays configured
WRITE TO IDF WHEN PATHNAME IS ALREADY SET		
30.1675	Press <COMND>	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathnames set)
30.1680	Press <ENTER>	AREA TREND CONFIG. data to IDF
STANDARD LOGS \$OLOG(n) Where n can be 1 to 100		
30.1900	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.1905	Select [STANDARD LOG]	STANDARD LOG CONFIGURATION display
30.1910	Key into LOG NUMBER port	<u>EG</u> : 1
30.1920	Key into LOG NAME port	<u>EG</u> : BLR1RPT
30.1930	Key into LOG DESCRIPTION port	<u>EG</u> : DAILY BOILER LOG
30.1940	Enter other data as required.	<u>EG</u> : Use default values.
30.1950	Key into NUM VALUES PER POINT port	<u>EG</u> : 1
30.1960	Key into NUMBER OF POINTS port	<u>EG</u> : 1
30.1965	Press <ENTER>; press <PAGE FWD>	INDEX ports appear.
30.1970	Key into INDEX ports until all Point IDs have been keyed in.	<u>EG</u> : HG1001 (for Index 001) (from step 21.30)
30.1980	Press <ENTER>	PED reappears with values in cyan

WRITE TO IDF		
30.2000	If this is not the first write to IDF, press <COMND> and go to 30.2005 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.2005	Press <ENTER>	Standard Log config data to IDF

PRINTED TREND \$OPTRND(n) where n can be 1 to 100		
30.2010	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.2012	Select [PRINTED TREND]	PRINTED TREND DISP CONFIG display
30.2014	Key into TREND NUMBER port	<u>EG</u> : 1
30.2016	Key into TREND NAME port	<u>EG</u> : BOILER 1
30.2018	Key into TREND TITLE port	<u>EG</u> : BOILER 1 TREND RECORD
30.2020	Select INVOCATION Type	<u>EG</u> : [PERIODIC]
30.2024	Select INITIALLY ACTIVE option	<u>EG</u> : [NO]
30.2028	Select START TIME port	<u>EG</u> : [0]:[10]
30.2032	Key into PERIOD port	<u>EG</u> : 1:30
30.2036	Key into PRINTER ID port	<u>EG</u> : 1
30.2040	Key into TIME SPAN port	<u>EG</u> : 2
30.2044	Key into GROUP NUMBER port	<u>EG</u> : 1
30.2050	Press <ENTER>	PED reappears with values in cyan

WRITE TO IDF		
30.2070	If this is not the first write to IDF, press <COMND> and go to 30.2080 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.2080	Press <ENTER>	PRINTED TREND CONFIG. data to IDF
30.2085	Repeat 30.2014 – 30.2080 for other Printed Trends.	All printed trends configured and stored in IDF

LCN NODE ANNUNCIATION POLICY \$ANPOLC		
30.2090	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.2094	Select [LCN NODE ANNUNC. POLICY]	LCN NODE ANNUNC. POLICY display
30.2102	Use <PAGE FWD> to get to all node numbers you wish annunciated. Select [ADDRESS ANNUNCIATED] option.	<u>EG</u> : 01, 02, 24, 25, 30, 38, 39, 40, 43, and 46 <u>EG</u> : [YES] for each of above nodes
30.2106	Press <ENTER>	PED reappears with values in cyan
30.2108	If this is not the first write to IDF, press <COMND> and go to 30.2112 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.2112	Press <ENTER>	LCN NODE ANNUN config data to IDF

SEQUENCE OF EVENTS JOURNAL \$SOEJOR(n)		
30.2120	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.2124	Select [SOE JOURNAL ASSIGNMENT]	SEQUENCE OF EVENTS JOURNAL display
30.2128	Key into JOURNAL NUMBER	<u>EG</u> : 1
30.2130	Key into JOURNAL NAME	<u>EG</u> : SOEJNL01
30.2134	Key into JOURNAL TITLE	<u>EG</u> : BOILER1 SEQ OF EVENTS
30.2138	Key into HM NODE NUMBER	<u>EG</u> : 43
30.2140	Press <ENTER>	PED reappears with values in cyan
30.2144	If this is not the first write to IDF, press <COMND> and go to 30.2148 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.2148	Press <ENTER>	SOE JOURNAL config data to IDF
REPORT DISPLAY \$OREPORT(n) where n can be 1 to 100		
30.2150	If PED display is still on the screen Hold <CTL> down and press <9>. If Engineering Main Menu is on the screen, select [AREA DATA BASE]	AREA CONFIGURATION Menu AREA CONFIGURATION Menu
30.2154	Select [REPORT]	REPORT DISPLAY CONFIGURATION display
30.2156	Key into REPORT NUMBER port	<u>EG</u> : 1
30.2160	Key into REPORT NAME port	<u>EG</u> : WESTRPT
30.2164	Key into REPORT TITLE port	<u>EG</u> : WEST AREA REPORT
30.2170	Select INVOCATION TYPE option	<u>EG</u> : [DEMAND]
30.2175	Key into NO. ITEMS IN REPORT port	<u>EG</u> : 1
30.2180	Press <ENTER>	ITEMS IN REPORT port appears.
30.2185	Key into Items In Report port Note: Items must be previously built as trends or logs.	<u>EG</u> : BLR1RPT (configured in 29.1920)
30.2190	Press <ENTER>	PED reappears with values in cyan
WRITE TO IDF		
30.2194	If this is not the first write to IDF, press <COMND> and go to 30.2198 If first write to IDF, go perform 30.80 – 30.110 and return here.	COMMAND DISPLAY, [WRITE TO IDF] selected; (with pathname set)
30.2198	Press <ENTER>	REPORT CONFIGURATION data to IDF
NOTE		
Nodes associated with the area being multiple loaded must have a status of OK before the load multiple can be successfully performed.		
LOAD MULTIPLE ENTITIES FROM IDF		
SELECT AN AREA		
30.2205	Press <COMND>	COMMAND DISPLAY
30.2208	Select [SELECT AREA]	Area Ports/Target appear

30.2212	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HMV1>
30.2216	Select an area Current AREA DATA BASE pathname port is filled in automatically	<u>EG</u> : [1] <u>EG</u> : NET>&D01>AREA01
30.2218	Press <ENTER>	"OPERATION COMPLETE"
30.2220	Select [LOAD MULTIPLE]	Additional Ports/Target appear
INSTALL THE AREA DATABASE		
NOTE The Area Database is installed to a volume and file with the volume name of &Dnn and with the file name of AREAnn. For &Dnn, the directory names of &D01 through &D10 were automatically created during the HM Volume Configuration task, Task 10. For AREAnn, nn is the index number from the Area Names configuration list that you configured in Task 7. For example, the first area name you entered into the list is automatically accessed by AREA01, the second name by AREA02, and so on.		
30.2230	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HMV1>
30.2240	Key into pathname for IDF port	<u>EG</u> : BOILER1
30.2250	Key into current AREA DATA BASE pathname port	<u>EG</u> : NET>&D01>AREA01
30.2260	Press <ENTER> Wait—Entities are listed twice	"PROCESSING ENTITIES..." "OPERATION COMPLETE"
30.2270	Press <COMND>	COMMAND DISPLAY
30.2280	Select [INSTALL AREA]	Additional targets appear.
30.2290	Key a (destination) pathname into Current AREA DATA BASE p/nm port.	<u>EG</u> : NET>&Dnn>AREAnn (where nn is equal to 01–10 in both cases)
30.2295	Key a (source pathname) into the Pathname for WORKFILE port.	<u>EG</u> : NET>&D01>AREA01
30.2300	Press <ENTER>	"OPERATION COMPLETE"
NOTE - Most of the Command Display functions deal with IDFs; however, Exception Build files are more portable. If you want to create Exception Build files from your IDFs, follow the instructions in Appendix E. - After the Area Database is completely built, the IDFs and Area Database files can be copied to emulated disk and then deleted from the User Volume NET>HMV1>Boiler1. - For security, you should copy the following onto disk: Area Database (&D01), Pictures, Free Format Logs, Button source files (.DS and .KS files), and custom files (CUS/CLX, and so on.). - Go perform the EC in Task 31, steps 31A.10 - 31A.50 to save all the &ASY files. Use the resulting disk as the new backup if you have to reload from disk or to perform On-Line Network Reconfiguration.		

Section 10 – Procedure 7/Load Area Database

10.1 Task 31 – Change Area Database on GUS Stations

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task 31 CHANGE AREA DATABASE ON GUS STATIONS

Step	Action	Response
31.150	To change the Area Database on a GUS station: From the <CONSOLE STATUS> display; select the [AREA CHANGE] target.	
31.151	Select the target for AREAnn	Selected Area Database targets appear on the display.
31.152	Select [DEFAULT SOURCE]	
31.153	Select [EXECUTE COMMAND]	Message "Busy US:xx, Area Change in Progress" appears.

Section 11 – Procedure 8/Modify the Database

11.1 Task 32 – Change Network Configuration

Keyboard: IKB or PC

Lock: ENGR

Reference: SW11-605

Personality Loaded: Universal

Task 32 CHANGE NETWORK CONFIGURATION

Step	Action	Response
MAKE A BACKUP COPY OF &ASY		
32.10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
32.30	Select [COMMAND PROCESSOR]	COMMAND PROCESSOR display
32.35	Place DISK_&Z1.LCN into \$F1	
32.36	Create a new emulated disk named ASY_BKUP.LCN and mount it in \$F2.	
32.40	Key in an EC command	<u>EG</u> : EC \$F1>&EC>ASY_BKUP.EC \$F2 \$P1 (printer \$P1 is optional)
32.45	Press <ENTER>	Message: "THIS EC WILL...." "....." "..... CONTINUE (Yes/No)?"
32.50	Key in YES , then press <ENTER>	Message: "THIS EC... IS THIS CORRECT?"
32.55	Key in YES , then press <ENTER>	Message: "MOUNT BACKUP MEDIA...."
32.60	Press <ENTER> (the EC prepares to update the Backup Log).	"UNPT NET>&ASY>BKUP_LOG.XX...." "DO YOU WANT TO RE-INITIALIZE...."
32.65	Key in YES , then press <ENTER> (the EC formats the emulated disk in \$F2, and begins copying files to it).	"CR \$F2>&ASY> -F -MF 500" "COPYING" "EC COMPLETE"
32.68	Dismount the emulated disk from \$F2.	
32.70	Make backup copies of other files that you created (for example, mount the AREA_DB.LCN emulated disk in \$F2 and copy the area files, button files, and custom software files).	<u>EG</u> : CP NET>&D01>*. * \$F2>&D01>= -D CP NET>&CUS>*. * \$F2>&CUS>= -D CP NET>&CLX>*. * \$F2>&CLX>= -D
NOTE		
Refer to the <i>Network Data Entry</i> manual (see References) for information on network reconfiguration.		
NOTE		
The GUS node you used for system startup may still be running the Universal Personality. You may wish to shut down the node, reset the LCNP4 card, and reload with the GUS personality.		

Section 12 – Error Recovery Procedures

12.1 Task 33 – Error Conditions, Reasons, and Recovery

Task ERROR CONDITIONS, REASONS, AND RECOVERY

Task #	Error Conditions	Reason and Recovery Procedures
Many	GUS Node fails with an Internal File Manager error. HG Fails	1. Copy to existing, protected file. 2. Installing to a protected NCF.CF. 3. HG checkpoint to a protected file. 4. Picture Editor/Button Configurator read or write to protected file. 5. Using DEB commands to write to a protected IDF or Entity List File. Manually reset the HG and then repeat Task 18.
2	GUS Personality will not load.	One of the nodes is active. Check that <u>all</u> on the NET are in the RESET state. Then repeat Task 2. You may have to press RESET on the HM an extra time because it may have autobooted.
12	1. A change to &ASY was made to a copy of &ASY disk is not the same as the time-stamp on the HM. 2. The changed &ASY on the HM <u>was not</u> copied to a cartridge. 3. A load of the GUS personality is being attempted, using emulated disks that include the CNCF.LCN emulated disk that existed before changing the &ASY on HM. 4. The CNCF.LCN emulated disk will not read in and the Station dies.	The time-stamp on the CNCF emulated for &ASY in one of the nodes. To recover, do the following: 1. Load the GUS Personality, using "W" and answering the NCF? Prompt with N (NET) for HM as the source of the NCF 2. If recovery step 1 works, copy the &ASY from HM to the CNCF.LCN you were trying to load from. 3. Try the new CNCF.LCN emulated disk to load the GUS Personality from emulated disks.
15	Got a message of "HM INIT DID RESPOND FOR 1 MINUTE(REVIEW)"	Power to the Winchester disks is off, even though power to the HM node is ON. Turn the power to Winchester disks to the "ON" position. Repeat Tasks 13 and 14.

Task #	Error Conditions	Reason and Recovery Procedures
18	On-Process Personality is not successfully loaded into a module or gateway.	Select the red failure message at the top of the display, to call up the module or gateway status display. On that display, select the node number for the node you are trying to load. This causes an error message to appear. If the message is TARGET NODE FAILED DURING LOAD, check the module or gateway to see if power is on, all boards have a green light on, and the node number appears in the red digits on the front of the unit. If, instead, the red digits show -189 or -190, reset the module or gateway and try to load it again. If another error code appears, call a hardware technician. If you were attempting to load a node or gateway using an HM as the program and data source, and the error message is FILES NOT FOUND, check the HM's status as described in the preceding paragraph. If it has failed, turn its power (power to both the module and disk drive) off, wait five seconds, and turn power back on. In about two minutes, its status on the module status display should become OK. If so, reset the node you were trying to load and try again. If you get the same error message again: c) Use the GUS Personality's Command Processor functions to make sure that the volumes mentioned in Task 18 are all on the source HM. d) Try to load the node using emulated disks. If this works, you may have to delete and recopy all the volumes on the HM. Get a hardware technician's assistance.
NOTE If you do not find a description of your problem in these error recovery procedures, see the <i>TPS Messages Directory</i> (see References).		

Appendix A – Load GUS Personality on Another GUS

A.1 Task A – Load the GUS Personality into Another Global User Station

Keyboard: IKB or PC

Lock: ENGR

Reference: This Document

Personality Loaded: Universal

Task A LOAD THE GUS PERSONALITY INTO ANOTHER GUS

Step	Action	Response
20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30	Select [CONSOLE STATUS]	Console Status display Other US Status seen on This US PWR_ON or QUALIF
40	Select the number of the other Universal Station.	<u>EG</u> : 2 (for STN #2, node no. 2) LOAD or DUMP NODE targets appear
50	Select [LOAD/DUMP]	LOAD/DUMP targets on screen
51	Select [AUTOLOAD.NET].	Press “Enter to Execute” appears.
55	Press <ENTER>	System Status Display appears on GUS when LOAD complete.

Appendix B – Save HG or PLCG Checkpoints

Perform the task described below if you wish to SAVE checkpoint data for an HG or PLCG and its Boxes. You must have created a Checkpoint emulated disk (CHKPOINT.LCN) in Task 4 before performing this task. You should save the HG and Box checkpoint data on a disk first, and then save it on the HM. No HM volume has to be created as the HG checkpoint volume is preconfigured.

B.1 Task B – Save Checkpoint Data for Hiway Gateway and Its Boxes

Keyboard: IKB or PC

Lock: ENGR

Reference: SW11-601 and SW09-605

Personality Loaded: Universal

Task B SAVE CHECKPOINT DATA FOR HIWAY GATEWAY AND ITS BOXES

Step	Action	Response
SAVE HG/BOX CHECKPOINT		
10	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Mount CHKPONT.LCN HG checkpoint emulated disk into \$F1.	<u>EG</u> : Volume ID = CKPT
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select HG Node	Node Box is selected
40	Select [NTWK/HWY STATUS] Hiway/box status must be running	HIWAY STATUS display
44	Select [BOX CMND]	
48	Enter Box Number	<u>EG</u> : 05
52	Press <ENTER>	BOX STATUS DISPLAY
56	Select [ENABLE PROC]	Target changes color
60	Select [EXECUTE COMMAND]	Wait...Box Status: OK
62	Repeat steps 44 - 60 for other boxes	All Box Status = OK
64	Select [SAVE DATA]	[ALL BOXES] target appears.
68	Select [ALL BOXES]	SELECT SAVE BOX PATH display
70	If you wish to save checkpoints on emulated disk, go to step 80. If you wish to save checkpoints on an HM, select [DEFAULT SOURCE] and then go to step 100.	
80	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear
90	Select [1]	<u>EG</u> : Volume ID = CKPT
100	Select [EXECUTE COMMAND]	HIWAY STATUS display
110	Select [START FUNCTION]	Asterisk appears by box number as data is saved. "Busy HWY nn"
120	Wait for status of last applicable box to change to "SAVED."	

Appendix C – Save AM Checkpoint

Perform parts of the task described below to SAVE checkpoint data for an Application Module, A^XM, or a Computing Module. You must have created a checkpoint emulated disk (CHKPOINT.LCN) in Task 4 before performing this task. You should always save the AM checkpoint data on an emulated disk first, and then save it on the HM. No HM volume has to be created as the AM and CM checkpoint volumes are preconfigured.

C.1 Task C – Save Checkpoint Data for AM or CM Modules

Keyboard: ENGR

Lock: ENGR

Reference: SW11-601

Personality Loaded: Universal

Task C SAVE CHECKPOINT DATA FOR AM OR CM MODULES

Step	Action	Response
10	Mount the checkpoint CHKPONT.LCN emulated disk into \$F1.	<u>EG</u> : Volume ID = CKPT
20	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
30	Select [SYSTEM STATUS]	SYSTEM STATUS display
34	Select AM Node	Node Box is selected (Similar for Computing Module)
40	Select [NODE STATUS]	Application Modules display
50	Select NODE number	<u>EG</u> : 40
60	Select [SAVE DATA]	CHKPT DEST FOR NODE NN display
70	If you are saving checkpoint data on an emulated disk, go to step 90.	
80	If you have saved checkpoint data on an emulated disk and now wish to save it on HM, select [DEFAULT SOURCE] and go to step 110.	
90	Select [ALTERNATE SOURCE]	Boxes [1] and [2] appear.
100	Select [1]	
110	Select [EXECUTE COMMAND]	STATUS changes from "OK" to "SAVE" "OK" for the selected node in step 50.
120	Repeat steps 60 – 110 to save checkpoint data to the HM.	

Appendix D – Exception Building

The steps listed describe the basic exception build procedure of cloning a point from a master point that you have previously built using Parameter Entry Displays. Refer to Section 7 in the Data Entity Builder manual for variations.

D.1 Task D – Build Points by Exception for HG, AM, A^XM, CM, PLCG, and PM/APM/HPM Points

Keyboard: Engr.

Lock: ENGR

Reference: SW11-506 and SW11-511

Personality Loaded: Universal

Task D BUILD POINTS BY EXCEPTION FOR HG, AM A^XM, CM, PLCG AND PM/APM/HPM POINTS

Step	Action	Response
CREATE THE EXCEPTION BUILD SOURCE FILE		
05	If the appropriate Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15.	<u>EG</u> : [HIWAY GATEWAY], [APPLICATION MODULE], [NETWORK INTERFACE MODULE], and so on.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HMV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into PATHNAME FOR IDF port	<u>EG</u> : HGPNTS (or AMPNTS, etc.)
70	Key master point ID into PATHNAME FOR SELECTION LIST	<u>EG</u> : HG1001 (or AM101, etc.)
80	Key a file name into PRINT DEVICE ID or DESTINATION PATHNAME	<u>EG</u> : HGEXCBLD.EB (or AMEXCBLD.EB, or PMEXCBLD.EB, and so on.)
100	Press <ENTER>	"BUSY...OPERATION COMPLETE"
110	Press <ESC>	"ESCAPED FROM DEB"
120	Key in an Edit command and the pathname of the source file	<u>EG</u> : ED NET>HMV1>HGEXCBLD.EB or ED NET>HMV1>AMEXCBLD.EB
130	Press <ENTER>	Point parameters displayed
140	Tab* down to &N item; tab over and type new point name over old point name.	<u>EG</u> : &N HG1002
150	Tab to other parameters that you want to change; key in changes.	<u>EG</u> : SLOTNUM = 02 <u>EG</u> : EUDESC = BBLS/HR
160	Press <F3>	JUMP Menu
170	Press <F2>	Jump to End-Of-File
180	Press <RETURN> three times	Inserts three blank lines
190	Key in an end of source file indicator	<u>EG</u> : &E
BUILD ADDITIONAL POINTS		
200	If you want to clone another similar point, copy the parameters as follows.	If no other points need to be cloned, go to step 310.
210	Press <F3>	JUMP Menu
220	Press <F1>	Jump to Beginning-Of-File

Task D BUILD POINTS BY EXCEPTION FOR HG, AM A^xM, CM, PLCG AND PM/APM/HPM POINTS

Step	Action	Response
230	Tab down to &T item	
240	Press <F2>	BLOCK Menu
250	Press <F1>	"MARK AT LINE: N"
260	<PAGE FWD>, <TAB> to end of parameters	<u>EG</u> : All parameters highlighted
*Press <HELP> to display the Text Editor Help functions; <Page FWD> to see more. Press <HELP> to return to the editor.		
270	Press <F2>	"BUSY"... "LINES WRITTEN: NN"
280	Place cursor on blank line just above the &E.	
285	Press <F4>	"LINES READ: NN" (parameters copied)
290	Tab down/over to &N item; Type new point name over old point name.	<u>EG</u> : &N HG1003
300	Tab to other parameters that you want to change; key in changes	<u>EG</u> : SLOTNUM = 03
305	If there are other similar points to clone, repeat Steps 210 – 300.	
WRITE SOURCE TO SOURCE FILE		
310	Press <F1>	QUIT Menu
320	Press <F2>	Message : "FILE UPDATED NN LINES WRITTEN"
PROCESS THE EB SOURCE FILE		
330	Hold <CTL> down and press <HELP>	COMMAND DISPLAY
340	Select [EXCEPTION BUILD]	Ports/targets appear
350	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HMV1>
360	Key into pathname for SOURCE file port	<u>EG</u> : HGEXCBLD.EB
370	Key into pathname for IDF port	<u>EG</u> : HGPNTS
375	Key into pathname for SELECTION LIST port	<u>EG</u> : blank to select all points in Exception Build File
380	Key into OPT path for TEMPLATE port	<u>EG</u> : blank to select all points in Exception Build File
390	Press <ENTER>	"PROCESSING ENTITY: nnnnnn"... "OPERATION COMPLETE"
395	Wait until the COMMAND DISPLAY reappears.	Points built by exception are added to the named IDF.

SAVE/LOAD CLONED POINTS		
460	Select [LOAD MULTIPLE]	Additional ports/target appear
470	Key into REFERENCE PATH NAME port	<u>EG</u> : NET>HMV1>
475	Select [with OVERWRITE]	
480	Key into pathname for IDF port	<u>EG</u> : HGPNTS, AMPNTS, APMPTNS, and so on.
485	Key into pathname for SELECTION LIST port	<u>EG</u> : blank for all points in IDF
487	Press <ENTER>	"WAIT" in yellow; "ENTITY NAMES". Pt. Config data written to IDF. "OPERATION COMPLETE"
495	If errors occur, select the point on the error display. Make corrections; then repeat Steps 460 – 495.	

Appendix E – Create Exception Build Files from IDFs

The steps listed describe the basic procedure to generate exception build files from Intermediate Data Files (IDFs). Refer to Section 7 in the *Data Entity Builder* manual for basic information on the Command Menu functions. This procedure does not destroy the IDFs. Intermediate Data Files have the suffix: .db. Exception Build Files have the suffix .eb.

E.1 Task E – Create Exception Build Files from IDFs

Keyboard: ENGR

Lock: ENGR

Reference: SW11-601

Personality Loaded: Universal

Task E CREATE EXCEPTION BUILD FILES FROM IDFS

Step	Action	Response
05	If any Data Entity Builder Menu function is already selected, go to step 10; otherwise go to step 15	<u>EG</u> : Any ped or the Command Display.
10	Press <COMND>, then go to step 30	COMMAND Display
15	Hold <CTL> down and press <HELP>	ENGINEERING MAIN MENU
20	Select [BUILDER COMMANDS]	COMMAND Display
30	Select [PRINT ENTITIES]	PATHNAMES display
40	Key into REFERENCE PATHNAME port	<u>EG</u> : NET>HMV1>
50	Select [PRINT IDF ENTITIES]	Additional pathnames ports
60	Key into pathname for IDF port	<u>EG</u> : HGPNTS (or AMPNTS, etc.)
65	If you do not remember all the IDF names, this procedure will list them, press <ESC>. Key in a Data Out command to your printer. Press <ENTER>. Key in a List command with a .db suffix, press <ENTER>. Key in another DO command to stop printing. Press <ENTER>. To return to the Command Display, hold <CTL> down and press <HELP>.	<u>EG</u> : DO \$P1 "DATA OUT COMPLETE" <u>EG</u> : LS NET>HMV1>*.db DB files are listed and printed. <u>EG</u> : DO "DATA OUT COMPLETE"
70	Leave Pathname for SELECTION LIST blank.	
80	Key a file name into PRINT device ID or DESTINATION pathname.	<u>EG</u> : HGPNTS.EB (or AMPNTS.EB, or PMPNTS.EB, HPMPNTS, and so on.)
100	Press <ENTER>	"WAIT...PROCESSING ENTITY: nnnn OPERATION COMPLETE"
110	Repeat steps 60 - 100 for other IDFs that you want to convert to .EB files. If you want to see which exception build files have been generated, press <ESC>. Key in a List command with an .eb suffix (EG: LS NET>HMV1>*.eb).	To review the contents of an .eb file, escape as before, then key in an edit command. <u>EG</u> : ED NET>HMV1>HGPNTS.EB
114	To exit the Text Editor, press F1 twice.	"Do you really wish to abort this edit?"
118	Key in the letter y	Command Processor Menu
122	To return to the Command Menu, hold <CTL> down and press <HELP>.	

Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA